

Project Data Page:

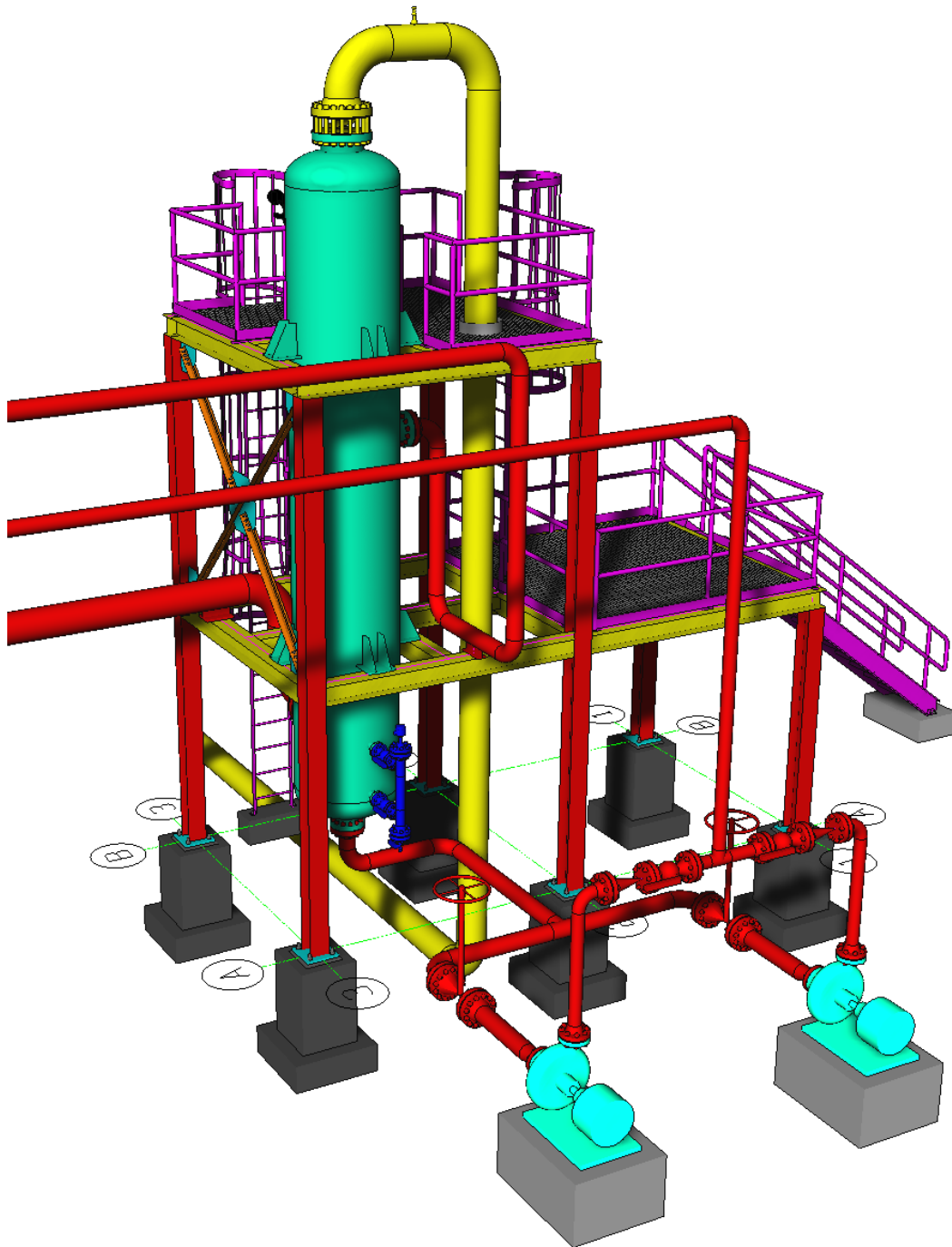




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Co2 Stripper

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Co2 StripperOpR4.pvdb

Date of Analysis : Mar 30,2026 12:22pm

PV Elite 28, January 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 4900.000 x 4906.000 mm ID	
Vessel Design Length, Tangent to Tangent	30050.00 mm
Distance of Bottom Tangent above Grade	26800.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	26500.00 mm
Internal Design Temperature	160 °C
Internal Design Pressure	410.000 kPa
External Design Temperature	160 °C
External Design Pressure	51.709 kPa
Maximum Allowable Working Pressure	523 kPa
External Max. Allowable Working Pressure	54 kPa
Shop Test Pressure	683.807 kPa
Field Test Pressure	533.000 kPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
K-10	...	...	...	1.6000	N/A	No
K-9	...	...	...	1.6000	N/A	No
K-8	...	...	...	1.6000	N/A	No
K-7	...	...	...	1.6000	N/A	No
K-6	...	...	...	1.6000	N/A	No
K-5	...	...	...	1.6000	N/A	No
K-4	...	...	...	1.6000	N/A	No
K-3	...	...	...	1.6000	N/A	No
K-2	...	...	...	1.6000	N/A	No
K-1	...	...	...	1.6000	N/A	No
H-1	453.780	51.71	745.000	6.0000	No	No
S-1	441.782	51.71	752.000	6.0000	N/A	No
S-2	413.552	51.71	780.000	6.0000	N/A	No



S-3	411.200	51.71	671.000	6.0000	N/A	No
S-4	411.200	51.71	671.000	6.0000	N/A	No
S-5	411.200	51.71	671.000	6.0000	N/A	No
S-6	411.200	51.71	671.000	6.0000	N/A	No
S-7	411.200	51.71	671.000	6.0000	N/A	No
S-8	411.200	51.71	671.000	6.0000	N/A	No
S-9	411.200	51.71	618.000	1.0000	N/A	No
S-10	411.200	51.71	618.000	1.0000	N/A	No
S-11	411.200	51.71	523.000	1.0000	N/A	No
H-2	411.200	51.71	617.000	1.0000	No	No

Liquid Level: 4525.00 mm Dens.: 959.560 kg/m³ Sp. Gr.: 0.960

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
1550.00	Bar 150.0 x 18.0	R1
5750.00	Bar 175.0 x 18.0	R2
10450.00	Bar 175.0 x 18.0	R3
14150.00	Bar 175.0 x 18.0	R4
18350.00	Bar 175.0 x 18.0	R5
22550.00	Bar 175.0 x 18.0	R6
26750.00	Bar 175.0 x 18.0	R7

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-24000.0	2500.0	22.0	22.0	...	...	0.70	0.70
Skirt	-21500.0	2500.0	20.0	20.0	...	...	0.70	0.70
Skirt	-19000.0	2500.0	20.0	20.0	...	...	0.70	0.70
Skirt	-16500.0	2500.0	20.0	20.0	...	...	0.70	0.70
Skirt	-14000.0	2500.0	20.0	20.0	...	...	0.70	0.70
Skirt	-11500.0	2500.0	18.0	18.0	...	...	0.70	0.70
Skirt	-9000.0	2500.0	18.0	18.0	...	...	0.70	0.70
Skirt	-6000.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	-3000.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	0.0	3000.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	24.0	20.0	14.1	...	1.00	1.00
Cylinder	3050.0	3000.0	20.0	20.0	13.9	...	1.00	1.00
Cylinder	6050.0	3000.0	20.0	20.0	13.4	...	1.00	1.00
Cylinder	9050.0	3000.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	12050.0	3000.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	15050.0	3000.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	18050.0	3000.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	21050.0	3000.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	23100.0	2050.0	18.0	18.0	13.3	...	1.00	1.00
Cylinder	26100.0	3000.0	14.0	14.0	9.6	...	0.85	0.85
Cylinder	29100.0	3000.0	14.0	14.0	9.6	...	0.85	0.85
Cylinder	30000.0	900.0	12.0	12.0	9.6	...	0.85	0.85
Ellipse	30050.0	50.0	14.0	12.0	8.3	...	1.00	0.85

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kPa	External M.A.W.P. kPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc
70	80	...	No Calc	...	No Calc
80	90	...	No Calc	...	No Calc
90	100	...	No Calc	...	No Calc



100	110	...	No Calc	...	No Calc
110	120	20	15.0757	51.7095	123.056
120	Ring	20	14.2482	51.7095	200.877
Ring	130	20	17.264	51.7095	90.0771
130	Ring	20	17.264	51.7095	90.0771
Ring	140	20	17.8098	51.7095	80.1962
140	150	18	17.8045	51.7095	53.7637
150	Ring	18	17.8045	51.7095	53.7637
Ring	160	18	16.6874	51.7095	68.7004
160	Ring	18	16.6874	51.7095	68.7004
Ring	170	18	17.2586	51.7095	60.2735
170	180	18	17.2586	51.7095	60.2735
180	Ring	18	17.2586	51.7095	60.2735
Ring	190	18	17.2586	51.7095	60.2735
190	Ring	18	17.2586	51.7095	60.2735
Ring	200	18	17.2586	51.7095	60.2734
200	210	14	12.2561	51.7095	74.3936
210	Ring	14	12.2561	51.7095	74.3936
Ring	220	14	11.6953	51.7095	84.6227
220	230	12	11.6902	51.7095	55.5818
230	240	12	10.0573	51.7095	76.2767

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	26500	No Calc	No Calc	No Calc
20	30	26500	No Calc	No Calc	No Calc
30	40	26500	No Calc	No Calc	No Calc
40	50	26500	No Calc	No Calc	No Calc
50	60	26500	No Calc	No Calc	No Calc
60	70	26500	No Calc	No Calc	No Calc
70	80	26500	No Calc	No Calc	No Calc
80	90	26500	No Calc	No Calc	No Calc
90	100	26500	No Calc	No Calc	No Calc
100	110	26500	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	Ring	1958.33	7183.09	No Calc	No Calc
Ring	130	4200	7183.97	1333.26	1602.14
130	Ring	4200	7183.97	No Calc	No Calc
Ring	140	4700	7185.14	1919.59	2392.84
140	150	4700	4881.03	No Calc	No Calc
150	Ring	4700	4881.03	No Calc	No Calc
Ring	160	3700	4879.07	1807.42	2198.2
160	Ring	3700	4879.07	No Calc	No Calc
Ring	170	4200	4881.16	1694.87	2198.2
170	180	4200	4881.16	No Calc	No Calc
180	Ring	4200	4881.16	No Calc	No Calc
Ring	190	4200	4881.16	1807.42	2198.2
190	Ring	4200	4881.16	No Calc	No Calc
Ring	200	4200	4881.16	1807.42	2198.21
200	210	4200	5980.12	No Calc	No Calc
210	Ring	4200	5980.12	No Calc	No Calc
Ring	220	3708.83	5979.03	1693.71	2297.61
220	230	3708.83	3972.4	No Calc	No Calc
230	240	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
C-30"	WRC-107/537	0.648	n/a	Passed
E1-36"	WRC-297	0.998	Shell	Passed
E2-36"	WRC-297	0.998	Shell	Passed



D-30"	WRC-297	0.998	Shell	Passed
A1-20"	WRC-297	0.998	Shell	Passed
A2-20"	WRC-297	0.998	Shell	Passed
B-36"	WRC-107/537	0.967	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	24078.5	20363.4	725234	882112
20	30	3750	22901.9	20361.1	666508	831206
30	40	6250	22115.9	20354.7	610236	780311
40	50	8750	21051	20337.9	556277	729446
50	60	11250	20265	20314.9	504632	678630
60	70	13750	19227.8	20272.2	455266	627896
70	80	16250	18158.8	20218.2	408533	577283
80	90	19000	17372.8	20162.9	364118	526806
90	100	22000	16167.9	20058.5	313807	466474
100	110	25000	14950.1	19926.8	267130	406496
110	120	26525	14006.8	19795.4	223694	346912
120	130	28050	13593.5	18952.9	223004	345944
130	140	31050	12618.9	17413	183686	291395
140	150	34050	11258.9	16310.9	147869	240809
150	160	37050	9886.99	14865	116150	194045
160	170	40050	8504.51	13259.3	88562.7	151859
170	180	43050	7530.69	11778.2	64509.9	114302
180	190	46050	6168.1	10228.3	43961.7	81292.5
190	200	48575	4796.37	8660.55	27515	52959.3
200	210	51100	4130.92	7412.22	18364.5	36484.7
210	220	54100	2767.74	5727.51	8016.51	16775.1
220	230	56050	1396.82	2828.06	1769.68	3941.75
230	240	56525	1105.01	2584.69	643.851	1506

Abs Max of the all of the Stress Ratio's : 0.6423

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	44.0000	mm
Inside Diameter of Basing	4752.0000	mm
Outside Diameter of Basing	5272.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	5104.0000	mm
Number of Bolts	32	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	168.0000	mm
Height of Gussets	256.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm

Basing Sketch for Cylindrical Skirts:**Basing Cross Section View****Loads for Foundation/Support Design:**

Total Wind Shear on Support	31213.	kgf
Total Earthquake Shear on Support	27498.	kgf
Wind Moment on Support	1321541.	kgf-m.
Earthquake Moment on Support	1478419.	kgf-m.
Maximum Initial Bolt Load	23469.	kgf



Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	184443.6	kg
Shop Test - Fabricated + water (full)	781611.4	kg
Shipping - Fab. + rem. intls.+ shipping app.	242080.2	kg
Erected - Fab. + rem. intls.+ insul. (etc)	253380.2	kg
Empty - Fab. + intls. + details + wghts.	284683.8	kg
Operating - Empty + operating liquid (no ca)	524604.3	kg
Field Test - Empty weight + test medium (full)	839247.9	kg

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Nozzle Calculation Summary (kPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
C-30"	745	OK	...	OK	14.33	OK	Passed
L1-2"	752	...	...	OK	9.42	OK	No Calc[*]
MH-4-24"	780	OK	...	OK	14.33	OK	Passed
E1-36"	671	OK	...	OK	14.33	OK	Passed
E2-36"	671	OK	...	OK	14.33	OK	Passed
D-30"	671	OK	...	OK	14.33	OK	Passed
MH-3-24"	671	OK	...	OK	14.33	OK	Passed
UC-2-24"	671	OK	...	OK	14.33	OK	Passed
MH-2-24"	671	OK	...	OK	14.33	OK	Passed
UC-1-24"	671	OK	...	OK	14.33	OK	Passed
MH-1-24"	618	OK	...	OK	9.33	OK	Passed
A1-20"	618	OK	...	OK	9.33	OK	Passed
A2-20"	618	OK	...	OK	9.33	OK	Passed
B-36"	617	OK	...	OK	9.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 617 Nozzle : B-36"
Minimum MAWP Shells/Flanges : 523 Element : S-11
Minimum MAPnc Shells/Flanges : 572 Element : S-11

Computed Vessel M.A.W.P. : 523 kPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
110	C-30"	0.000	0.000	1468.000
120	L1-2"	549.999	0.000	105.890
130	MH-4-24"	4550.001	0.000	1171.200
140	E1-36"	7400.000	45.000	1400.000
140	E2-36"	7400.000	225.000	1400.000
150	D-30"	9600.000	0.000	1200.000
160	MH-3-24"	13550.001	0.000	1171.200
160	UC-2-24"	12800.000	180.000	1171.200
180	MH-2-24"	18550.002	0.000	1171.200
180	UC-1-24"	19600.004	180.000	1171.200
200	MH-1-24"	25300.000	0.000	1151.200
210	A1-20"	28599.994	148.196	948.000
210	A2-20"	28599.994	328.196	948.000
230	B-36"	0.000	0.000	1760.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
L1-2"	2.000 in	XXS	WNF	2.375	11.074	...	...	270.19	150
A1-20"	20.000 in	Actual	WNF	20.000	18.000	1208.00	16.00	1604.79	150
A2-20"	20.000 in	Actual	WNF	20.000	18.000	1208.00	16.00	1604.79	150
MH-4-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	539.03	150
MH-3-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	537.03	150
UC-2-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	537.03	150
MH-2-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	537.03	150
UC-1-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	537.03	150
MH-1-24"	24.000 in	Actual	WNF	24.000	18.000	1009.60	18.00	533.01	150
C-30"	30.000 in	Actual	WNF	30.000	20.000	1362.00	16.00	536.43	150
D-30"	30.000 in	Actual	WNF	30.000	26.000	...	...	561.81	150
E1-36"	36.000 in	Actual	WNF	36.000	28.000	...	...	579.04	150
E2-36"	36.000 in	Actual	WNF	36.000	28.000	...	...	579.04	150
B-36"	36.000 in	Actual	WNF	36.000	18.000	1314.40	18.00	635.68	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
L1-2"	SA-105	18.000	10.000	...	...	...
A1-20"	SA-516 70	16.000	12.000	14.000	16.000	...
A2-20"	SA-516 70	16.000	12.000	14.000	16.000	...
MH-4-24"	SA-516 70	18.000	10.000	14.000	18.000	...
MH-3-24"	SA-516 70	18.000	10.000	14.000	18.000	...
UC-2-24"	SA-516 70	18.000	10.000	14.000	18.000	...
MH-2-24"	SA-516 70	18.000	10.000	14.000	18.000	...
UC-1-24"	SA-516 70	18.000	10.000	14.000	18.000	...
MH-1-24"	SA-516 70	18.000	10.000	14.000	18.000	...
C-30"	SA-516 70	16.000	18.000	12.000	16.000	...
D-30"	SA-516 70	16.000	18.000	...	...	...
E1-36"	SA-516 70	16.000	18.000	...	...	...
E2-36"	SA-516 70	16.000	18.000	...	...	...
B-36"	SA-516 70	12.000	18.000	14.000	18.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
L1-2"	549.999	0.0	250.00	0.00	S-1
A1-20"	28599.992	148.2	1000.00	0.00	S-10
A2-20"	28599.992	328.2	1000.00	0.00	S-10
MH-4-24"	4550.001	0.0	500.00	0.00	S-2
MH-3-24"	13550.000	0.0	500.00	0.00	S-5
UC-2-24"	12800.000	180.0	500.00	0.00	S-5
MH-2-24"	18550.002	0.0	500.00	0.00	S-7
UC-1-24"	19600.002	180.0	500.00	0.00	S-7



MH-1-24"	25299.998	0.0	500.00	0.00	S-9
C-30"	...	0.0	500.00	0.00	H-1
D-30"	9600.000	0.0	500.00	0.00	S-4
E1-36"	7399.999	45.0	500.00	0.00	S-3
E2-36"	7399.999	225.0	500.00	0.00	S-3
B-36"	...	0.0	600.00	0.00	H-2

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 24.0mm THK X 4900.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 4906.0mm ID X 50.0mm	SA-516 70
14	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2500.0mm	...
14	LADDER: 4000.0mm	...
1	INSULATION: 1275mm X 80mm THK	
9	INSULATION: 3000mm X 80mm THK	
1	BAR RING STIFFENER: 4940.0mm ID X 5240.0mm OD X 18.0mm	SA-516 70
1	BAR RING STIFFENER: 4940.0mm ID X 5290.0mm OD X 18.0mm	SA-516 70
3	BAR RING STIFFENER: 4936.0mm ID X 5286.0mm OD X 18.0mm	SA-516 70
1	PACKING: 3000.000mm	...
1	PACKING: 2000.000mm	...
1	PACKING: 1950.000mm	...
1	PACKING: 2049.998mm	...
1	INSULATION: 2050mm X 80mm THK	
1	BAR RING STIFFENER: 4942.0mm ID X 5292.0mm OD X 18.0mm	SA-516 70
1	PACKING: 999.999mm	...
2	LINING: 3000.000mm X 3.000mm THK	...
1	BAR RING STIFFENER: 4930.0mm ID X 5280.0mm OD X 18.0mm	SA-516 70
1	INSULATION: 900mm X 80mm THK	
1	LINING: 900.000mm X 3.000mm THK	...
1	PLATFORM: 1000mm X 6999mm WIDE	
1	INSULATION: 1276mm X 80mm THK	
1	LINING: 1276.500mm X 3.000mm THK	...
6	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 5272.0mm OD X 4752.0 ID X 44.0mm	SA-516 70
32	BASERING BOLTS: 48.0mm DIA	SA-36
32	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 5272.0mm OD X 4936.0 ID X 44.0mm	...
64	BASERING GUSSET PLATES: 168.0mm X 168.0mm X 256.0mm X 16.0mm	...
4	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 3020.8mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 1510.4mm X 2500.0mm	SA-516 70



1	CYLINDRICAL SKIRT SEGMENT 22.0mm THK X 1774.9mm X 2500.0mm	SA-516 70
16	CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 3023.3mm X 2500.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 1511.6mm X 2500.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 20.0mm THK X 1776.3mm X 2500.0mm	SA-516 70
8	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3022.0mm X 2500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1511.0mm X 2500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1794.3mm X 2500.0mm	SA-516 70
8	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3022.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1511.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1794.3mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1524.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 1677.8mm X 3000.0mm	SA-516 70
8	CYLINDRICAL SEGMENT 20.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1524.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1677.8mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 13869.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 10667.8mm X 3000.0mm	SA-516 70
15	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
5	CYLINDRICAL SEGMENT 18.0mm THK X 12345.6mm X 3000.0mm	SA-516 70
5	CYLINDRICAL SEGMENT 18.0mm THK X 1524.0mm X 3000.0mm	SA-516 70
5	CYLINDRICAL SEGMENT 18.0mm THK X 7619.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 12345.6mm X 2050.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 2050.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1524.0mm X 2050.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 7619.9mm X 2050.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 12360.7mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 14.0mm THK X 3051.7mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 1525.8mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 7629.2mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 12366.3mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 14.0mm THK X 3046.1mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 13889.4mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 19981.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 12364.4mm X 900.0mm	SA-516 70



2	CYLINDRICAL SEGMENT 12.0mm THK X 3048.0mm X 900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 1524.0mm X 900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 13888.4mm X 900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 18460.3mm X 900.0mm	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
H-1	[10]	D	-40	-48	-29	0.719	20.000	1.00	No
H-1	[7]	D	-36	-48	-29	0.562	24.000	1.00	No
S-1	[8]	D	-40	-48	-29	0.707	20.000	1.00	No
S-2	[8]	D	-40	-48	-29	0.671	20.000	1.00	No
S-3	[8]	D	-43	-48	-29	0.779	18.000	1.00	No
S-4	[8]	D	-43	-48	-29	0.779	18.000	1.00	No
S-5	[8]	D	-43	-48	-29	0.779	18.000	1.00	No
S-6	[8]	D	-43	-48	-29	0.779	18.000	1.00	No
S-7	[8]	D	-43	-48	-29	0.779	18.000	1.00	No
S-8	[8]	B	-11	-23	-29	0.779	18.000	1.00	No
S-9	[8]	D	-48	-48	-29	0.719	14.000	0.85	No
S-10	[8]	D	-48	-48	-29	0.719	14.000	0.85	No
S-11	[8]	D	-48	-48	-29	0.850	12.000	0.85	No
H-2	[10]	B	-23	-32	-29	0.848	12.000	1.00	No
H-2	[7]	B	-18	-34	-29	0.719	14.000	1.00	No
R1	[23]	D	-43	-48	-29	0.707	18.000	1.00	No
R2	[23]	D	-43	-48	-29	0.671	18.000	1.00	No
R3	[23]	D	-43	-48	-29	0.779	18.000	1.00	No
R4	[23]	D	-43	-48	-29	0.779	18.000	1.00	No
R5	[23]	D	-43	-48	-29	0.779	18.000	1.00	No
R6	[23]	B	-11	-23	-29	0.779	18.000	1.00	No
R7	[23]	D	-48	-48	-29	0.719	14.000	0.85	No
C-30"	[1]	B	-14	-34	-29	0.647	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
L1-2"	[1]	A	-8	-104	-29	0.033	9.690	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH-4-24"	[1]	B	-11	-29	-29	0.668	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
E1-36"	[1]	D	22	-104		0.079	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
E2-36"	[1]	D	22	-104		0.079	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
D-30"	[1]	D	20	-104		0.072	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH-3-24"	[1]	B	-11	-29	-29	0.779	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
UC-2-24"	[1]	B	-11	-29	-29	0.779	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH-2-24"	[1]	B	-11	-29	-29	0.779	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
UC-1-24"	[1]	B	-11	-29	-29	0.779	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH-1-24"	[1]	B	-18	-34	-29	0.719	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A1-20"	[1]	B	-14	-30	-29	0.719	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
A2-20"	[1]	B	-14	-30	-29	0.719	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
B-36"	[1]	B	-23	-36	-29	0.763	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Noz1. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: 22 -23

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.

[4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).

[5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).



- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 100 120 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (QEUP):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	kPa
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Co2 Stripper

Design Internal Pressure (for Hydrotest)	410	kPa
Design Internal Temperature	160.0	°C
Projection of Nozzle from Vessel Top	600	mm
Projection of Nozzle from Vessel Bottom	600	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	Amine	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/23	
Element From Node	10	
Element To Node	20	
Element Type	Skirt Support	
Description	K-10	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm
Skirt Thickness	22	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2473.7	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Platform	
Detail ID	PL-14	
Dist. from "FROM" Node / Offset dist	2500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	25160	cm ²
Platform Weight	1425.1	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm
Element From Node	10	
Detail Type	Weight	
Detail ID	Skirt Bracing	
Dist. from "FROM" Node / Offset dist	9.9994	mm
Miscellaneous Weight	2500	kgf
Offset from Element Centerline	0	mm

Element#	2/23	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	K-9	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm



Skirt Thickness	20	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Weight	
Detail ID	Skirt MW	
Dist. from "FROM" Node / Offset dist	9.9994	mm
Miscellaneous Weight	500	kgf
Offset from Element Centerline	0	mm

Element#	3/23	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	K-8	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Platform	
Detail ID	PL-13	
Dist. from "FROM" Node / Offset dist	1500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	31256	cm ²
Platform Weight	1479.8	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element#	4/23	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	K-7	
Distance "FROM" to "TO"	2500	mm



Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	5/23	
Element From Node	50	
Element To Node	60	
Element Type	Skirt Support	
Description	K-6	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PL-12	
Dist. from "FROM" Node / Offset dist	500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	31256	cm ²
Platform Weight	1479.8	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element#	6/23	
Element From Node	60	
Element To Node	70	
Element Type	Skirt Support	
Description	K-5	
Distance "FROM" to "TO"	2500	mm
Skirt Outside Diameter	4936	mm
Diameter of Skirt at Base	4936	mm
Skirt Thickness	18	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	18	mm



External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Platform	
Detail ID	PL-11	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	31256	cm ²
Platform Weight	1479	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element#	7/23
Element From Node	70
Element To Node	80
Element Type	Skirt Support
Description	K-4
Distance "FROM" to "TO"	2500 mm
Skirt Outside Diameter	4936 mm
Diameter of Skirt at Base	4936 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	8/23
Element From Node	80
Element To Node	90
Element Type	Skirt Support
Description	K-3
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	4936 mm
Diameter of Skirt at Base	4936 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70



Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	80
Detail Type	Platform
Detail ID	PL-10
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	90.0
Platform Wind Area	31256 cm ²
Platform Weight	1479 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm

Element#	9/23
Element From Node	90
Element To Node	100
Element Type	Skirt Support
Description	K-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	4936 mm
Diameter of Skirt at Base	4936 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	90
Detail Type	Platform
Detail ID	PL-9
Dist. from "FROM" Node / Offset dist	2000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	90.0
Platform Wind Area	31256 cm ²
Platform Weight	1479 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm



Element From Node	100
Element To Node	110
Element Type	Skirt Support
Description	K-1
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	4936 mm
Diameter of Skirt at Base	4936 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	160 °C
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	11/23
Element From Node	110
Element To Node	120
Element Type	Elliptical
Description	H-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	4900 mm
Element Thickness	20 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Platform
Detail ID	PL-8
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	46256 cm ²
Platform Weight	5446.2 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm

Element From Node	110
Detail Type	Liquid
Detail ID	L-1
Dist. from "FROM" Node / Offset dist	-1225 mm
Height/Length of Liquid	1275 mm
Liquid Density	959.56 kg/m ³



Element From Node	110
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	-1225 mm
Height/Length of Insulation	1275 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³

Element From Node	110
Detail Type	Nozzle
Detail ID	C-30"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1941 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	110
Detail Type	Weight
Detail ID	Vortex at C
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	300 kgf
Offset from Element Centerline	0 mm

Element From Node	110
Detail Type	Weight
Detail ID	PPC
Dist. from "FROM" Node / Offset dist	-1225 mm
Miscellaneous Weight	7815 kgf
Offset from Element Centerline	3890 mm

Element#	12/23
Element From Node	120
Element To Node	130
Element Type	Cylinder
Description	S-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	20 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	120
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	959.56 kg/m ³

Element From Node	120
Detail Type	Insulation
Detail ID	Insulation



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³

Element From Node	120	
Detail Type	Ring	
Detail ID	R1	
Dist. from "FROM" Node / Offset dist	1500	mm
Inside Diameter of Ring	4940	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	5240	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	120	
Detail Type	Nozzle	
Detail ID	L1-2"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8.152	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	13/23	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	S-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4900	mm
Element Thickness	20	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	410	kPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	51.709	kPa
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	130	
Detail Type	Platform	
Detail ID	PL-7	
Dist. from "FROM" Node / Offset dist	550	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	46256	cm ²
Platform Weight	2797.1	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm



Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	130	
Detail Type	Liquid	
Detail ID	L-3	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	250	mm
Liquid Density	959.56	kg/m ³

Element From Node	130	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³

Element From Node	130	
Detail Type	Ring	
Detail ID	R2	
Dist. from "FROM" Node / Offset dist	2700	mm
Inside Diameter of Ring	4940	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	5290	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	130	
Detail Type	Nozzle	
Detail ID	MH-4-24"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	528.61	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	130	
Detail Type	Weight	
Detail ID	Weld Clip	
Dist. from "FROM" Node / Offset dist	2950	mm
Miscellaneous Weight	11500	kgf
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	Margine Weight	
Dist. from "FROM" Node / Offset dist	2950	mm
Miscellaneous Weight	10000	kgf
Offset from Element Centerline	-1.0101	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	Small Nozzle	
Dist. from "FROM" Node / Offset dist	2950	mm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	0	mm

Element#	14/23
Element From Node	140
Element To Node	150
Element Type	Cylinder



Description	S-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	140
Detail Type	Platform
Detail ID	PL-6
Dist. from "FROM" Node / Offset dist	1550 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	46256 cm ²
Platform Weight	2795.5 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm
Element From Node	140
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³
Element From Node	140
Detail Type	Nozzle
Detail ID	E1-36"
Dist. from "FROM" Node / Offset dist	1350 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	45.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	453.33 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	140
Detail Type	Nozzle
Detail ID	E2-36"
Dist. from "FROM" Node / Offset dist	1350 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	225.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	453.33 kgf
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-516 70
Element From Node	140
Detail Type	Weight
Detail ID	PPE1
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	15795 kgf
Offset from Element Centerline	4085 mm
Element From Node	140
Detail Type	Weight
Detail ID	PPE2
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	15795 kgf
Offset from Element Centerline	4085 mm
Element From Node	140
Detail Type	Weight
Detail ID	Insert pad E1.E
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	500 kgf
Offset from Element Centerline	0 mm

Element#	15/23
Element From Node	150
Element To Node	160
Element Type	Cylinder
Description	S-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	150
Detail Type	Platform
Detail ID	PL-5
Dist. from "FROM" Node / Offset dist	2550 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	46256 cm ²
Platform Weight	2799.5 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.038 kgf/mm
Platform Length (top head platform)	-1.0101 mm

Element From Node	150
Detail Type	Insulation
Detail ID	Insulation



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³

Element From Node	150	
Detail Type	Ring	
Detail ID	R3	
Dist. from "FROM" Node / Offset dist	1400	mm
Inside Diameter of Ring	4936	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	5286	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	150	
Detail Type	Nozzle	
Detail ID	D-30"	
Dist. from "FROM" Node / Offset dist	550	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	330.23	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	150	
Detail Type	Weight	
Detail ID	TAD0-1	
Dist. from "FROM" Node / Offset dist	250	mm
Miscellaneous Weight	350	kgf
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	OPD0-1	
Dist. from "FROM" Node / Offset dist	750	mm
Miscellaneous Weight	15800	kgf
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	DLD0-1	
Dist. from "FROM" Node / Offset dist	250	mm
Miscellaneous Weight	3000	kgf
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Inset pad-D	
Dist. from "FROM" Node / Offset dist	550	mm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	2450	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	PPD	
Dist. from "FROM" Node / Offset dist	550	mm
Miscellaneous Weight	11664	kgf
Offset from Element Centerline	4009	mm



Element Type	Cylinder
Description	S-5
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	160
Detail Type	Packing
Detail ID	Bed#2
Dist. from "FROM" Node / Offset dist	0.001161 mm
Height of Packed Section	3000 mm
Density	166 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.95999998
Element From Node	160
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³
Element From Node	160
Detail Type	Ring
Detail ID	R4
Dist. from "FROM" Node / Offset dist	2100 mm
Inside Diameter of Ring	4936 mm
Thickness of Ring	18 mm
Outside Diameter of Ring	5286 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No
Element From Node	160
Detail Type	Nozzle
Detail ID	MH-3-24"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	528.61 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	160
Detail Type	Nozzle
Detail ID	UC-2-24"
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	528.61 kgf



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	160
Detail Type	Weight
Detail ID	TABS-2
Dist. from "FROM" Node / Offset dist	0.001161 mm
Miscellaneous Weight	300 kgf
Offset from Element Centerline	0 mm
Element From Node	160
Detail Type	Weight
Detail ID	DL-BS-2
Dist. from "FROM" Node / Offset dist	0.001161 mm
Miscellaneous Weight	2000 kgf
Offset from Element Centerline	0 mm

Element#	17/23
Element From Node	170
Element To Node	180
Element Type	Cylinder
Description	S-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	170
Detail Type	Platform
Detail ID	PL-4
Dist. from "FROM" Node / Offset dist	550 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	46256 cm ²
Platform Weight	2795.5 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm

Element From Node	170
Detail Type	Packing
Detail ID	Bed#2
Dist. from "FROM" Node / Offset dist	0.001161 mm
Height of Packed Section	2000 mm
Density	166 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.95999998



Element From Node	170
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³

Element From Node	170
Detail Type	Weight
Detail ID	TALD-2
Dist. from "FROM" Node / Offset dist	2600 mm
Miscellaneous Weight	300 kgf
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	TABL-2
Dist. from "FROM" Node / Offset dist	2000 mm
Miscellaneous Weight	150 kgf
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	OPLD-2
Dist. from "FROM" Node / Offset dist	2750 mm
Miscellaneous Weight	3900 kgf
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	DL-LD-2
Dist. from "FROM" Node / Offset dist	2600 mm
Miscellaneous Weight	1500 kgf
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	DL-BL-2
Dist. from "FROM" Node / Offset dist	2000 mm
Miscellaneous Weight	450 kgf
Offset from Element Centerline	0 mm

Element#	18/23
Element From Node	180
Element To Node	190
Element Type	Cylinder
Description	S-7
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	180
Detail Type	Platform
Detail ID	PL-3



Dist. from "FROM" Node / Offset dist	1550	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	46256	cm ²
Platform Weight	2795.5	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	180	
Detail Type	Packing	
Detail ID	Bed#1	
Dist. from "FROM" Node / Offset dist	1050	mm
Height of Packed Section	1950	mm
Density	166	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.95999998	

Element From Node	180	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³

Element From Node	180	
Detail Type	Ring	
Detail ID	R5	
Dist. from "FROM" Node / Offset dist	300	mm
Inside Diameter of Ring	4936	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	5286	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	180	
Detail Type	Nozzle	
Detail ID	MH-2-24"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	528.61	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	180	
Detail Type	Nozzle	
Detail ID	UC-1-24"	
Dist. from "FROM" Node / Offset dist	1550	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	528.61	kgf
Grade of Attached Flange	GR 1.1	



Nozzle Matl	SA-516 70
Element From Node	180
Detail Type	Weight
Detail ID	TABS1
Dist. from "FROM" Node / Offset dist	1050 mm
Miscellaneous Weight	300 kgf
Offset from Element Centerline	0 mm
Element From Node	180
Detail Type	Weight
Detail ID	DLBS-1
Dist. from "FROM" Node / Offset dist	1050 mm
Miscellaneous Weight	2000 kgf
Offset from Element Centerline	0 mm

Element#	19/23
Element From Node	190
Element To Node	200
Element Type	Cylinder
Description	S-8
Distance "FROM" to "TO"	2050 mm
Inside Diameter	4900 mm
Element Thickness	18 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	190
Detail Type	Packing
Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	2050 mm
Density	166 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.95999998

Element From Node	190
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2050 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³

Element From Node	190
Detail Type	Ring
Detail ID	R6
Dist. from "FROM" Node / Offset dist	1500 mm
Inside Diameter of Ring	4942 mm
Thickness of Ring	18 mm
Outside Diameter of Ring	5292 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No



Element#	20/23
Element From Node	200
Element To Node	210
Element Type	Cylinder
Description	S-9
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4906 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	200
Detail Type	Platform
Detail ID	PL-2
Dist. from "FROM" Node / Offset dist	500 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	46256 cm ²
Platform Weight	2794.7 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm
Element From Node	200
Detail Type	Packing
Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	0.002326 mm
Height of Packed Section	1000 mm
Density	166 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.95999998
Element From Node	200
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³
Element From Node	200
Detail Type	Lining
Detail ID	Cald
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³
Element From Node	200
Detail Type	Nozzle
Detail ID	MH-1-24"



Dist. from "FROM" Node / Offset dist	2200	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	528.61	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	200	
Detail Type	Weight	
Detail ID	TAGLR	
Dist. from "FROM" Node / Offset dist	2700	mm
Miscellaneous Weight	1500	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	TALD-1	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	300	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	TABL-1	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	OPLD-1	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	3900	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	DLBL-1	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	450	kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	DLDD-1	
Dist. from "FROM" Node / Offset dist	1600	mm
Miscellaneous Weight	1900	kgf
Offset from Element Centerline	0	mm

Element#	21/23	
Element From Node	210	
Element To Node	220	
Element Type	Cylinder	
Description	S-10	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4906	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	410	kPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	51.709	kPa
Design Temperature External Pressure	160	°C



Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	
Element From Node	210	
Detail Type	Platform	
Detail ID	PL-1	
Dist. from "FROM" Node / Offset dist	1500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	46256	cm ²
Platform Weight	2794.7	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	210	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³
Element From Node	210	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³
Element From Node	210	
Detail Type	Ring	
Detail ID	R7	
Dist. from "FROM" Node / Offset dist	650	mm
Inside Diameter of Ring	4930	mm
Thickness of Ring	18	mm
Outside Diameter of Ring	5280	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	A1-20"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	148.196	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	377.11	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	A2-20"	



Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	328.19601	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	431.6	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	210	
Detail Type	Weight	
Detail ID	Shroud	
Dist. from "FROM" Node / Offset dist	2600	mm
Miscellaneous Weight	3900	kgf
Offset from Element Centerline	0	mm

Element From Node	210	
Detail Type	Weight	
Detail ID	OPGLR	
Dist. from "FROM" Node / Offset dist	-0.002326	mm
Miscellaneous Weight	2700	kgf
Offset from Element Centerline	0	mm

Element From Node	210	
Detail Type	Weight	
Detail ID	Trunnion	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	2000	kgf
Offset from Element Centerline	0	mm

Element From Node	210	
Detail Type	Weight	
Detail ID	PPA1	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	10923	kgf
Offset from Element Centerline	3882	mm

Element From Node	210	
Detail Type	Weight	
Detail ID	PPA2	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	10923	kgf
Offset from Element Centerline	3882	mm

Element#	22/23	
Element From Node	220	
Element To Node	230	
Element Type	Cylinder	
Description	S-11	
Distance "FROM" to "TO"	900	mm
Inside Diameter	4906	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	410	kPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	51.709	kPa
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	220	
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Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	900 mm
Thickness of Insulation	80 mm
Density	352 kg/m ³

Element From Node	220
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	900 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	220
Detail Type	Weight
Detail ID	Pipe Davit
Dist. from "FROM" Node / Offset dist	450 mm
Miscellaneous Weight	1300 kgf
Offset from Element Centerline	2450 mm

Element#	23/23
Element From Node	230
Element To Node	240
Element Type	Elliptical
Description	H-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	4906 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	410 kPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.709 kPa
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	230
Detail Type	Platform
Detail ID	TOP PL
Dist. from "FROM" Node / Offset dist	1500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	86256 cm ²
Platform Weight	8482.6 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	7000 mm
Platform Height	1000 mm
Platform Clearance or End Offset	3500 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	7000 mm

Element From Node	230
Detail Type	Insulation
Detail ID	Insulation



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1276.5	mm
Thickness of Insulation	80	mm
Density	352	kg/m ³

Element From Node	230	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1276.5	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	230	
Detail Type	Nozzle	
Detail ID	B-36"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	470.35	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	230	
Detail Type	Weight	
Detail ID	PPB	
Dist. from "FROM" Node / Offset dist	1225	mm
Miscellaneous Weight	20966	kgf
Offset from Element Centerline	4085	mm

Element From Node	230	
Detail Type	For./Mom.	
Detail ID	BFxFz	
Dist. from "FROM" Node / Offset dist	1225	mm
Force in X Direction	5045	kgf
Force in Y Direction	-7568	kgf
Force in Z Direction	5045	kgf
Moment about X Axis	12716	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	12716	kgf-m.
Force/Moment Combination Method	SRSS	

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	K-10	...	2500	...	2500
	K-9	...	5000	...	2500
	K-8	...	7500	...	2500
	K-7	...	10000	...	2500
	K-6	...	12500	...	2500
	K-5	...	15000	...	2500
	K-4	...	17500	...	2500
	K-3	...	20500	...	3000
	K-2	...	23500	...	3000
	K-1	...	26500	...	3000
	H-1	...	26550	...	50
	S-1	...	29550	...	3000
	S-2	...	32550	...	3000
	S-3	...	35550	...	3000
	S-4	...	38550	...	3000
	S-5	...	41550	...	3000
	S-6	...	44550	...	3000
	S-7	...	47550	...	3000
	S-8	...	49600	...	2050
	S-9	...	52600	...	3000
	S-10	...	55600	...	3000
	S-11	...	56500	...	900
	H-2	...	56550	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd kPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
	K-10	...	22	1.6	4936	...	1
	K-9	...	20	1.6	4936	...	2
	K-8	...	20	1.6	4936	...	3
	K-7	...	20	1.6	4936	...	4
	K-6	...	20	1.6	4936	...	5
	K-5	...	18	1.6	4936	...	6
	K-4	...	18	1.6	4936	...	7
	K-3	...	18	1.6	4936	...	8
	K-2	...	18	1.6	4936	...	9
	K-1	...	18	1.6	4936	...	10
	H-1	453.78	24	6	4900	1407	11
	S-1	441.78	20	6	4900	1407	12
	S-2	413.55	20	6	4900	1407	13
	S-3	411.2	18	6	4900	1407	14
	S-4	411.2	18	6	4900	1407	15
	S-5	411.2	18	6	4900	1407	16
	S-6	411.2	18	6	4900	1407	17
	S-7	411.2	18	6	4900	1407	18
	S-8	411.2	18	6	4900	1407.1	19
	S-9	411.2	14	1	4906	1196	20
	S-10	411.2	14	1	4906	1196	21
	S-11	411.2	12	1	4906	1196	22
	H-2	411.2	14	1	4906	1407.1	23

Element Required Thickness and MAWP :

From	To	Design Pressure kPa	M.A.W.P. Corroded kPa	M.A.P. New & Cold kPa	Minimum Thickness mm	Required Thickness mm	Element#
	K-10	...	No Calc	No Calc	22	No Calc	1
	K-9	...	No Calc	No Calc	20	No Calc	2
	K-8	...	No Calc	No Calc	20	No Calc	3
	K-7	...	No Calc	No Calc	20	No Calc	4
	K-6	...	No Calc	No Calc	20	No Calc	5
	K-5	...	No Calc	No Calc	18	No Calc	6
	K-4	...	No Calc	No Calc	18	No Calc	7
	K-3	...	No Calc	No Calc	18	No Calc	8
	K-2	...	No Calc	No Calc	18	No Calc	9
	K-1	...	No Calc	No Calc	18	No Calc	10
	H-1	410	744.84	1125.42	20	14.0538	11
	S-1	410	752.047	1120.85	20	13.8789	12
	S-2	410	780.277	1120.85	20	13.3746	13
	S-3	410	670.981	1009.25	18	13.3325	14
	S-4	410	670.981	1009.25	18	13.3325	15
	S-5	410	670.981	1009.25	18	13.3325	16
	S-6	410	670.981	1009.25	18	13.3325	17
	S-7	410	670.981	1009.25	18	13.3325	18
	S-8	410	671.045	1009.35	18	13.3318	19
	S-9	410	618.119	667.067	14	9.62221	20
	S-10	410	618.119	667.067	14	9.62221	21
	S-11	410	523.094	572.051	12	9.62221	22
	H-2	410	617.393	674.71	12	8.31097	23

Minimum

523

572

MAWP: 523 kPa, limited by: S-11.

Elements Suitable for Design Internal Pressure.



**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: H-1**

Elliptical Head Design Information:							
Design Pressure	P	453.780	kPa	Int. Design Temperature		160	°C
Material		SA-516	70	Ext. Design Temperature		160	°C
Length		50.000	mm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	6.0000	mm	Allowable Stress (ope)	S	1407.0	kgf/cm ²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1407.0	kgf/cm ²
Inside Diameter	Di	4900.000	mm	Actual Stress		809.6	kgf/cm ²
Spec. Min. Thk t _e , or s _a		20.0000	mm	Spec. Nominal Thk.		24.000	mm
Surface Area		273161.3	cm ²	SG of Contents		0.960	
Element Volume		16343090.0	cm ³	Weight of Contents		15682.3	kgf
Empty Weight		5541.3	kgf	Operating Weight		21223.6	kgf
				Comp. All. Ope. Stress		860.8	kgf/cm ²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (453.78 \cdot 4912.0 \cdot 0.997) / (2 \cdot 1407.0 \cdot 1.0 - 0.2 \cdot 453.78)$$

$$= 8.0538 + 6.0000 = 14.0538 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 43.780 kPa](#)

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.0 \cdot 1.0 \cdot 14.0) / (0.997 \cdot 4912.0 + 0.2 \cdot 14.0)$$

$$= 789 - 44 = 745 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.0 \cdot 1.0 \cdot 20.0) / (1.0 \cdot 4900.0 + 0.2 \cdot 20.0)$$

$$= 1125 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (453.78 \cdot (0.997 \cdot 4912.0 + 0.2 \cdot 14.0)) / (2 \cdot 1.0 \cdot 14.0)$$

$$= 809.602 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (453.78 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 453.78) + 6.0$$

$$= 14.093 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 32.253 kPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2456.0 + 0.6 \cdot 18.0)$$

$$= 1007 - 32 = 975 \text{ kPa}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (4912.0 / (2 \cdot 1231.0))^2) / 6$$

$$= 0.996755$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.130 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 20.0$, $t_r = 10.062$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 24.0$, $t_r = 10.115$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.562$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	566.874 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-40 °C
Stress/Thk. Ratio	0.719	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.625 mm

EXTERNAL PRESSURE RESULTS for: H-1

Elliptical Head From 110 to 120 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
14.000	4940.00	352.86	0.0003936	398.50

$$MAEP = B / (K_0 \cdot Do / t) = 398.5048 / (0.9 \cdot 352.8571) = 123.056 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
9.076	4940.00	544.31	0.0002552	258.34

$$MAEP = B / (K_0 \cdot Do / t) = 258.3374 / (0.9 \cdot 544.3085) = 51.7143 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-1**

Cylindrical Shell Design Information:					
Design Pressure	P	441.782 kPa	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	793.0 kgf/cm²	
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		465584.1 cm²	SG of Contents	0.960	
Element Volume		56572236.0 cm³	Weight of Contents	54285.0 kgf	
Empty Weight		7187.3 kgf	Operating Weight	61472.2 kgf	
			Comp. All. Ope. Stress	860.8 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (441.782 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 441.782)$$

$$= 7.8789 + 6.0000 = 13.8789 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 31.782 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 14.0) / (2456.0 + 0.6 \cdot 14.0)$$

$$= 784 - 32 = 752 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 20.0) / (2450.0 + 0.6 \cdot 20.0)$$

$$= 1121 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (441.782 \cdot (2456.0 + 0.6 \cdot 14.0)) / (1.0 \cdot 14.0)$$

$$= 793.014 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.407 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20.0, t_r = 9.901, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.707, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	554.877 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-40 °C

Stress/Thk. Ratio	0.707	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-1

Cylindrical Shell From 120 to R1 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	1958.33	352.86	0.3964	0.0005354	542.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*542.1006)/(3*352.8571) = 200.877 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.248	4940.00	1958.33	598.92	0.3964	0.0002340	236.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*236.8718)/(3*598.9164) = 51.7126 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	7183.09	352.86	1.4541	0.0001380	139.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*139.677)/(3*352.8571) = 51.7577 \text{ kPa}$$

Cylindrical Shell From R1 to 130 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	4200.00	352.86	0.8502	0.0002401	243.09

$$MAEP = (4*B)/(3*(Do/t)) = (4*243.0882)/(3*352.8571) = 90.0771 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.264	4940.00	4200.00	438.57	0.8502	0.0001713	173.45

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.4488)/(3*438.5662) = 51.7113 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	7183.97	352.86	1.4542	0.0001379	139.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*139.6594)/(3*352.8571) = 51.7512 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-2**

Cylindrical Shell Design Information:					
Design Pressure	P	413.552 kPa	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	742.3 kgf/cm²	
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		465584.1 cm²	SG of Contents	0.960	
Element Volume		56572236.0 cm³	Weight of Contents	4523.7 kgf	
Empty Weight		7187.3 kgf	Operating Weight	11711.0 kgf	
			Comp. All. Ope. Stress	860.8 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (413.552 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 413.552)$$

$$= 7.3746 + 6.0000 = 13.3746 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 3.552 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 14.0) / (2456.0 + 0.6 \cdot 14.0)$$

$$= 784 - 4 = 780 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 20.0) / (2450.0 + 0.6 \cdot 20.0)$$

$$= 1121 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (413.552 \cdot (2456.0 + 0.6 \cdot 14.0)) / (1.0 \cdot 14.0)$$

$$= 742.341 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.407 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 20.0, t_r = 9.396, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.671, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	526.647 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-40 °C

Stress/Thk. Ratio	0.671	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-2

Cylindrical Shell From 130 to R2 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	4200.00	352.86	0.8502	0.0002401	243.09

$$MAEP = (4*B)/(3*(Do/t)) = (4*243.0882)/(3*352.8571) = 90.0771 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.264	4940.00	4200.00	438.57	0.8502	0.0001713	173.45

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.4488)/(3*438.5662) = 51.7113 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	7183.97	352.86	1.4542	0.0001379	139.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*139.6594)/(3*352.8571) = 51.7512 \text{ kPa}$$

Cylindrical Shell From R2 to 140 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	4700.00	352.86	0.9514	0.0002138	216.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*216.423)/(3*352.8571) = 80.1962 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.810	4940.00	4700.00	418.30	0.9514	0.0001634	165.43

$$MAEP = (4*B)/(3*(Do/t)) = (4*165.4327)/(3*418.2965) = 51.7114 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.000	4940.00	7185.14	352.86	1.4545	0.0001379	139.64

$$MAEP = (4*B)/(3*(Do/t)) = (4*139.6358)/(3*352.8571) = 51.7425 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-3**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		160 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress		860.7 kgf/cm²
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		465207.1 cm²	SG of Contents		0.000
Element Volume		56572236.0 cm³	Weight of Contents		0.0 kgf
Empty Weight		6466.0 kgf	Operating Weight		6466.0 kgf
			Comp. All. Ope. Stress		738.4 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3325 + 6.0000 = 13.3325 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-43 °C



Stress/Thk. Ratio	0.779	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-3

Cylindrical Shell From 140 to 150 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4700.00	411.33	0.9522	0.0001671	169.13

$$MAEP = (4*B)/(3*(Do/t)) = (4*169.135)/(3*411.3333) = 53.7637 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.804	4936.00	4700.00	418.15	0.9522	0.0001633	165.38

$$MAEP = (4*B)/(3*(Do/t)) = (4*165.3787)/(3*418.1459) = 51.7131 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.03	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7626)/(3*411.3333) = 51.7381 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-4**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	860.7 kgf/cm²	
Spec. Min. Thk t _e , or sa		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		465207.1 cm²	SG of Contents	0.000	
Element Volume		56572236.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		6466.0 kgf	Operating Weight	6466.0 kgf	
			Comp. All. Ope. Stress	738.4 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3325 + 6.0000 = 13.3325 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-43 °C

Stress/Thk. Ratio	0.779	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-4

Cylindrical Shell From 150 to R3 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4700.00	411.33	0.9522	0.0001671	169.13

$$MAEP = (4*B)/(3*(Do/t)) = (4*169.135)/(3*411.3333) = 53.7637 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.804	4936.00	4700.00	418.15	0.9522	0.0001633	165.38

$$MAEP = (4*B)/(3*(Do/t)) = (4*165.3787)/(3*418.1459) = 51.7131 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.03	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7626)/(3*411.3333) = 51.7381 \text{ kPa}$$

Cylindrical Shell From R3 to 160 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	3700.00	411.33	0.7496	0.0002135	216.12

$$MAEP = (4*B)/(3*(Do/t)) = (4*216.1244)/(3*411.3333) = 68.7004 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.687	4936.00	3700.00	461.85	0.7496	0.0001804	182.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*182.6573)/(3*461.8515) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4879.07	411.33	0.9885	0.0001608	162.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.8291)/(3*411.3333) = 51.7592 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-5**

Cylindrical Shell Design Information:				
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C
Material		SA-516 70	Ext. Design Temperature	160 °C
Length		3000.000 mm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope) S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	860.7 kgf/cm²
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area		465207.1 cm²	SG of Contents	0.000
Element Volume		56572236.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		6466.0 kgf	Operating Weight	6466.0 kgf
			Comp. All. Ope. Stress	738.4 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3325 + 6.0000 = 13.3325 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-43 °C

Stress/Thk. Ratio	0.779	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-5

Cylindrical Shell From 160 to R4 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	3700.00	411.33	0.7496	0.0002135	216.12

$$MAEP = (4*B)/(3*(Do/t)) = (4*216.1244)/(3*411.3333) = 68.7004 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.687	4936.00	3700.00	461.85	0.7496	0.0001804	182.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*182.6573)/(3*461.8515) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4879.07	411.33	0.9885	0.0001608	162.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.8291)/(3*411.3333) = 51.7592 \text{ kPa}$$

Cylindrical Shell From R4 to 170 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6142)/(3*411.3333) = 60.2735 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3902)/(3*438.4197) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7584)/(3*411.3333) = 51.7367 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-6**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		160 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress		860.7 kgf/cm²
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		465207.1 cm²	SG of Contents		0.000
Element Volume		56572236.0 cm³	Weight of Contents		0.0 kgf
Empty Weight		6466.0 kgf	Operating Weight		6466.0 kgf
			Comp. All. Ope. Stress		738.4 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3325 + 6.0000 = 13.3325 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-43 °C



Stress/Thk. Ratio	0.779	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-6

Cylindrical Shell From 170 to 180 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6142)/(3*411.3333) = 60.2735 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3902)/(3*438.4197) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7584)/(3*411.3333) = 51.7367 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-7**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	860.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		465207.1 cm²	SG of Contents	0.000	
Element Volume		56572236.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		6466.0 kgf	Operating Weight	6466.0 kgf	
			Comp. All. Ope. Stress	738.4 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.0 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3325 + 6.0000 = 13.3325 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-43 °C

Stress/Thk. Ratio	0.779	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-7

Cylindrical Shell From 180 to R5 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6142)/(3*411.3333) = 60.2735 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3902)/(3*438.4197) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7584)/(3*411.3333) = 51.7367 \text{ kPa}$$

Cylindrical Shell From R5 to 190 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6143)/(3*411.3333) = 60.2735 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3903)/(3*438.4198) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7583)/(3*411.3333) = 51.7367 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-8**

Cylindrical Shell Design Information:				
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C
Material		SA-516 70	Ext. Design Temperature	160 °C
Length		2050.000 mm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope) S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm²
Inside Diameter	Di	4900.000 mm	Actual Stress	860.7 kgf/cm²
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area		317891.5 cm²	SG of Contents	0.000
Element Volume		38657696.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		4418.7 kgf	Operating Weight	4418.7 kgf
			Comp. All. Ope. Stress	738.4 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2456.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 411.2)$$

$$= 7.3318 + 6.0000 = 13.3318 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (2456.0 + 0.6 \cdot 12.0)$$

$$= 672 - 1 = 671 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 18.0) / (2450.0 + 0.6 \cdot 18.0)$$

$$= 1009 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2456.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 860.718 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.366 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 9.353, c = 6.0 mm, E* = 1.0

Thickness Ratio = t_r * E* / (t_g - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C



Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.779	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: S-8**Cylindrical Shell From 190 to R6 Ext. Chart: CS-2 at 160 °C**Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6143)/(3*411.3333) = 60.2735 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3903)/(3*438.4198) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7583)/(3*411.3333) = 51.7367 \text{ kPa}$$

Cylindrical Shell From R6 to 200 Ext. Chart: CS-2 at 160 °CElastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4200.00	411.33	0.8509	0.0001873	189.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*189.6141)/(3*411.3333) = 60.2734 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.259	4936.00	4200.00	438.42	0.8509	0.0001713	173.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3901)/(3*438.4196) = 51.7111 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	4936.00	4881.16	411.33	0.9889	0.0001608	162.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*162.7583)/(3*411.3333) = 51.7367 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-9**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		160 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm ²
Inside Diameter	Di	4906.000 mm	Actual Stress		934.2 kgf/cm ²
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		465018.5 cm ²	SG of Contents		0.000
Element Volume		56710864.0 cm ³	Weight of Contents		0.0 kgf
Empty Weight		5031.1 kgf	Operating Weight		5031.1 kgf
			Comp. All. Ope. Stress		800.3 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2454.0) / (1407.0 \cdot 0.85 - 0.6 \cdot 411.2)$$

$$= 8.6222 + 1.0000 = 9.6222 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 13.0) / (2454.0 + 0.6 \cdot 13.0)$$

$$= 619 - 1 = 618 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 14.0) / (2453.0 + 0.6 \cdot 14.0)$$

$$= 667 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2454.0 + 0.6 \cdot 13.0)) / (0.85 \cdot 13.0)$$

$$= 934.185 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.285 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14.0, t_r = 11.0, c = 1.0 mm, E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.719, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D

-48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.719	Temperature Difference	0 °C

UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: S-9

Cylindrical Shell From 200 to 210 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	4200.00	379.54	0.8512	0.0002133	215.94

$$MAEP = (4*B)/(3*(Do/t)) = (4*215.9445)/(3*379.5384) = 74.3936 \text{ kPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.256	4934.00	4200.00	438.34	0.8512	0.0001712	173.36

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3642)/(3*438.3404) = 51.7127 \text{ kPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	5980.12	379.54	1.2120	0.0001483	150.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*150.1795)/(3*379.5384) = 51.7373 \text{ kPa}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-10**

Cylindrical Shell Design Information:				
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C
Material		SA-516 70	Ext. Design Temperature	160 °C
Length		3000.000 mm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.0 kgf/cm²
Inside Diameter	Di	4906.000 mm	Actual Stress	934.2 kgf/cm²
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area		465018.5 cm²	SG of Contents	0.000
Element Volume		56710864.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		5031.1 kgf	Operating Weight	5031.1 kgf
			Comp. All. Ope. Stress	800.3 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2454.0) / (1407.0 \cdot 0.85 - 0.6 \cdot 411.2)$$

$$= 8.6222 + 1.0000 = 9.6222 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 13.0) / (2454.0 + 0.6 \cdot 13.0)$$

$$= 619 - 1 = 618 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 14.0) / (2453.0 + 0.6 \cdot 14.0)$$

$$= 667 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2454.0 + 0.6 \cdot 13.0)) / (0.85 \cdot 13.0)$$

$$= 934.185 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.285 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14.0, t_r = 11.0, c = 1.0 mm, E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.719, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D

-48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.719	Temperature Difference	0 °C

UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: S-10

Cylindrical Shell From 210 to R7 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	4200.00	379.54	0.8512	0.0002133	215.94

$$MAEP = (4*B)/(3*(Do/t)) = (4*215.9445)/(3*379.5384) = 74.3936 \text{ kPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.256	4934.00	4200.00	438.34	0.8512	0.0001712	173.36

$$MAEP = (4*B)/(3*(Do/t)) = (4*173.3642)/(3*438.3404) = 51.7127 \text{ kPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	5980.12	379.54	1.2120	0.0001483	150.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*150.1795)/(3*379.5384) = 51.7373 \text{ kPa}$$

Cylindrical Shell From R7 to 220 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	3708.83	379.54	0.7517	0.0002426	245.64

$$MAEP = (4*B)/(3*(Do/t)) = (4*245.6367)/(3*379.5384) = 84.6227 \text{ kPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.695	4934.00	3708.83	461.33	0.7517	0.0001802	182.45

$$MAEP = (4*B)/(3*(Do/t)) = (4*182.4528)/(3*461.3255) = 51.7121 \text{ kPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4934.00	5979.03	379.54	1.2118	0.0001484	150.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*150.2078)/(3*379.5384) = 51.7471 \text{ kPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: S-11**

Cylindrical Shell Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		160 °C
Length		900.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1407.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.0 kgf/cm²
Inside Diameter	Di	4906.000 mm	Actual Stress		1103.5 kgf/cm²
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		139392.5 cm²	SG of Contents		0.000
Element Volume		17013260.0 cm³	Weight of Contents		0.0 kgf
Empty Weight		1293.2 kgf	Operating Weight		1293.2 kgf
			Comp. All. Ope. Stress		677.7 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2454.0) / (1407.0 \cdot 0.85 - 0.6 \cdot 411.2)$$

$$= 8.6222 + 1.0000 = 9.6222 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 11.0) / (2454.0 + 0.6 \cdot 11.0)$$

$$= 524 - 1 = 523 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.0 \cdot 0.85 \cdot 12.0) / (2453.0 + 0.6 \cdot 12.0)$$

$$= 572 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (411.2 \cdot (2454.0 + 0.6 \cdot 11.0)) / (0.85 \cdot 11.0)$$

$$= 1103.499 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.244 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 11.0, c = 1.0 mm, E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D

-48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	524.294 kPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.850	Temperature Difference	0 °C

UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: S-11

Cylindrical Shell From 220 to 230 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4930.00	3708.83	448.18	0.7523	0.0001882	190.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*190.5188)/(3*448.1818) = 55.5818 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.690	4930.00	3708.83	461.17	0.7523	0.0001802	182.39

$$MAEP = (4*B)/(3*(Do/t)) = (4*182.3936)/(3*461.1719) = 51.7125 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	4930.00	3972.40	448.18	0.8058	0.0001752	177.41

$$MAEP = (4*B)/(3*(Do/t)) = (4*177.4142)/(3*448.1818) = 51.7587 \text{ kPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: H-2

Elliptical Head Design Information:					
Design Pressure	P	411.200 kPa	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	4906.000 mm	Actual Stress	935.4 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		271639.6 cm ²	SG of Contents	0.000	
Element Volume		16402043.0 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		3227.4 kgf	Operating Weight	3227.4 kgf	
			Comp. All. Ope. Stress	677.7 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.325 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.663 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (411.2 \cdot 4908.0 \cdot 0.999) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 411.2)$$

$$= 7.3110 + 1.0000 = 8.3110 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 11.0) / (0.999 \cdot 4908.0 + 0.2 \cdot 11.0)$$

$$= 619 - 1 = 617 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 4906.0 + 0.2 \cdot 12.0)$$

$$= 675 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (411.2 \cdot (0.999 \cdot 4908.0 + 0.2 \cdot 11.0)) / (2 \cdot 1.0 \cdot 11.0)$$

$$= 935.370 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (411.2 \cdot 2454.0) / (1407.13 \cdot 1.0 - 0.6 \cdot 411.2) + 1.0$$

$$= 8.326 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 1.200 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 13.0) / (2454.0 + 0.6 \cdot 13.0)$$

$$= 729 - 1 = 727 \text{ kPa}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (4908.0 / (2 \cdot 1227.5))^2) / 6$$

$$= 0.999457$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.248 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 12.0$, $t_r = 9.323$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.848$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 14.0$, $t_r = 9.345$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	524.294 kPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.848	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-32 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 61.325 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 30.663 mm

EXTERNAL PRESSURE RESULTS for: H-2

Elliptical Head From 230 to 240 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
11.000	4930.00	448.18	0.0003099	313.75

$$MAEP = B / (K_0 \cdot Do / t) = 313.746 / (0.9 \cdot 448.1818) = 76.2767 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
9.057	4930.00	544.31	0.0002552	258.34

$$MAEP = B / (K_0 \cdot Do / t) = 258.3369 / (0.9 \cdot 544.3098) = 51.714 \text{ kPa}$$

**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	680.023 kPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	533.000 kPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	683.807 kPa
UG100	= 1.10 * M.A.W.P. * Sa/S	575.404 kPa
PED	= max(1.43*DP, 1.25*DP*ratio)	586.300 kPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	586.300 kPa
App 27-4	= M.A.W.P.	523.094 kPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	914.9	2386.0	0.383	731.84
S-1	918.7	2386.0	0.385	731.84
S-2	918.7	2386.0	0.385	731.84
S-3	1020.3	2386.0	0.428	731.84
S-4	1020.3	2386.0	0.428	731.84
S-5	1020.3	2386.0	0.428	731.84
S-6	1020.3	2386.0	0.428	731.84
S-7	1020.3	2386.0	0.428	731.84
S-8	1020.3	2386.0	0.428	731.84
S-9	1543.7	2386.0	0.647	731.90
S-10	1543.7	2386.0	0.647	731.90
S-11	1800.2	2386.0	0.754	731.90
H-2	1526.4	2386.0	0.640	731.90

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
C-30"	Nozzle	1407.00	1407.00	1.000
C-30"	Pad	1407.00	1407.00	1.000
L1-2"	Nozzle	1407.00	1407.00	1.000
MH-4-24"	Nozzle	1407.00	1407.00	1.000
MH-4-24"	Pad	1407.00	1407.00	1.000
E1-36"	Nozzle	1407.00	1407.00	1.000
E2-36"	Nozzle	1407.00	1407.00	1.000
D-30"	Nozzle	1407.00	1407.00	1.000
MH-3-24"	Nozzle	1407.00	1407.00	1.000
MH-3-24"	Pad	1407.00	1407.00	1.000
UC-2-24"	Nozzle	1407.00	1407.00	1.000
UC-2-24"	Pad	1407.00	1407.00	1.000
MH-2-24"	Nozzle	1407.00	1407.00	1.000
MH-2-24"	Pad	1407.00	1407.00	1.000
UC-1-24"	Nozzle	1407.00	1407.00	1.000
UC-1-24"	Pad	1407.00	1407.00	1.000
MH-1-24"	Nozzle	1407.00	1407.00	1.000
MH-1-24"	Pad	1407.00	1407.00	1.000
A1-20"	Nozzle	1407.00	1407.00	1.000
A1-20"	Pad	1407.00	1407.00	1.000
A2-20"	Nozzle	1407.00	1407.00	1.000
A2-20"	Pad	1407.00	1407.00	1.000
B-36"	Nozzle	1407.00	1407.00	1.000
B-36"	Pad	1407.00	1407.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (kgf/cm²):



Description	Ambient	Operating	Ratio
R1	1407.00	1407.00	1.000
R2	1407.00	1407.00	1.000
R3	1407.00	1407.00	1.000
R4	1407.00	1407.00	1.000
R5	1407.00	1407.00	1.000
R6	1407.00	1407.00	1.000
R7	1407.00	1407.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1407.00	1407.00	1.000
S-1	1407.00	1407.00	1.000
S-2	1407.00	1407.00	1.000
S-3	1407.00	1407.00	1.000
S-4	1407.00	1407.00	1.000
S-5	1407.00	1407.00	1.000
S-6	1407.00	1407.00	1.000
S-7	1407.00	1407.00	1.000
S-8	1407.13	1407.13	1.000
S-9	1407.00	1407.00	1.000
S-10	1407.00	1407.00	1.000
S-11	1407.00	1407.00	1.000
H-2	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
C-30"	139.18	2386.00	0.058
L1-2"	20.24	2294.00	0.009
MH-4-24"	123.39	2386.00	0.052
E1-36"	118.87	2386.00	0.050
E2-36"	118.87	2386.00	0.050
D-30"	106.37	2386.00	0.045
MH-3-24"	123.39	2386.00	0.052
UC-2-24"	123.39	2386.00	0.052
MH-2-24"	123.39	2386.00	0.052
UC-1-24"	123.39	2386.00	0.052
MH-1-24"	123.40	2386.00	0.052
A1-20"	102.33	2386.00	0.043
A2-20"	102.33	2386.00	0.043
B-36"	186.59	2386.00	0.078

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	680.023 kPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	533.000 kPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	716.442 kPa
UG100	= 1.10 * M.A.W.P. * Sa/S	575.404 kPa
PED	= max(1.43*DP, 1.25*DP*ratio)	586.300 kPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	586.300 kPa
App 27-4	= M.A.W.P.	523.094 kPa

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 410.0 • 1.0

= 533.000 kPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	1529.8	2386.0	0.641	857.46
S-1	1516.7	2386.0	0.636	844.96
S-2	1464.0	2386.0	0.614	815.56
S-3	1645.6	2386.0	0.690	786.15
S-4	1584.0	2386.0	0.664	756.75
S-5	1522.5	2386.0	0.638	727.34
S-6	1460.9	2386.0	0.612	697.93
S-7	1399.4	2386.0	0.586	668.53
S-8	1337.8	2386.0	0.561	639.12
S-9	1406.3	2386.0	0.589	619.03
S-10	1339.5	2386.0	0.561	589.62
S-11	1503.4	2386.0	0.630	560.22
H-2	1254.3	2386.0	0.526	551.39

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
C-30"	Nozzle	1407.00	1407.00	1.000
C-30"	Pad	1407.00	1407.00	1.000
L1-2"	Nozzle	1407.00	1407.00	1.000
MH-4-24"	Nozzle	1407.00	1407.00	1.000
MH-4-24"	Pad	1407.00	1407.00	1.000
E1-36"	Nozzle	1407.00	1407.00	1.000
E2-36"	Nozzle	1407.00	1407.00	1.000
D-30"	Nozzle	1407.00	1407.00	1.000
MH-3-24"	Nozzle	1407.00	1407.00	1.000
MH-3-24"	Pad	1407.00	1407.00	1.000
UC-2-24"	Nozzle	1407.00	1407.00	1.000
UC-2-24"	Pad	1407.00	1407.00	1.000
MH-2-24"	Nozzle	1407.00	1407.00	1.000
MH-2-24"	Pad	1407.00	1407.00	1.000
UC-1-24"	Nozzle	1407.00	1407.00	1.000
UC-1-24"	Pad	1407.00	1407.00	1.000
MH-1-24"	Nozzle	1407.00	1407.00	1.000
MH-1-24"	Pad	1407.00	1407.00	1.000
A1-20"	Nozzle	1407.00	1407.00	1.000
A1-20"	Pad	1407.00	1407.00	1.000
A2-20"	Nozzle	1407.00	1407.00	1.000
A2-20"	Pad	1407.00	1407.00	1.000
B-36"	Nozzle	1407.00	1407.00	1.000
B-36"	Pad	1407.00	1407.00	1.000



Minimum 1.000

Stress ratios for Stiffening Ring Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
R1	1407.00	1407.00	1.000
R2	1407.00	1407.00	1.000
R3	1407.00	1407.00	1.000
R4	1407.00	1407.00	1.000
R5	1407.00	1407.00	1.000
R6	1407.00	1407.00	1.000
R7	1407.00	1407.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1407.00	1407.00	1.000
S-1	1407.00	1407.00	1.000
S-2	1407.00	1407.00	1.000
S-3	1407.00	1407.00	1.000
S-4	1407.00	1407.00	1.000
S-5	1407.00	1407.00	1.000
S-6	1407.00	1407.00	1.000
S-7	1407.00	1407.00	1.000
S-8	1407.13	1407.13	1.000
S-9	1407.00	1407.00	1.000
S-10	1407.00	1407.00	1.000
S-11	1407.00	1407.00	1.000
H-2	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
C-30"	163.07	2386.00	0.068
L1-2"	23.37	2294.00	0.010
MH-4-24"	137.50	2386.00	0.058
E1-36"	127.69	2386.00	0.054
E2-36"	127.69	2386.00	0.054
D-30"	109.99	2386.00	0.046
MH-3-24"	122.63	2386.00	0.051
UC-2-24"	122.63	2386.00	0.051
MH-2-24"	112.71	2386.00	0.047
UC-1-24"	112.71	2386.00	0.047
MH-1-24"	104.37	2386.00	0.044
A1-20"	82.44	2386.00	0.035
A2-20"	82.44	2386.00	0.035
B-36"	140.57	2386.00	0.059

Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****MDMT Calculations for: R1, Shell to Ring Junction, Curve: D**

Govrn. thk, tg = 18.0, c = 6.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.707$, Temp. Reduction = 16 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: R1, SA-516 70 , Bar Ring: 150.0 x 18.0 mm**Material treatment options:**

Effective Length of Shell: 289 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	40.499	7.0000	283.496
Ring :	27.000	89.0000	2402.999
Total:	67.499		2686.494

Centroid of Ring plus Shell: 40 mm

	Inertia	Distance	A*Dist ²
Shell:	6.615	32.8003	43571.609
Ring :	506.249	-49.1997	65356.418
Total:	512.864		108928.031

Available Moment of Inertia, Ring + Shell: 0.160E+04 cm⁴Required Stress in Ring plus Shell Breq 131.32 kgf/cm²

Required Strain in Ring plus Shell Areq 0.0001300

Required Moment of Inertia, Ring plus Shell:

$$= (Do^2 \cdot Ls(t + As/Ls)A) / 10.9$$

$$= (4940.0^2 \cdot 3079.1667(14.0 + 27.0/3079.1667)0.00013) / 10.9$$

$$= 1333 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style		INTERMITTENT	
Location of Stiffening Ring		EXTERNAL	
Radial Pressure Load	Pext*Slen	16.24	kgf/mm
The Radial Shear Load	V	802.08	kgf
The First Moment of the Area (Ring + Shell)	Q	132839.09	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	6.65	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		112.00	mm
The Weld Allowable Load	WLeg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	17.55	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	2.79	kgf/cm ²

MDMT Calculations for: R2, Shell to Ring Junction, Curve: D

Govrn. thk, tg = 18.0, c = 6.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.671$, Temp. Reduction = 18 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

**Stiffening Ring Calculations for: R2, SA-516 70 , Bar Ring: 175.0 x 18.0 mm****Material treatment options:**

Effective Length of Shell: 289 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	40.499	7.0000	283.496
Ring :	31.500	101.5000	3197.249
Total:	71.999		3480.745

Centroid of Ring plus Shell: 48 mm

	Inertia	Distance	A*Dist ²
Shell:	6.615	41.3441	69226.977
Ring :	803.905	-53.1559	89004.734
Total:	810.520		158231.719

Available Moment of Inertia, Ring + Shell: 0.239E+04 cm⁴

Required Stress in Ring plus Shell Breq 132.83 kgf/cm²
 Required Strain in Ring plus Shell Areq 0.0001310

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4940.0^2 \cdot 4450.0(14.0 + 31.5/4450.0)0.000131) / 10.9$$

$$= 1920 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	23.46	kgf/mm
The Radial Shear Load	V	1159.17	kgf
The First Moment of the Area (Ring + Shell)	Q	167440.98	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	8.11	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		112.00	mm
The Weld Allowable Load	WLeg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	24.83	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	1.97	kgf/cm ²

MDMT Calculations for: R3, Shell to Ring Junction, Curve: D

Govern. thk, tg = 18.0, tr = 9.354, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: R3, SA-516 70 , Bar Ring: 175.0 x 18.0 mm**Material treatment options:**

Effective Length of Shell: 268 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	32.126	6.0000	192.754
Ring :	31.500	99.5000	3134.249
Total:	63.626		3327.003

Centroid of Ring plus Shell: 52 mm



	Inertia	Distance	A*Dist ²
Shell:	3.855	46.2903	68838.539
Ring :	803.905	-47.2097	70205.844
Total:	807.760		139044.375

Available Moment of Inertia, Ring + Shell: 0.220E+04 cm⁴

Required Stress in Ring plus Shell Breq 153.10 kgf/cm²
 Required Strain in Ring plus Shell Areq 0.0001510

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4936.0^2 \cdot 4200.0(12.0 + 31.5/4200.0)0.000151) / 10.9$$

$$= 1807 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	22.15	kgf/mm
The Radial Shear Load	V	1093.16	kgf
The First Moment of the Area (Ring + Shell)	Q	148710.58	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	7.40	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		96.00	mm
The Weld Allowable Load	WLeg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	23.35	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	2.10	kgf/cm ²

MDMT Calculations for: R4, Shell to Ring Junction, Curve: D

Govrn. thk, tg = 18.0, tr = 9.354, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: R4, SA-516 70 , Bar Ring: 175.0 x 18.0 mm

Material treatment options:

Effective Length of Shell: 268 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	32.126	6.0000	192.754
Ring :	31.500	99.5000	3134.249
Total:	63.626		3327.003

Centroid of Ring plus Shell: 52 mm

	Inertia	Distance	A*Dist ²
Shell:	3.855	46.2903	68838.539
Ring :	803.905	-47.2097	70205.844
Total:	807.760		139044.375

Available Moment of Inertia, Ring + Shell: 0.220E+04 cm⁴

Required Stress in Ring plus Shell Breq 152.54 kgf/cm²
 Required Strain in Ring plus Shell Areq 0.0001500

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4936.0^2 \cdot 3950.0(12.0 + 31.5/3950.0)0.00015) / 10.9$$

$$= 1695 \text{ cm}^4$$

**Results for Stiffening Ring Weld Calculations per UG-30:**

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	20.83	kgf/mm
The Radial Shear Load	V	1028.09	kgf
The First Moment of the Area (Ring + Shell)	Q	148710.58	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	6.96	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		96.00	mm
The Weld Allowable Load	WLeg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	21.96	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	2.23	kgf/cm ²

MDMT Calculations for: R5, Shell to Ring Junction, Curve: D

Govern. thk, tg = 18.0, tr = 9.354, c = 6.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.779, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Stiffening Ring Calculations for: R5, SA-516 70 , Bar Ring: 175.0 x 18.0 mm**Material treatment options:**

Effective Length of Shell: 268 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	32.126	6.0000	192.754
Ring :	31.500	99.5000	3134.249
Total:	63.626		3327.003

Centroid of Ring plus Shell: 52 mm

	Inertia	Distance	A*Dist ²
Shell:	3.855	46.2903	68838.539
Ring :	803.905	-47.2097	70205.844
Total:	807.760		139044.375

Available Moment of Inertia, Ring + Shell: 0.220E+04 cm⁴

Required Stress in Ring plus Shell	Breq	153.10	kgf/cm ²
Required Strain in Ring plus Shell	Areq	0.0001510	

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4936.0^2 \cdot 4200.0(12.0 + 31.5/4200.0)0.000151) / 10.9$$

$$= 1807 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	22.15	kgf/mm
The Radial Shear Load	V	1093.16	kgf
The First Moment of the Area (Ring + Shell)	Q	148710.58	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	7.40	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		96.00	mm
The Weld Allowable Load	WLeg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	23.35	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	2.10	kgf/cm ²

**MDMT Calculations for: R6, Shell to Ring Junction, min(Curve:D, Curve:B)**

Govrn. thk, tg = 18.0, tr = 9.353, c = 6.0 mm, E* = 1.0

Thickness Ratio = $tr * E^*/(tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: R6, SA-516 70 , Bar Ring: 175.0 x 18.0 mm**Material treatment options:**

Effective Length of Shell: 268 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	32.126	6.0000	192.754
Ring :	31.500	99.5001	3134.256
Total:	63.626		3327.010

Centroid of Ring plus Shell: 52 mm

	Inertia	Distance	A*Dist ²
Shell:	3.855	46.2903	68838.734
Ring :	803.908	-47.2097	70206.008
Total:	807.763		139044.750

Available Moment of Inertia, Ring + Shell: 0.220E+04 cm⁴

Required Stress in Ring plus Shell Breq 153.10 kgf/cm²
 Required Strain in Ring plus Shell Areq 0.0001510

Required Moment of Inertia, Ring plus Shell:= (Do² • Ls(t + As/Ls)A)/10.9= (4936.0²•4200.002(12.0+31.5/4200.002)0.000151)/10.9= 1807 cm⁴**Results for Stiffening Ring Weld Calculations per UG-30:**

Given Stiffening Ring Fillet Weld Size Wleg 12.000 mm
 Stiffening Ring Attachment Style INTERMITTENT
 Location of Stiffening Ring EXTERNAL
 Radial Pressure Load Pext*Slen 22.15 kgf/mm
 The Radial Shear Load V 1093.16 kgf
 The First Moment of the Area (Ring + Shell) Q 148710.80 mm³
 Weld Shear Flow due to Rad. Shear Load VQ/I 7.40 kgf/mm
 The Weld Allowable Stress 0.55*S 773.85 kgf/cm²
 Minimum Weld Leg Size Min(6mm, t, tw) Wldmin 6.00 mm
 Max Clear Spacing bet Toes of Adj Weld Segments, 8t 96.00 mm
 The Weld Allowable Load WLeg*0.55*S 92.86 kgf/mm
 The Combined Weld Load SRSS of VQ/I and Pext*Slen 23.35 kgf/mm
 MAEP of Stiffening Ring Weld Weld MAEP 2.10 kgf/cm²

MDMT Calculations for: R7, Shell to Ring Junction, Curve: D

Govrn. thk, tg = 14.0, tr = 11.0, c = 1.0 mm, E* = 0.85

Thickness Ratio = $tr * E^*/(tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Stiffening Ring Calculations for: R7, SA-516 70 , Bar Ring: 175.0 x 18.0 mm**Material treatment options:**

Effective Length of Shell: 279 mm



	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	36.217	6.5000	235.408
Ring :	31.500	100.5000	3165.749
Total:	67.717		3401.157

Centroid of Ring plus Shell: 50 mm

	Inertia	Distance	A*Dist ²
Shell:	5.100	43.7264	69245.898
Ring :	803.905	-50.2736	79614.273
Total:	809.006		148860.172

Available Moment of Inertia, Ring + Shell: 0.230E+04 cm⁴

Required Stress in Ring plus Shell Breq 141.43 kgf/cm²

Required Strain in Ring plus Shell Areq 0.0001390

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (4934.0^2 \cdot 3954.4167(13.0 + 31.5/3954.4167)0.000139) / 10.9$$

$$= 1694 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	12.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	20.85	kgf/mm
The Radial Shear Load	V	1028.82	kgf
The First Moment of the Area (Ring + Shell)	Q	158361.89	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	7.09	kgf/mm
The Weld Allowable Stress	0.55*S	773.85	kgf/cm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		104.00	mm
The Weld Allowable Load	Wleg*0.55*S	92.86	kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	22.02	kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	2.22	kgf/cm ²

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	26500	...	...	No Calc	No Calc
20	30	26500	...	...	No Calc	No Calc
30	40	26500	...	...	No Calc	No Calc
40	50	26500	...	...	No Calc	No Calc
50	60	26500	...	...	No Calc	No Calc
60	70	26500	...	...	No Calc	No Calc
70	80	26500	...	...	No Calc	No Calc
80	90	26500	...	...	No Calc	No Calc
90	100	26500	...	...	No Calc	No Calc
100	110	26500	...	...	No Calc	No Calc
110	120	No Calc	4940	14	0.00039361	398.505
120	Ring	1958.33	4940	14	0.00053545	542.101
Ring	130	4200	4940	14	0.0002401	243.088
130	Ring	4200	4940	14	0.0002401	243.088
Ring	140	4700	4940	14	0.00021377	216.423
140	150	4700	4936	12	0.00016706	169.135
150	Ring	4700	4936	12	0.00016706	169.135
Ring	160	3700	4936	12	0.00021347	216.124
160	Ring	3700	4936	12	0.00021347	216.124
Ring	170	4200	4936	12	0.00018729	189.614
170	180	4200	4936	12	0.00018729	189.614
180	Ring	4200	4936	12	0.00018729	189.614
Ring	190	4200	4936	12	0.00018729	189.614



190	Ring	4200	4936	12	0.00018729	189.614
Ring	200	4200	4936	12	0.00018729	189.614
200	210	4200	4934	13	0.00021329	215.944
210	Ring	4200	4934	13	0.00021329	215.944
Ring	220	3708.83	4934	13	0.00024262	245.637
220	230	3708.83	4930	11	0.00018818	190.519
230	240	No Calc	4930	11	0.00030989	313.746

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kPa	External M.A.W.P. kPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc
70	80	...	No Calc	...	No Calc
80	90	...	No Calc	...	No Calc
90	100	...	No Calc	...	No Calc
100	110	...	No Calc	...	No Calc
110	120	20	15.0757	51.7095	123.056
120	Ring	20	14.2482	51.7095	200.877
Ring	130	20	17.264	51.7095	90.0771
130	Ring	20	17.264	51.7095	90.0771
Ring	140	20	17.8098	51.7095	80.1962
140	150	18	17.8045	51.7095	53.7637
150	Ring	18	17.8045	51.7095	53.7637
Ring	160	18	16.6874	51.7095	68.7004
160	Ring	18	16.6874	51.7095	68.7004
Ring	170	18	17.2586	51.7095	60.2735
170	180	18	17.2586	51.7095	60.2735
180	Ring	18	17.2586	51.7095	60.2735
Ring	190	18	17.2586	51.7095	60.2735
190	Ring	18	17.2586	51.7095	60.2735
Ring	200	18	17.2586	51.7095	60.2734
200	210	14	12.2561	51.7095	74.3936
210	Ring	14	12.2561	51.7095	74.3936
Ring	220	14	11.6953	51.7095	84.6227
220	230	12	11.6902	51.7095	55.5818
230	240	12	10.0573	51.7095	76.2767
R1					274
R2					193
R3					206
R4					219
R5					206
R6					206
R7					218

Minimum

54

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	26500	No Calc	No Calc	No Calc
20	30	26500	No Calc	No Calc	No Calc
30	40	26500	No Calc	No Calc	No Calc
40	50	26500	No Calc	No Calc	No Calc
50	60	26500	No Calc	No Calc	No Calc
60	70	26500	No Calc	No Calc	No Calc
70	80	26500	No Calc	No Calc	No Calc



80	90	26500	No Calc	No Calc	No Calc
90	100	26500	No Calc	No Calc	No Calc
100	110	26500	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	Ring	1958.33	7183.09	No Calc	No Calc
Ring	130	4200	7183.97	1333.26	1602.14
130	Ring	4200	7183.97	No Calc	No Calc
Ring	140	4700	7185.14	1919.59	2392.84
140	150	4700	4881.03	No Calc	No Calc
150	Ring	4700	4881.03	No Calc	No Calc
Ring	160	3700	4879.07	1807.42	2198.2
160	Ring	3700	4879.07	No Calc	No Calc
Ring	170	4200	4881.16	1694.87	2198.2
170	180	4200	4881.16	No Calc	No Calc
180	Ring	4200	4881.16	No Calc	No Calc
Ring	190	4200	4881.16	1807.42	2198.2
190	Ring	4200	4881.16	No Calc	No Calc
Ring	200	4200	4881.16	1807.42	2198.21
200	210	4200	5980.12	No Calc	No Calc
210	Ring	4200	5980.12	No Calc	No Calc
Ring	220	3708.83	5979.03	1693.71	2297.61
220	230	3708.83	3972.4	No Calc	No Calc
230	240	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm³	Corroded Metal Wgt. kg	Corroded ID Volume cm³	Extra due Misc % kg
10	20	9236.72	...	8758.22	...	...
20	30	5984.86	...	5506.11	...	...
30	40	5984.86	...	5506.11	...	...
40	50	5984.86	...	5506.11	...	...
50	60	5984.86	...	5506.11	...	...
60	70	5388.6	...	4909.66	...	...
70	80	5388.6	...	4909.66	...	...
80	90	6466.31	...	5891.58	...	...
90	100	6466.31	...	5891.58	...	...
100	110	6465.97	...	5891.27	...	...
110	120	5541.3	16343090	4155.98	16461138	...
120	130	7187.27	56572236	5037.2	56849668	...
130	140	7187.27	56572236	5037.2	56849668	...
140	150	6465.97	56572236	4315.91	56849668	...
150	160	6465.97	56572236	4315.91	56849668	...
160	170	6465.97	56572236	4315.91	56849668	...
170	180	6465.97	56572236	4315.91	56849668	...
180	190	6465.97	56572236	4315.91	56849668	...
190	200	4418.66	38657696	2949.36	38847272	...
200	210	5031.1	56710864	4672.68	56757112	...
210	220	5031.1	56710864	4672.68	56757112	...
220	230	1293.19	17013260	1185.67	17027134	...
230	240	3227.41	16402043	2996.88	16421724	...
Total		134599	0.598E+09	110563	0.600E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	1425.07	2125.13	2500	-2125.13	PL-14
10	Wght	2500	...	9.99939	...	Skirt Bracing
20	Wght	500	...	9.99939	...	Skirt MW
30	Plat	1479.77	2126.38	1500	-2126.38	PL-13
50	Plat	1479.77	2126.38	500.001	-2126.38	PL-12
60	Plat	1478.96	2127.63	2000	-2127.63	PL-11
80	Plat	1478.96	2127.63	999.999	-2127.63	PL-10
90	Plat	1478.96	2127.63	2000	-2127.63	PL-9
110	Plat	5446.21	0.29593E-04	...	...	PL-8
110	Liqd	15682.3	...	-612.5	...	L-1
110	Insl	1027.21	...	-587.5	...	Insulation
110	Nozl	1941	...	-1515	...	C-30"
110	Wght	300	...	...	...	Vortex at C
110	Wght	7815	3890	-1225	...	PP_C
120	Liqd	54285	...	1500	...	Liquid 20
120	Insl	1332.33	...	1500	...	Insulation
120	Ring	334.604	...	1500	...	R1
120	Nozl	8.15198	2575.01	500	...	L1-2"
130	Plat	2797.1	...	550	-2127.63	PL-7
130	Liqd	4523.75	...	125	...	L-3
130	Insl	1332.33	...	1500	...	Insulation
130	Ring	392.289	...	2700	...	R2
130	Nozl	528.615	2700	1500	...	MH-4-24"
130	Wght	11500	...	2950	...	Weld Clip
130	Wght	10000	-1.0101	2950	...	Margine Weight
130	Wght	200	...	2950	...	Small Nozzle
140	Plat	2795.5	...	1550	2127.63	PL-6
140	Insl	1331.27	...	1500	...	Insulation
140	Nozl	453.331	1909.19	1350	-1909.19	E1-36"
140	Nozl	453.331	-1909.19	1350	1909.19	E2-36"



140	Wght	15795	2888.53	1350	-2888.53	PP_E1
140	Wght	15795	-2888.53	1350	2888.53	PP_E2
140	Wght	500	...	1350	...	Insert pad E1.E
150	Plat	2799.5	...	2550	-2127.63	PL-5
150	Insl	1331.27	...	1500	...	Insulation
150	Ring	391.982	...	1400	...	R3
150	Nozl	330.225	2700	550	...	D-30"
150	Wght	350	...	250.002	...	TA_D0-1
150	Wght	15800	...	750.002	...	OP_D0-1
150	Wght	3000	...	250.002	...	DL_D0-1
150	Wght	200	2450	550.002	...	Inset pad-D
150	Wght	11664	4009	550.002	...	PP_D
160	Pack	9391.08	...	1500	...	Bed#2
160	Insl	1331.27	...	1500	...	Insulation
160	Ring	391.982	...	2100	...	R4
160	Nozl	528.615	2700	1500	...	MH-3-24"
160	Nozl	528.615	-2700	749.998	...	UC-2-24"
160	Wght	300	...	0.0011613	...	TA_BS-2
160	Wght	2000	...	0.0011613	...	DL-BS-2
170	Plat	2795.5	...	550.002	2127.63	PL-4
170	Pack	6260.72	...	1000	...	Bed#2
170	Insl	1331.27	...	1500	...	Insulation
170	Wght	300	...	2600	...	TA_LD-2
170	Wght	150	...	2000	...	TA_BL-2
170	Wght	3900	...	2750	...	OP_LD-2
170	Wght	1500	...	2600	...	DL-LD-2
170	Wght	450	...	2000	...	DL-BL-2
180	Plat	2795.5	...	1550	-2127.63	PL-3
180	Pack	6104.2	...	2025	...	Bed#1
180	Insl	1331.27	...	1500	...	Insulation
180	Ring	391.982	...	300	...	R5
180	Nozl	528.615	2700	500.002	...	MH-2-24"
180	Nozl	528.615	-2700	1550	...	UC-1-24"
180	Wght	300	...	1050	...	TA_BS_1
180	Wght	2000	...	1050	...	DL_BS-1
190	Pack	6417.23	...	1025	...	Bed#1
190	Insl	909.7	...	1025	...	Insulation
190	Ring	392.443	...	1500	...	R6
200	Plat	2794.7	...	500.002	2129.51	PL-2
200	Pack	3130.36	...	500.002	...	Bed#1
200	Insl	1330.74	...	1500	...	Insulation
200	Lini	1112.83	...	1500	...	Cald
200	Nozl	528.615	2703	2200	...	MH-1-24"
200	Wght	1500	...	2700	...	TA_GLR
200	Wght	300	...	1600	...	TA_LD-1
200	Wght	150	...	1000	...	TA_BL-1
200	Wght	3900	...	1750	...	OP_LD-1
200	Wght	450	...	1000	...	DL_BL-1
200	Wght	1900	...	1600	...	DL_LD-1
210	Plat	2794.7	...	1500	-2129.51	PL-1
210	Insl	1330.74	...	1500	...	Insulation
210	Lini	1112.83	...	1500	...	Clad
210	Ring	391.522	...	650.004	...	R7
210	Nozl	377.106	-2509.62	2500	-1556.28	A1-20"
210	Nozl	431.604	2509.62	2500	1556.27	A2-20"
210	Wght	3900	...	2600	...	Shroud
210	Wght	2700	...	-0.0023256	...	OP_GLR
210	Wght	2000	...	1500	...	Trunnion
210	Wght	10923	...	2500	-3882	PP_A1
210	Wght	10923	...	2500	3882	PP_A2
220	Insl	398.903	...	450	...	Insulation
220	Lini	333.849	...	450	...	Clad
220	Wght	1300	2450	449.996	...	Pipe Davit
230	Plat	8482.56	...	1499.99	...	TOP PL
230	Insl	995.49	...	638.25	...	Insulation
230	Lini	711.403	...	638.25	...	Clad
230	Nozl	470.351	...	1570.5	...	B-36"
230	Wght	20966	4085	1225	...	PP_B
230	Forc	...	...	1225	...	B_FxFz



160 Pliq	13571.25	...	1500.001	...
170 Pliq	9047.50	...	1000.001	...
180 Pliq	8821.31	...	2025.000	...
190 Pliq	9273.68	...	1024.999	...
200 Pliq	4534.83	...	500.002	...

Total Weight of Each Detail Type:

Platforms	42322.8
Packing	31303.6
Packing Liquid	45248.6
Liquid	74491.1
Insulation	15313.8
Lining	3270.9
Stiffeners	2686.8
Nozzles	7636.8
Weights	167731.0

Sum of the Detail Weights 390005.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	134599.1	134599.1	134599.1	134599.1	134599.1	134599.1
Stif. Rings	2686.8	2686.8	2686.8	2686.8	2686.8	2686.8
Nozzles	7636.8	7636.8	7636.8	7636.8	7636.8	7636.8
Wld Weights	36250.0	36250.0	36250.0	36250.0	36250.0	36250.0
Platforms	...	...	42322.8	42322.8	42322.8	42322.8
Packing	...	...	...	...	31303.6	31303.6
Insulation	...	...	15313.8	15313.8	15313.8	15313.8
Lining	3270.9	3270.9	3270.9	3270.9	3270.9	3270.9
Empty Weights	...	...	...	11300.0	11300.0	...
Ope Weights	...	...	...	...	...	131481.0
Ope. Liquid	...	...	...	...	...	74491.1
Packing Liq	...	...	...	...	...	45248.6
Test Liquid	...	597167.8	...	...	...	...
Totals	184443.6	781611.4	242080.2	253380.2	284683.8	524604.3

Field Installation Options:

* Packing installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	184443.6 kg
Shop Test Wt.	- Fabricated weight + water (full)	781611.4 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	242080.2 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	253380.2 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	284683.8 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	524604.3 kg
Field Test Wt.	- Empty weight + test medium (full)	839247.9 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	219579.7 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	491611.3 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	130602.7 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	387673	384217
20	30	387673	384531
30	40	387673	384531



40	50	387673	384531
50	60	387673	384531
60	70	387673	384845
70	80	387673	384845
80	90	465207	461814
90	100	465207	461814
100	110	465207	461814
110	120	273161	267962
120	130	465584	461814
130	140	465584	461814
140	150	465207	461814
150	160	465207	461814
160	170	465207	461814
170	180	465207	461814
180	190	465207	461814
190	200	317892	315573
200	210	465019	462380
210	220	465019	462380
220	230	139392	138714
230	240	271640	268609

Total 9298655.0 9225790.0 cm²
Total 10009.0 9930.6 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	13161.8	13161.8	13161.8	3028.46	13161.8
20	30	6484.86	6484.86	6484.86	...	6484.86
30	40	7464.63	7464.63	7464.63	3146.56	7464.63
40	50	5984.86	5984.86	5984.86	...	5984.86
50	60	7464.63	7464.63	7464.63	3146.55	7464.63
60	70	6867.56	6867.56	6867.56	3146.69	6867.56
70	80	5388.6	5388.6	5388.6	...	5388.6
80	90	7945.27	7945.27	7945.27	3146.69	7945.27
90	100	7945.27	7945.27	7945.27	3146.69	7945.27
100	110	6465.97	6465.97	6465.97	...	6465.97
110	120	14255.7	37753	30591.5	30400.4	22070.7
120	130	8862.35	63147.3	65409.2	20.9914	8862.35
130	140	33937.6	38461.3	90484.4	1417.16	33937.6
140	150	11999.4	43589.4	68546.2	...	43589.4
150	160	14868.9	42332.9	68415.8	48142.6	42332.9
160	170	11546.5	34508.8	66093.3	...	20937.5
170	180	12992.7	32201	67589.6	...	23153.5
180	190	14342	29267.5	68888.8	...	20446.2
190	200	5720.8	21411.7	44361.1	...	12138
200	210	15098	26663.2	69433.4	1428.85	22128.3
210	220	17369.6	41915.6	74055	136.77	41915.6
220	230	3325.95	3325.95	20331.6	3185	3325.95
230	240	13887.2	34853.2	30281.9	85646.2	34853.2

Empty Support Force + the Sum of the Y forces 215699.62 kgf
Operating Support Force + the Sum of the Y forces 532172.38 kgf
Hydro Support Force + the Sum of the Y forces 847223.25 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	404865	524604	839655
20	30	391703	511443	826493
30	40	385218	504958	820009
40	50	377754	497493	812544



50	60	371769	491508	806559
60	70	364304	484044	799094
70	80	357436	477176	792227
80	90	352048	471787	786838
90	100	344103	463842	778893
100	110	336157	455897	770948
110	120	329691	449431	764482
120	130	307621	411678	733890
130	140	298758	348531	668481
140	150	264821	310069	577997
150	160	221231	266480	509450
160	170	178898	224147	441035
170	180	157961	189638	374941
180	190	134807	157437	307352
190	200	114361	128170	238463
200	210	102223	106758	194102
210	220	80094.8	80094.8	124668
220	230	38179.2	38179.2	50613.5
230	240	34853.2	34853.2	30281.9

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	189140	189140	189140
20	30	186111	186111	186111
30	40	186111	186111	186111
40	50	182965	182965	182965
50	60	182965	182965	182965
60	70	179818	179818	179818
70	80	176671	176671	176671
80	90	176671	176671	176671
90	100	173525	173525	173525
100	110	170378	170378	170378
110	120	170378	170378	170378
120	130	139978	139978	139978
130	140	139957	139957	139957
140	150	138539	138539	138539
150	160	138539	138539	138539
160	170	90396.8	90396.8	90396.8
170	180	90396.8	90396.8	90396.8
180	190	90396.8	90396.8	90396.8
190	200	90396.8	90396.8	90396.8
200	210	90396.8	90396.8	90396.8
210	220	88968	88968	88968
220	230	88831.2	88831.2	88831.2
230	240	85646.2	85646.2	85646.2



Nozzle Flange MAWP Results: (kPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
C-30"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
L1-2"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
MH-4-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
E1-36"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
E2-36"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
D-30"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
MH-3-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
UC-2-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
MH-2-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
UC-1-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
MH-1-24"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
A1-20"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
A2-20"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
B-36"	1540.0	1960.0	160	150	GR 1.1	...	...	...	...
Min Rating	1540.0	1960.0, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.65201 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.59718 Hz.

The Natural Frequency for the Vessel (Filled) is 0.44576 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	7134.71	26723.2

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	...	...
230	240	7134.71	26723.2

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
230	1225.00	5045.	-7568.	5045.	12716.		12716.



**Wind Load Calculation Results:****Input Values:**

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_z \bar{V} \sqrt{(gQ^2 + Q^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_z \bar{V})]$$

$$= 0.925[(1+1.7 \cdot 0.163 \cdot \sqrt{(3.4^2 \cdot 0.875^2 + 4.065^2 \cdot 0.537^2)}) / (1+1.7 \cdot 3.4 \cdot 0.163)]$$

$$= 0.964$$

Natural Frequency of Vessel (Operating)	0.597 Hz
Natural Frequency of Vessel (Empty)	0.652 Hz
Natural Frequency of Vessel (Test)	0.446 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.625
Structure Height to Diameter ratio	11.589
Height to top of Structure	57868.496 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1250.0 + 300.0 = 1550.0$ mm
 $= 5.085$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (5.085 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

= 16.0 psf [78.119] kgf/m²

Force on the first element [F]:

$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$

= 16.0 • 0.964 • 0.625 • 163.377

= 1575.4 lbs. [714.6] kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
K-10	1550.0	0.000	0.000	0.000	0.849	1.000	78.119
K-9	4050.0	0.000	0.000	0.000	0.849	1.000	78.119
K-8	6550.0	0.000	0.000	0.000	0.916	1.000	78.119
K-7	9050.0	0.000	0.000	0.000	0.980	1.000	78.119
K-6	11550.0	0.000	0.000	0.000	1.032	1.000	78.119
K-5	14050.0	0.000	0.000	0.000	1.075	1.000	78.119
K-4	16550.0	0.000	0.000	0.000	1.113	1.000	78.119
K-3	19300.0	0.000	0.000	0.000	1.150	1.000	78.119
K-2	22300.0	0.000	0.000	0.000	1.185	1.000	78.119
K-1	25300.0	0.000	0.000	0.000	1.217	1.000	78.119
H-1	26825.0	0.000	0.000	0.000	1.232	1.000	78.119
S-1	28350.0	0.000	0.000	0.000	1.246	1.000	78.119
S-2	31350.0	0.000	0.000	0.000	1.273	1.000	78.119
S-3	34350.0	0.000	0.000	0.000	1.298	1.000	78.119
S-4	37350.0	0.000	0.000	0.000	1.321	1.000	78.119
S-5	40350.0	0.000	0.000	0.000	1.343	1.000	78.119
S-6	43350.0	0.000	0.000	0.000	1.363	1.000	78.119
S-7	46350.0	0.000	0.000	0.000	1.382	1.000	78.119
S-8	48875.0	0.000	0.000	0.000	1.398	1.000	78.119
S-9	51400.0	0.000	0.000	0.000	1.413	1.000	78.119
S-10	54400.0	0.000	0.000	0.000	1.430	1.000	78.119
S-11	56350.0	0.000	0.000	0.000	1.440	1.000	78.119
H-2	57382.7	0.000	0.000	0.000	1.446	1.000	78.119

Note:

The height/(avg. diameter) ratio [11.3], was less than or equal to 15, therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf	Cf
PL-14	25160.00	2800.00	78.12	196.55	1.00
PL-13	31256.00	6800.00	78.12	244.17	1.00
PL-12	31256.00	10800.00	78.12	244.17	1.00
PL-11	31256.00	14800.00	78.12	244.17	1.00
PL-10	31256.00	18800.00	78.12	244.17	1.00
PL-9	31256.00	22800.00	78.12	244.17	1.00
PL-8	46256.00	26800.00	78.12	361.35	1.00
PL-7	46256.00	30400.00	78.12	361.35	1.00
PL-6	46256.00	34400.00	78.12	361.35	1.00
PL-5	46256.00	38400.00	78.12	361.35	1.00
PL-4	46256.00	42400.00	78.12	361.35	1.00
PL-3	46256.00	46400.00	78.12	361.35	1.00
PL-2	46256.00	50400.00	78.12	361.35	1.00
PL-1	46256.00	54399.99	78.12	361.35	1.00



TOP PL	86256.00	58299.99	78.12	673.82	1.00
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Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf
Skirt Bracing	0.00	310.00	78.12	0.00
Skirt MW	0.00	2810.00	78.12	0.00
Vortex at C	0.00	26800.00	78.12	0.00
PPC	0.00	25575.00	78.12	0.00
Weld Clip	0.00	32800.00	78.12	0.00
Margine Weigh	0.00	32800.00	78.12	0.00
Small Nozzle	0.00	32800.00	78.12	0.00
PPE1	0.00	34200.00	78.12	0.00
PPE2	0.00	34200.00	78.12	0.00
Insert pad E1	0.00	34200.00	78.12	0.00
TADO-1	0.00	36100.00	78.12	0.00
OPD0-1	0.00	36600.00	78.12	0.00
DLD0-1	0.00	36100.00	78.12	0.00
Inset pad-D	0.00	36400.00	78.12	0.00
PPD	0.00	36400.00	78.12	0.00
TABS-2	0.00	38850.00	78.12	0.00
DL-BS-2	0.00	38850.00	78.12	0.00
TALD-2	0.00	44450.01	78.12	0.00
TABL-2	0.00	43850.00	78.12	0.00
OPLD-2	0.00	44600.01	78.12	0.00
DL-LD-2	0.00	44450.01	78.12	0.00
DL-BL-2	0.00	43850.00	78.12	0.00
TABS1	0.00	45900.00	78.12	0.00
DLBS-1	0.00	45900.00	78.12	0.00
TAGLR	0.00	52600.00	78.12	0.00
TALD-1	0.00	51500.00	78.12	0.00
TABL-1	0.00	50900.00	78.12	0.00
OPLD-1	0.00	51650.00	78.12	0.00
DLBL-1	0.00	50900.00	78.12	0.00
DLLD-1	0.00	51500.00	78.12	0.00
Shroud	0.00	55500.00	78.12	0.00
OPGLR	0.00	52899.99	78.12	0.00
Trunnion	0.00	54399.99	78.12	0.00
PPA1	0.00	55399.99	78.12	0.00
PPA2	0.00	55399.99	78.12	0.00
Pipe Davit	0.00	56349.99	78.12	0.00
PPB	0.00	58025.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 0.59718 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	1550	6071.28	151782	78.1189	1176.6
20	30	4050	6071.28	151782	78.1189	786.039
30	40	6550	6071.28	151782	78.1189	1064.88
40	50	9050	6071.28	151782	78.1189	786.039
50	60	11550	6071.28	151782	78.1189	1037.18
60	70	14050	6071.28	151782	78.1189	1068.96
70	80	16550	6071.28	151782	78.1189	786.039
80	90	19300	6071.28	182138	78.1189	1204.87
90	100	22300	6071.28	182138	78.1189	1217.85
100	110	25300	6071.28	182138	78.1189	943.247
110	120	26825	6273	3136.5	78.1189	413.354
120	130	28350	6273	188190	78.1189	974.586
130	140	31350	6273	188190	78.1189	1360.02
140	150	34350	6268.08	188042	78.1189	1371.88
150	160	37350	6268.08	188042	78.1189	1382.48



160	170	40350	6268.08	188042	78.1189	973.822
170	180	43350	6268.08	188042	78.1189	1362.59
180	190	46350	6268.08	188042	78.1189	1371.73
190	200	48875	6268.08	128496	78.1189	665.445
200	210	51400	6265.62	187969	78.1189	1363.19
210	220	54400	6265.62	187969	78.1189	1370.92
220	230	56350	6260.7	56346.3	78.1189	291.803
230	240	57382.7	6260.7	67962.9	78.1189	1105.01

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.3 * 0.147 = 0.191 \\ SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\ SDS &= 2/3 * Sms = 2/3 * 0.191 = 0.127 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(189.55550.75)$
 $= 1.022$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/0.597)$
 $= 1.675$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 1.022, 1.675)$ per 12.8.2
 $= 1.675$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.127 / (1.25 / 1.25)$
 $= 0.127$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / Ie))$
 $= 0.065 / (1.675(1.25 / 1.25))$
 $= 0.039$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot Ie, 0.030)$
 $= \max(0.044 \cdot 0.1274 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs \cdot W$ (Equation 12.8-1):
 $= 0.0388 \cdot 524604$
 $= 20363.449$ kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$



$$= 0.2 \cdot 0.127 \cdot 524604$$

$$= 13366.9 \text{ kgf}$$

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.5873

The Natural Frequency for the Vessel (Ope...) is 0.59718 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	1250	13161.8	2.3016	3.63656	5798.92
20	30	3750	6484.86	6.48536	9.48419	5463.56
30	40	6250	7464.63	16.7948	23.693	5298.32
40	50	8750	5984.86	22.9702	31.6461	5108.13
50	60	11250	7464.63	42.6934	57.7871	4955.63
60	70	13750	6867.56	54.0114	72.0808	4765.43
70	80	16250	5388.6	55.2479	72.8688	4590.45
80	90	19000	7945.27	104.406	136.197	4453.15
90	100	22000	7945.27	131.76	170.116	4250.7
100	110	25000	6465.97	131.35	168.067	4048.25
110	120	26525	37753	842.492	405.363	3883.5
120	130	28050	63147.3	1539.94	274.303	2921.55
130	140	31050	38461.3	1102.09	1225.46	1312.56
140	150	34050	43589.4	1445.95	498.351	332.564
150	160	37050	42332.9	1605.67	701.909	-778.095
160	170	40050	34508.8	1481.09	613.396	-1856.74
170	180	43050	32201	1549.93	770.154	-2736.02
180	190	46050	29267.5	1567.72	941.592	-3556.5
190	200	48575	21411.7	1248.34	407.265	-4302.24
200	210	51100	26663.2	1684.71	1160.71	-4847.81
210	220	54100	41915.6	2899.45	1456.05	-5527.19
220	230	56050	3325.95	243.368	294.189	-6595.19
230	240	56525	34853.2	2584.69	1244.19	-6679.94



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	7134.71	7134.71	429835	429835
20	30	3750	7134.71	7134.71	411998	411998
30	40	6250	7134.71	7134.71	394161	394161
40	50	8750	7134.71	7134.71	376324	376324
50	60	11250	7134.71	7134.71	358487	358487
60	70	13750	7134.71	7134.71	340651	340651
70	80	16250	7134.71	7134.71	322814	322814
80	90	19000	7134.71	7134.71	304977	304977
90	100	22000	7134.71	7134.71	283573	283573
100	110	25000	7134.71	7134.71	262169	262169
110	120	26525	7134.71	7134.71	240765	240765
120	130	28050	7134.71	7134.71	240408	240408
130	140	31050	7134.71	7134.71	219004	219004
140	150	34050	7134.71	7134.71	197600	197600
150	160	37050	7134.71	7134.71	176195	176195
160	170	40050	7134.71	7134.71	154791	154791
170	180	43050	7134.71	7134.71	133387	133387
180	190	46050	7134.71	7134.71	111983	111983
190	200	48575	7134.71	7134.71	90578.9	90578.9
200	210	51100	7134.71	7134.71	75952.7	75952.7
210	220	54100	7134.71	7134.71	54548.6	54548.6
220	230	56050	7134.71	7134.71	33144.4	33144.4
230	240	56525	7134.71	7134.71	26723.2	26723.2

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;

Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	24078.5	20363.4	725234	882112
20	30	3750	22901.9	20361.1	666508	831206
30	40	6250	22115.9	20354.7	610236	780311
40	50	8750	21051	20337.9	556277	729446
50	60	11250	20265	20314.9	504632	678630
60	70	13750	19227.8	20272.2	455266	627896
70	80	16250	18158.8	20218.2	408533	577283
80	90	19000	17372.8	20162.9	364118	526806
90	100	22000	16167.9	20058.5	313807	466474
100	110	25000	14950.1	19926.8	267130	406496
110	120	26525	14006.8	19795.4	223694	346912
120	130	28050	13593.5	18952.9	223004	345944
130	140	31050	12618.9	17413	183686	291395
140	150	34050	11258.9	16310.9	147869	240809
150	160	37050	9886.99	14865	116150	194045
160	170	40050	8504.51	13259.3	88562.7	151859
170	180	43050	7530.69	11778.2	64509.9	114302
180	190	46050	6168.1	10228.3	43961.7	81292.5
190	200	48575	4796.37	8660.55	27515	52959.3
200	210	51100	4130.92	7412.22	18364.5	36484.7
210	220	54100	2767.74	5727.51	8016.51	16775.1
220	230	56050	1396.82	2828.06	1769.68	3941.75
230	240	56525	1105.01	2584.69	643.851	1506

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;

Wind: 1.100; Earthquake: 1.000.

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**Wind Deflection Calculations:**

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	7134.71	0.016409	0.065177	0.51776E-04
20	30	7134.71	0.14724	0.26347	0.00010646
30	40	7134.71	0.41313	0.59544	0.00015872
40	50	7134.71	0.80967	1.05504	0.00020856
50	60	7134.71	1.33081	1.63622	0.00025598
60	70	7134.71	1.97216	2.33945	0.00030615
70	80	7134.71	2.73726	3.16474	0.00035363
80	90	7134.71	3.7157	4.30672	0.00040704
90	100	7134.71	4.93633	5.60309	0.00045656
100	110	7134.71	6.30553	7.04221	0.0005022
110	120	7134.71	7.05477	7.06734	0.00050309
120	130	7134.71	7.84169	8.65427	0.00055406
130	140	7134.71	9.50329	10.387	0.00060029
140	150	7134.71	11.3063	12.2619	0.00064873
150	160	7134.71	13.2518	14.2738	0.00069162
160	170	7134.71	15.3259	16.4061	0.00072897
170	180	7134.71	17.5122	18.6421	0.00076077
180	190	7134.71	19.7938	20.9652	0.00078702
190	200	7134.71	21.7759	22.5941	0.00080177
200	210	7134.71	23.8033	25.0242	0.00081741
210	220	7134.71	26.2549	27.4935	0.00082792
220	230	7134.71	27.8663	28.2398	0.00083047
230	240	7134.71	28.2605	28.2813	0.00083047

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)

Allowable deflection : 282.750 Actual deflection : 28.281 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	24078.5	0.027844	0.10984	0.86656E-04
20	30	22901.9	0.24637	0.43769	0.00017433
30	40	22115.9	0.68141	0.97519	0.00025443
40	50	21051	1.31674	1.70381	0.00032728
50	60	20265	2.13422	2.60582	0.00039319
60	70	19227.8	3.11872	3.67294	0.00045931
70	80	18158.8	4.26627	4.89657	0.00051846
80	90	17372.8	5.69876	6.54762	0.0005807
90	100	16167.9	7.43976	8.37188	0.00063404
100	110	14950.1	9.34126	10.3452	0.00068018
110	120	14006.8	10.3623	10.3793	0.00068103
120	130	13593.5	11.4195	12.4947	0.00072777
130	140	12618.9	13.6017	14.7372	0.00076583
140	150	11258.9	15.9003	17.0899	0.00080122
150	160	9886.99	18.3029	19.5365	0.00082863
160	170	8504.51	20.788	22.0548	0.00084914
170	180	7530.69	23.3346	24.6253	0.00086362
180	190	6168.1	25.9249	27.2316	0.00087314
190	200	4796.37	28.1277	29.0261	0.00087733
200	210	4130.92	30.3436	31.6635	0.00088052
210	220	2767.74	32.9849	34.3072	0.00088165
220	230	1396.82	34.7039	35.1007	0.00088176



230	240	1105.01	35.1227	35.1448	0.00088176
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Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed m/sec	2nd Crit. Wind Speed m/sec
10	20	18.0795	112.997
20	30	18.0795	112.997
30	40	18.0795	112.997
40	50	18.0795	112.997
50	60	18.0795	112.997
60	70	18.0795	112.997
70	80	18.0795	112.997
80	90	18.0795	112.997
90	100	18.0795	112.997
100	110	18.0795	112.997
110	120	18.6802	116.751
120	130	18.6802	116.751
130	140	18.6802	116.751
140	150	18.6656	116.66
150	160	18.6656	116.66
160	170	18.6656	116.66
170	180	18.6656	116.66
180	190	18.6656	116.66
190	200	18.6656	116.66
200	210	18.6582	116.614
210	220	18.6582	116.614
220	230	18.6436	116.522
230	240	18.6436	116.522

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 282.750 Actual deflection : 35.145 mm

Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 282.750 Actual deflection : 63.426 mm

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**Longitudinal Stress Constants:**

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	3396.33	3149.35	415387488	385303328
20	30	3088.82	2841.72	378084000	347949632
30	40	3088.82	2841.72	378084000	347949632
40	50	3088.82	2841.72	378084000	347949632
50	60	3088.82	2841.72	378084000	347949632
60	70	2781.07	2533.89	340690464	310510016
70	80	2781.07	2533.89	340690464	310510016
80	90	2781.07	2533.89	340690464	310510016
90	100	2781.07	2533.89	340690464	310510016
100	110	2781.07	2533.89	340690464	310510016
110	120	3091.31	2166.55	378697984	266056560
120	130	3091.31	2166.55	378697984	266056560
130	140	3091.31	2166.55	378697984	266056560
140	150	2781.07	1856.31	340690464	227957744
150	160	2781.07	1856.31	340690464	227957744
160	170	2781.07	1856.31	340690464	227957744
170	180	2781.07	1856.31	340690464	227957744
180	190	2781.07	1856.31	340690464	227957744
190	200	2781.07	1856.31	340690464	227957744
200	210	2163.92	2009.77	265409024	246601840
210	220	2163.92	2009.77	265409024	246601840
220	230	1854.05	1699.9	227402320	208579888
230	240	1854.05	1699.9	227402320	208579888

**Longitudinal Allowable Stresses:**

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1181.99	2004.25	-1056.08	-1056.08
20	30	1181.99	2004.25	-1016.99	-1016.99
30	40	1181.99	2004.25	-1016.99	-1016.99
40	50	1181.99	2004.25	-1016.99	-1016.99
50	60	1181.99	2004.25	-1016.99	-1016.99
60	70	1181.99	2004.25	-971.561	-971.561
70	80	1181.99	2004.25	-971.561	-971.561
80	90	1181.99	2004.25	-971.561	-971.561
90	100	1181.99	2004.25	-971.561	-971.561
100	110	844.2	1431.6	-955.054	-971.561
110	120	1688.4	2863.2	-860.77	-866.74
120	130	1688.4	2863.2	-860.77	-866.74
130	140	1688.4	2863.2	-860.77	-866.74
140	150	1688.4	2863.2	-738.401	-743.522
150	160	1688.4	2863.2	-738.401	-743.522
160	170	1688.4	2863.2	-738.401	-743.522
170	180	1688.4	2863.2	-738.401	-743.522
180	190	1688.4	2863.2	-738.401	-743.522
190	200	1688.56	2863.21	-738.401	-743.522
200	210	1435.14	2433.72	-800.259	-805.809
210	220	1435.14	2433.72	-800.259	-805.809
220	230	1435.14	2433.72	-677.691	-682.391
230	240	1435.28	2433.73	-677.691	-682.391

**Longitudinal Stress Report**

Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	...	...	...
60	70	...	...	...
70	80	...	...	...
80	90	...	...	...
90	100	...	...	...
100	110	...	...	...
110	120	366.962	-46.6482	475.659
120	130	366.962	-46.6482	475.659
130	140	366.962	-46.6482	475.659
140	150	428.263	-54.3562	555.117
150	160	428.263	-54.3562	555.117
160	170	428.263	-54.3562	555.117
170	180	428.263	-54.3562	555.117
180	190	428.263	-54.3562	555.117
190	200	428.263	-54.3562	555.117
200	210	394.932	-50.1651	511.914
210	220	394.932	-50.1651	511.914
220	230	466.891	-59.2133	605.186
230	240	466.891	-59.2133	605.186

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-128.555	-166.576	-266.613	49.0885	49.0885
20	30	-137.84	-179.977	-290.843	53.488	53.488
30	40	-135.558	-177.695	-288.561	53.488	53.488
40	50	-132.932	-175.068	-285.934	52.5837	52.5837
50	60	-130.825	-172.962	-283.828	52.5837	52.5837
60	70	-143.773	-191.028	-315.363	57.9106	57.9106
70	80	-141.062	-188.318	-312.652	56.8972	56.8972
80	90	-138.936	-186.191	-310.526	56.8972	56.8972
90	100	-135.8	-183.055	-307.39	55.8838	55.8838
100	110	-132.665	-179.92	-304.255	54.8704	54.8704
110	120	-152.174	-173.059	-152.174	64.0383	64.0383
120	130	-141.987	-162.872	-141.987	52.612	52.612
130	140	-137.896	-158.781	-137.896	52.6041	52.6041
140	150	-142.66	-167.035	-142.66	60.7742	60.7742
150	160	-119.178	-143.554	-119.178	60.7742	60.7742
160	170	-96.3732	-120.749	-96.3732	39.6551	39.6551
170	180	-85.094	-102.159	-85.094	39.6551	39.6551
180	190	-72.6212	-84.8119	-72.6212	39.6551	39.6551
190	200	-61.6068	-69.0454	-61.6068	39.6551	39.6551
200	210	-50.8632	-53.1196	-50.8632	36.657	36.657
210	220	-39.8528	-39.8528	-39.8528	36.0776	36.0776
220	230	-22.4596	-22.4596	-22.4596	42.5886	42.5886
230	240	-20.5031	-20.5031	-20.5031	41.0616	41.0616

**Longitudinal Stresses due to Weight Loads and Bending for these Conditions:**

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	49.0885	188.224	228.94	94.1121	...
20	30	53.488	191.553	238.887	95.7765	...
30	40	53.488	175.381	224.26	87.6903	...
40	50	52.5837	159.873	209.641	79.9364	...
50	60	52.5837	145.03	195.037	72.5151	...
60	70	57.9106	146.619	202.214	73.3094	...
70	80	56.8972	131.568	185.914	65.7841	...
80	90	56.8972	117.264	169.658	58.6322	...
90	100	55.8838	101.062	150.228	50.5309	...
100	110	54.8704	86.0293	130.912	43.0147	...
110	120	64.0383	84.0777	130.391	42.0389	...
120	130	52.612	83.8184	130.026	41.9092	...
130	140	52.6041	69.0401	109.524	34.52	...
140	150	60.7742	64.8668	105.638	32.4334	...
150	160	60.7742	50.9524	85.1232	25.4762	...
160	170	39.6551	38.8505	66.617	19.4253	...
170	180	39.6551	28.2991	50.1419	14.1495	...
180	190	39.6551	19.285	35.6612	9.64251	...
190	200	39.6551	12.0702	23.2321	6.03511	...
200	210	36.657	7.44703	14.795	3.72351	...
210	220	36.0776	3.25079	6.80251	1.6254	...
220	230	42.5886	0.84844	1.8898	0.42422	...
230	240	41.0616	0.30868	0.72203	0.15434	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	119.323
20	30	...	...	...	124.418
30	40	...	...	...	116.709
40	50	...	...	...	109.011
50	60	...	...	...	101.333
60	70	...	...	...	104.984
70	80	...	...	...	96.4688
80	90	...	...	...	88.012
90	100	...	...	...	77.9647
100	110	...	...	...	68.0655
110	120	...	...	...	68.0754
120	130	...	...	...	67.8915
130	140	...	...	...	57.2354
140	150	...	...	...	55.351
150	160	...	...	...	45.0351
160	170	...	...	...	35.509
170	180	...	...	...	26.8483
180	190	...	...	...	19.0981
190	200	...	...	...	12.4743
200	210	...	...	...	7.90774
210	220	...	...	...	3.5589
220	230	...	...	...	0.94789
230	240	...	...	...	0.34756

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
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10	20	-2.40304	1.84131
20	30	-2.66318	1.92263
30	40	-2.66318	1.86448
40	50	-2.66318	1.79755
50	60	-2.66318	1.74389
60	70	-2.98671	1.88068
70	80	-2.98671	1.81162
80	90	-2.98671	1.75743
90	100	-2.98671	1.67754
100	110	-2.98671	1.59764
110	120	-3.49312	1.79248
120	130	-3.49312	1.34848
130	140	-3.49312	0.60583
140	150	-4.07691	0.17915
150	160	-4.07691	-0.41916
160	170	-4.07691	-1.00023
170	180	-4.07691	-1.4739
180	190	-4.07691	-1.9159
190	200	-4.07691	-2.31763
200	210	-3.76561	-2.41213
210	220	-3.76561	-2.75016
220	230	-4.45202	-3.87975
230	240	-4.45202	-3.9296

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	111.557	111.557	103.478	103.478
20	30	118.407	118.407	108.97	108.97
30	40	113.281	113.281	104.252	104.252
40	50	108.155	108.155	99.5346	99.5346
50	60	103.029	103.029	94.8169	94.8169
60	70	109.707	109.707	99.9883	99.9883
70	80	103.962	103.962	94.7528	94.7528
80	90	98.2181	98.2181	89.5174	89.5174
90	100	91.3249	91.3249	83.2348	83.2348
100	110	84.4317	84.4317	76.9522	76.9522
110	120	90.4938	90.4938	63.577	63.577
120	130	90.3597	90.3597	63.4828	63.4828
130	140	82.3147	82.3147	57.8307	57.8307
140	150	86.6826	86.6826	57.9997	57.9997
150	160	77.293	77.293	51.7172	51.7172
160	170	67.9035	67.9035	45.4346	45.4346
170	180	58.514	58.514	39.152	39.152
180	190	49.1245	49.1245	32.8694	32.8694
190	200	39.7349	39.7349	26.5869	26.5869
200	210	30.7997	30.7997	28.6172	28.6172
210	220	22.1201	22.1201	20.5526	20.5526
220	230	15.8905	15.8905	14.5752	14.5752
230	240	12.812	12.812	11.7515	11.7515

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...



70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...
230	240	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	217.91	1181.99	-479.83	1056.08	0.1844	0.4543
20	222.94	1181.99	-503.95	1016.99	0.1886	0.4955
30	203.93	1181.99	-480.37	1016.99	0.1725	0.4723
40	185.02	1181.99	-456.21	1016.99	0.1565	0.4486
50	167.15	1181.99	-434.13	1016.99	0.1414	0.4269
60	167.48	1181.99	-461.00	971.56	0.1417	0.4745
70	148.38	1181.99	-436.48	971.56	0.1255	0.4493
80	130.46	1181.99	-414.30	971.56	0.1104	0.4264
90	109.48	1181.99	-387.06	971.56	0.0926	0.3984
100	89.68	844.20	-360.98	955.05	0.1062	0.3780
110	82.94	1688.40	-394.28	860.77	0.0491	0.4581
120	81.31	1688.40	-372.27	860.77	0.0482	0.4325
130	62.57	1688.40	-345.35	860.77	0.0371	0.4012
140	65.59	1688.40	-359.06	738.40	0.0388	0.4863
150	65.76	1688.40	-312.27	738.40	0.0390	0.4229
160	45.96	1688.40	-246.86	738.40	0.0272	0.3343
170	37.30	1688.40	-215.64	738.40	0.0221	0.2920
180	31.37	1688.40	-184.76	738.40	0.0186	0.2502
190	25.78	1688.56	-157.14	738.40	0.0153	0.2128
200	20.27	1435.14	-129.53	800.26	0.0141	0.1619
210	17.83	1435.14	-105.07	800.26	0.0124	0.1313
220	32.42	1435.14	-86.24	677.69	0.0226	0.1273
230	29.23	1435.28	-79.14	677.69	0.0204	0.1168



Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	153.25	1181.99	-406.68	1056.08	0.1297	0.3851
20	160.40	1181.99	-432.23	1016.99	0.1357	0.4250
30	149.78	1181.99	-417.17	1016.99	0.1267	0.4102
40	138.62	1181.99	-400.88	1016.99	0.1173	0.3942
50	127.86	1181.99	-386.03	1016.99	0.1082	0.3796
60	130.71	1181.99	-414.49	971.56	0.1106	0.4266
70	118.08	1181.99	-396.58	971.56	0.0999	0.4082
80	105.95	1181.99	-380.31	971.56	0.0896	0.3914
90	91.05	1181.99	-359.30	971.56	0.0770	0.3698
100	76.30	844.20	-338.43	955.05	0.0904	0.3544
110	72.23	1688.40	-372.99	860.77	0.0428	0.4333
120	70.23	1688.40	-351.50	860.77	0.0416	0.4084
130	54.86	1688.40	-329.44	860.77	0.0325	0.3827
140	60.33	1688.40	-345.29	738.40	0.0357	0.4676
150	63.51	1688.40	-302.70	738.40	0.0376	0.4099
160	45.69	1688.40	-240.44	738.40	0.0271	0.3256
170	38.45	1688.40	-211.59	738.40	0.0228	0.2865
180	33.34	1688.40	-182.41	738.40	0.0197	0.2470
190	27.94	1688.56	-155.79	738.40	0.0165	0.2110
200	22.09	1435.14	-128.64	800.26	0.0154	0.1607
210	19.15	1435.14	-104.36	800.26	0.0133	0.1304
220	33.09	1435.14	-85.77	677.69	0.0231	0.1266
230	29.79	1435.28	-78.65	677.69	0.0208	0.1161

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	179.89	1181.99	-517.85	1056.08	0.1522	0.4903
20	180.81	1181.99	-546.09	1016.99	0.1530	0.5370
30	161.79	1181.99	-522.51	1016.99	0.1369	0.5138
40	142.88	1181.99	-498.34	1016.99	0.1209	0.4900
50	125.02	1181.99	-476.27	1016.99	0.1058	0.4683
60	120.22	1181.99	-508.25	971.56	0.1017	0.5231
70	101.12	1181.99	-483.73	971.56	0.0856	0.4979
80	83.20	1181.99	-461.56	971.56	0.0704	0.4751
90	62.23	1181.99	-434.31	971.56	0.0526	0.4470
100	42.42	844.20	-408.24	955.05	0.0503	0.4275
110	62.06	1688.40	-415.16	860.77	0.0368	0.4823
120	60.43	1688.40	-393.15	860.77	0.0358	0.4567
130	41.68	1688.40	-366.23	860.77	0.0247	0.4255
140	41.21	1688.40	-383.44	738.40	0.0244	0.5193
150	41.39	1688.40	-336.65	738.40	0.0245	0.4559
160	21.58	1688.40	-271.23	738.40	0.0128	0.3673
170	20.23	1688.40	-232.70	738.40	0.0120	0.3151
180	19.18	1688.40	-196.95	738.40	0.0114	0.2667
190	18.34	1688.56	-164.58	738.40	0.0109	0.2229
200	18.02	1435.14	-131.79	800.26	0.0126	0.1647
210	17.83	1435.14	-105.07	800.26	0.0124	0.1313
220	32.42	1435.14	-86.24	677.69	0.0226	0.1273
230	29.23	1435.28	-79.14	677.69	0.0204	0.1168

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	224.85	1181.99	-554.32	1056.08	0.1902	0.5249
20	232.73	1181.99	-588.84	1016.99	0.1969	0.5790
30	215.20	1181.99	-566.86	1016.99	0.1821	0.5574
40	197.11	1181.99	-543.65	1016.99	0.1668	0.5346
50	179.43	1181.99	-521.87	1016.99	0.1518	0.5132
60	180.68	1181.99	-558.98	971.56	0.1529	0.5753
70	160.27	1181.99	-533.28	971.56	0.1356	0.5489
80	140.34	1181.99	-509.21	971.56	0.1187	0.5241



90	116.06	1181.99	-478.81	971.56	0.0982	0.4928
100	91.89	844.20	-448.54	955.05	0.1089	0.4696
110	113.66	1688.40	-456.19	860.77	0.0673	0.5300
120	111.47	1688.40	-434.52	860.77	0.0660	0.5048
130	86.27	1688.40	-402.62	860.77	0.0511	0.4677
140	86.24	1688.40	-419.95	738.40	0.0511	0.5687
150	79.22	1688.40	-367.16	738.40	0.0469	0.4972
160	52.43	1688.40	-295.92	738.40	0.0311	0.4008
170	44.68	1688.40	-251.94	738.40	0.0265	0.3412
180	37.71	1688.40	-211.17	738.40	0.0223	0.2860
190	31.26	1688.56	-173.99	738.40	0.0185	0.2356
200	26.72	1435.14	-137.78	800.26	0.0186	0.1722
210	22.40	1435.14	-107.60	800.26	0.0156	0.1345
220	34.03	1435.14	-86.71	677.69	0.0237	0.1279
230	30.16	1435.28	-79.03	677.69	0.0210	0.1166

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-409.81	1056.08		0.3881
20		2004.25	-440.11	1016.99		0.4328
30		2004.25	-429.74	1016.99		0.4226
40		2004.25	-418.45	1016.99		0.4115
50		2004.25	-408.93	1016.99		0.4021
60		2004.25	-446.58	971.56		0.4597
70		2004.25	-435.33	971.56		0.4481
80		2004.25	-426.06	971.56		0.4385
90		2004.25	-413.80	971.56		0.4259
100		1431.60	-402.14	971.56		0.4139
110		2863.20	-258.25	866.74		0.2980
120		2863.20	-236.51	866.74		0.2729
130		2863.20	-225.02	866.74		0.2596
140		2863.20	-235.87	743.52		0.3172
150		2863.20	-205.43	743.52		0.2763
160		2863.20	-155.45	743.52		0.2091
170		2863.20	-138.90	743.52		0.1868
180		2863.20	-121.92	743.52		0.1640
190		2863.21	-107.30	743.52		0.1443
200		2433.72	-91.24	805.81		0.1132
210		2433.72	-77.56	805.81		0.0962
220	20.55	2433.72	-65.47	682.39	0.0084	0.0959
230	20.71	2433.73	-61.72	682.39	0.0085	0.0904

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-315.70	1056.08		0.2989
20		2004.25	-344.33	1016.99		0.3386
30		2004.25	-342.05	1016.99		0.3363
40		2004.25	-338.52	1016.99		0.3329
50		2004.25	-336.41	1016.99		0.3308
60		2004.25	-373.27	971.56		0.3842
70		2004.25	-369.55	971.56		0.3804
80		2004.25	-367.42	971.56		0.3782
90		2004.25	-363.27	971.56		0.3739
100		1431.60	-359.12	971.56		0.3696
110		2863.20	-216.21	866.74		0.2495
120		2863.20	-194.60	866.74		0.2245
130		2863.20	-190.50	866.74		0.2198
140		2863.20	-203.43	743.52		0.2736
150		2863.20	-179.95	743.52		0.2420
160		2863.20	-136.03	743.52		0.1830
170		2863.20	-124.75	743.52		0.1678
180		2863.20	-112.28	743.52		0.1510
190		2863.21	-101.26	743.52		0.1362
200		2433.72	-87.52	805.81		0.1086
210		2433.72	-75.93	805.81		0.0942



220	20.13	2433.72	-65.05	682.39	0.0083	0.0953
230	20.56	2433.73	-61.56	682.39	0.0084	0.0902

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	179.89	1181.99	-517.85	1056.08	0.1522	0.4903
20	180.81	1181.99	-546.09	1016.99	0.1530	0.5370
30	161.79	1181.99	-522.51	1016.99	0.1369	0.5138
40	142.88	1181.99	-498.34	1016.99	0.1209	0.4900
50	125.02	1181.99	-476.27	1016.99	0.1058	0.4683
60	120.22	1181.99	-508.25	971.56	0.1017	0.5231
70	101.12	1181.99	-483.73	971.56	0.0856	0.4979
80	83.20	1181.99	-461.56	971.56	0.0704	0.4751
90	62.23	1181.99	-434.31	971.56	0.0526	0.4470
100	42.42	844.20	-408.24	955.05	0.0503	0.4275
110	429.02	1688.40	-48.20	860.77	0.2541	0.0560
120	427.39	1688.40	-26.19	860.77	0.2531	0.0304
130	408.65	1688.40		860.77	0.2420	
140	469.47	1688.40		738.40	0.2781	
150	469.65	1688.40		738.40	0.2782	
160	449.85	1688.40		738.40	0.2664	
170	448.50	1688.40		738.40	0.2656	
180	447.44	1688.40		738.40	0.2650	
190	446.60	1688.56		738.40	0.2645	
200	412.95	1435.14		800.26	0.2877	
210	412.76	1435.14		800.26	0.2876	
220	499.31	1435.14		677.69	0.3479	
230	496.12	1435.28		677.69	0.3457	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	224.85	1181.99	-554.32	1056.08	0.1902	0.5249
20	232.73	1181.99	-588.84	1016.99	0.1969	0.5790
30	215.20	1181.99	-566.86	1016.99	0.1821	0.5574
40	197.11	1181.99	-543.65	1016.99	0.1668	0.5346
50	179.43	1181.99	-521.87	1016.99	0.1518	0.5132
60	180.68	1181.99	-558.98	971.56	0.1529	0.5753
70	160.27	1181.99	-533.28	971.56	0.1356	0.5489
80	140.34	1181.99	-509.21	971.56	0.1187	0.5241
90	116.06	1181.99	-478.81	971.56	0.0982	0.4928
100	91.89	844.20	-448.54	955.05	0.1089	0.4696
110	480.62	1688.40	-89.23	860.77	0.2847	0.1037
120	478.44	1688.40	-67.56	860.77	0.2834	0.0785
130	453.23	1688.40	-35.66	860.77	0.2684	0.0414
140	514.50	1688.40		738.40	0.3047	
150	507.48	1688.40		738.40	0.3006	
160	480.69	1688.40		738.40	0.2847	
170	472.94	1688.40		738.40	0.2801	
180	465.98	1688.40		738.40	0.2760	
190	459.52	1688.56		738.40	0.2721	
200	421.65	1435.14		800.26	0.2938	
210	417.33	1435.14		800.26	0.2908	
220	500.92	1435.14		677.69	0.3490	
230	497.05	1435.28		677.69	0.3463	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	179.89	1181.99	-517.85	1056.08	0.1522	0.4903
20	180.81	1181.99	-546.09	1016.99	0.1530	0.5370
30	161.79	1181.99	-522.51	1016.99	0.1369	0.5138
40	142.88	1181.99	-498.34	1016.99	0.1209	0.4900
50	125.02	1181.99	-476.27	1016.99	0.1058	0.4683
60	120.22	1181.99	-508.25	971.56	0.1017	0.5231



70	101.12	1181.99	-483.73	971.56	0.0856	0.4979
80	83.20	1181.99	-461.56	971.56	0.0704	0.4751
90	62.23	1181.99	-434.31	971.56	0.0526	0.4470
100	42.42	844.20	-408.24	955.05	0.0503	0.4275
110	15.41	1688.40	-461.81	860.77	0.0091	0.5365
120	13.78	1688.40	-439.80	860.77	0.0082	0.5109
130		1688.40	-412.88	860.77		0.4797
140		1688.40	-437.79	738.40		0.5929
150		1688.40	-391.01	738.40		0.5295
160		1688.40	-325.59	738.40		0.4409
170		1688.40	-287.06	738.40		0.3888
180		1688.40	-251.31	738.40		0.3403
190		1688.56	-218.94	738.40		0.2965
200		1435.14	-181.95	800.26		0.2274
210		1435.14	-155.23	800.26		0.1940
220		1435.14	-145.45	677.69		0.2146
230		1435.28	-138.35	677.69		0.2041

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	224.85	1181.99	-554.32	1056.08	0.1902	0.5249
20	232.73	1181.99	-588.84	1016.99	0.1969	0.5790
30	215.20	1181.99	-566.86	1016.99	0.1821	0.5574
40	197.11	1181.99	-543.65	1016.99	0.1668	0.5346
50	179.43	1181.99	-521.87	1016.99	0.1518	0.5132
60	180.68	1181.99	-558.98	971.56	0.1529	0.5753
70	160.27	1181.99	-533.28	971.56	0.1356	0.5489
80	140.34	1181.99	-509.21	971.56	0.1187	0.5241
90	116.06	1181.99	-478.81	971.56	0.0982	0.4928
100	91.89	844.20	-448.54	955.05	0.1089	0.4696
110	67.01	1688.40	-502.84	860.77	0.0397	0.5842
120	64.83	1688.40	-481.17	860.77	0.0384	0.5590
130	39.62	1688.40	-449.27	860.77	0.0235	0.5219
140	31.88	1688.40	-474.31	738.40	0.0189	0.6423
150	24.86	1688.40	-421.52	738.40	0.0147	0.5709
160		1688.40	-350.28	738.40		0.4744
170		1688.40	-306.30	738.40		0.4148
180		1688.40	-265.52	738.40		0.3596
190		1688.56	-228.34	738.40		0.3092
200		1435.14	-187.95	800.26		0.2349
210		1435.14	-157.77	800.26		0.1971
220		1435.14	-145.92	677.69		0.2153
230		1435.28	-138.24	677.69		0.2040

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-409.81	1056.08		0.3881
20		2004.25	-440.11	1016.99		0.4328
30		2004.25	-429.74	1016.99		0.4226
40		2004.25	-418.45	1016.99		0.4115
50		2004.25	-408.93	1016.99		0.4021
60		2004.25	-446.58	971.56		0.4597
70		2004.25	-435.33	971.56		0.4481
80		2004.25	-426.06	971.56		0.4385
90		2004.25	-413.80	971.56		0.4259
100		1431.60	-402.14	971.56		0.4139
110	429.56	2863.20		866.74	0.1500	
120	428.19	2863.20		866.74	0.1496	
130	424.89	2863.20		866.74	0.1484	
140	505.66	2863.20		743.52	0.1766	
150	522.19	2863.20		743.52	0.1824	
160	517.82	2863.20		743.52	0.1809	
170	523.83	2863.20		743.52	0.1830	
180	531.79	2863.20		743.52	0.1857	
190	539.20	2863.21		743.52	0.1883	



200	501.43	2433.72	805.81	0.2060
210	509.76	2433.72	805.81	0.2095
220	625.74	2433.72	682.39	0.2571
230	625.90	2433.73	682.39	0.2572

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-315.70	1056.08		0.2989
20		2004.25	-344.33	1016.99		0.3386
30		2004.25	-342.05	1016.99		0.3363
40		2004.25	-338.52	1016.99		0.3329
50		2004.25	-336.41	1016.99		0.3308
60		2004.25	-373.27	971.56		0.3842
70		2004.25	-369.55	971.56		0.3804
80		2004.25	-367.42	971.56		0.3782
90		2004.25	-363.27	971.56		0.3739
100		1431.60	-359.12	971.56		0.3696
110	387.52	2863.20		866.74	0.1353	
120	386.28	2863.20		866.74	0.1349	
130	390.37	2863.20		866.74	0.1363	
140	473.23	2863.20		743.52	0.1653	
150	496.71	2863.20		743.52	0.1735	
160	498.40	2863.20		743.52	0.1741	
170	509.68	2863.20		743.52	0.1780	
180	522.15	2863.20		743.52	0.1824	
190	533.16	2863.21		743.52	0.1862	
200	497.71	2433.72		805.81	0.2045	
210	508.14	2433.72		805.81	0.2088	
220	625.32	2433.72		682.39	0.2569	
230	625.74	2433.73		682.39	0.2571	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-177.64	1056.08		0.1682
20		1181.99	-191.33	1016.99		0.1881
30		1181.99	-189.05	1016.99		0.1859
40		1181.99	-185.52	1016.99		0.1824
50		1181.99	-183.41	1016.99		0.1803
60		1181.99	-201.68	971.56		0.2076
70		1181.99	-197.96	971.56		0.2038
80		1181.99	-195.83	971.56		0.2016
90		1181.99	-191.68	971.56		0.1973
100		844.20	-187.53	955.05		0.1964
110	278.83	1688.40		860.77	0.1651	
120	277.59	1688.40		860.77	0.1644	
130	281.67	1688.40		860.77	0.1668	
140	346.38	1688.40		738.40	0.2052	
150	369.86	1688.40		738.40	0.2191	
160	371.54	1688.40		738.40	0.2201	
170	382.82	1688.40		738.40	0.2267	
180	395.30	1688.40		738.40	0.2341	
190	406.31	1688.56		738.40	0.2406	
200	380.73	1435.14		800.26	0.2653	
210	391.16	1435.14		800.26	0.2726	
220	487.02	1435.14		677.69	0.3394	
230	487.45	1435.28		677.69	0.3396	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-154.46	1056.08		0.1463
20		1181.99	-165.58	1016.99		0.1628
30		1181.99	-163.48	1016.99		0.1607
40		1181.99	-161.06	1016.99		0.1584



50		1181.99	-159.13	1016.99	0.1565
60		1181.99	-174.05	971.56	0.1791
70		1181.99	-171.58	971.56	0.1766
80		1181.99	-169.64	971.56	0.1746
90		1181.99	-166.79	971.56	0.1717
100		844.20	-163.93	955.05	0.1716
110	245.67	1688.40		860.77	0.1455
120	252.81	1688.40		860.77	0.1497
130	255.68	1688.40		860.77	0.1514
140	316.77	1688.40		738.40	0.1876
150	332.44	1688.40		738.40	0.1969
160	347.67	1688.40		738.40	0.2059
170	360.07	1688.40		738.40	0.2133
180	371.65	1688.40		738.40	0.2201
190	382.18	1688.56		738.40	0.2263
200	345.60	1435.14		800.26	0.2408
210	357.92	1435.14		800.26	0.2494
220	446.30	1435.14		677.69	0.3110
230	448.09	1435.28		677.69	0.3122

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-215.66	1056.08		0.2042
20		1181.99	-233.46	1016.99		0.2296
30		1181.99	-231.18	1016.99		0.2273
40		1181.99	-227.65	1016.99		0.2238
50		1181.99	-225.55	1016.99		0.2218
60		1181.99	-248.94	971.56		0.2562
70		1181.99	-245.21	971.56		0.2524
80		1181.99	-243.09	971.56		0.2502
90		1181.99	-238.94	971.56		0.2459
100		844.20	-234.79	955.05		0.2458
110	257.94	1688.40		860.77	0.1528	
120	256.70	1688.40		860.77	0.1520	
130	260.79	1688.40		860.77	0.1545	
140	322.00	1688.40		738.40	0.1907	
150	345.48	1688.40		738.40	0.2046	
160	347.17	1688.40		738.40	0.2056	
170	365.76	1688.40		738.40	0.2166	
180	383.11	1688.40		738.40	0.2269	
190	398.87	1688.56		738.40	0.2362	
200	378.47	1435.14		800.26	0.2637	
210	391.16	1435.14		800.26	0.2726	
220	487.02	1435.14		677.69	0.3394	
230	487.45	1435.28		677.69	0.3396	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-177.64	1056.08		0.1682
20		1181.99	-191.33	1016.99		0.1881
30		1181.99	-189.05	1016.99		0.1859
40		1181.99	-185.52	1016.99		0.1824
50		1181.99	-183.41	1016.99		0.1803
60		1181.99	-201.68	971.56		0.2076
70		1181.99	-197.96	971.56		0.2038
80		1181.99	-195.83	971.56		0.2016
90		1181.99	-191.68	971.56		0.1973
100		844.20	-187.53	955.05		0.1964
110	278.83	1688.40		860.77	0.1651	
120	277.59	1688.40		860.77	0.1644	
130	281.67	1688.40		860.77	0.1668	
140	346.38	1688.40		738.40	0.2052	
150	369.86	1688.40		738.40	0.2191	
160	371.54	1688.40		738.40	0.2201	
170	382.82	1688.40		738.40	0.2267	



180	395.30	1688.40	738.40	0.2341
190	406.31	1688.56	738.40	0.2406
200	380.73	1435.14	800.26	0.2653
210	391.16	1435.14	800.26	0.2726
220	487.02	1435.14	677.69	0.3394
230	487.45	1435.28	677.69	0.3396

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-215.66	1056.08		0.2042
20		1181.99	-233.46	1016.99		0.2296
30		1181.99	-231.18	1016.99		0.2273
40		1181.99	-227.65	1016.99		0.2238
50		1181.99	-225.55	1016.99		0.2218
60		1181.99	-248.94	971.56		0.2562
70		1181.99	-245.21	971.56		0.2524
80		1181.99	-243.09	971.56		0.2502
90		1181.99	-238.94	971.56		0.2459
100		844.20	-234.79	955.05		0.2458
110		1688.40	-237.10	860.77		0.2754
120		1688.40	-215.48	860.77		0.2503
130		1688.40	-211.39	860.77		0.2456
140		1688.40	-227.81	738.40		0.3085
150		1688.40	-204.33	738.40		0.2767
160		1688.40	-160.40	738.40		0.2172
170		1688.40	-141.81	738.40		0.1921
180		1688.40	-124.47	738.40		0.1686
190		1688.56	-108.70	738.40		0.1472
200		1435.14	-89.78	800.26		0.1122
210		1435.14	-75.93	800.26		0.0949
220	20.13	1435.14	-65.05	677.69	0.0140	0.0960
230	20.56	1435.28	-61.56	677.69	0.0143	0.0908

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		984.99	-325.38	880.07		0.3697
20		984.99	-349.95	847.49		0.4129
30		984.99	-342.60	847.49		0.4043
40		984.99	-334.01	847.49		0.3941
50		984.99	-326.83	847.49		0.3856
60		984.99	-356.76	809.63		0.4406
70		984.99	-347.37	809.63		0.4290
80		984.99	-339.55	809.63		0.4194
90		984.99	-328.59	809.63		0.4058
100		703.50	-317.62	795.88		0.3991
110	350.23	1407.00		717.31	0.2489	
120	348.41	1407.00		717.31	0.2476	
130	343.71	1407.00		717.31	0.2443	
140	408.86	1407.00		615.33	0.2906	
150	422.36	1407.00		615.33	0.3002	
160	414.07	1407.00		615.33	0.2943	
170	422.80	1407.00		615.33	0.3005	
180	430.31	1407.00		615.33	0.3058	
190	436.29	1407.13		615.33	0.3101	
200	406.86	1195.95		666.88	0.3402	
210	410.53	1195.95		666.88	0.3433	
220	499.03	1195.95		564.74	0.4173	
230	496.33	1196.06		564.74	0.4150	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		984.99	-325.38	880.07		0.3697
20		984.99	-349.95	847.49		0.4129



30	984.99	-342.60	847.49	0.4043
40	984.99	-334.01	847.49	0.3941
50	984.99	-326.83	847.49	0.3856
60	984.99	-356.76	809.63	0.4406
70	984.99	-347.37	809.63	0.4290
80	984.99	-339.55	809.63	0.4194
90	984.99	-328.59	809.63	0.4058
100	703.50	-317.62	795.88	0.3991
110	1407.00	-372.45	717.31	0.5192
120	1407.00	-351.14	717.31	0.4895
130	1407.00	-339.74	717.31	0.4736
140	1407.00	-368.67	615.33	0.5991
150	1407.00	-336.40	615.33	0.5467
160	1407.00	-283.66	615.33	0.4610
170	1407.00	-256.16	615.33	0.4163
180	1407.00	-229.86	615.33	0.3736
190	1407.13	-205.11	615.33	0.3333
200	1195.95	-173.15	666.88	0.2596
210	1195.95	-150.97	666.88	0.2264
220	1195.95	-144.03	564.74	0.2550
230	1196.06	-137.52	564.74	0.2435

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-215.66	1056.08		0.2042
20		1181.99	-233.46	1016.99		0.2296
30		1181.99	-231.18	1016.99		0.2273
40		1181.99	-227.65	1016.99		0.2238
50		1181.99	-225.55	1016.99		0.2218
60		1181.99	-248.94	971.56		0.2562
70		1181.99	-245.21	971.56		0.2524
80		1181.99	-243.09	971.56		0.2502
90		1181.99	-238.94	971.56		0.2459
100		844.20	-234.79	955.05		0.2458
110	257.94	1688.40		860.77	0.1528	
120	256.70	1688.40		860.77	0.1520	
130	260.79	1688.40		860.77	0.1545	
140	322.00	1688.40		738.40	0.1907	
150	345.48	1688.40		738.40	0.2046	
160	347.17	1688.40		738.40	0.2056	
170	365.76	1688.40		738.40	0.2166	
180	383.11	1688.40		738.40	0.2269	
190	398.87	1688.56		738.40	0.2362	
200	378.47	1435.14		800.26	0.2637	
210	391.16	1435.14		800.26	0.2726	
220	487.02	1435.14		677.69	0.3394	
230	487.45	1435.28		677.69	0.3396	

Absolute Maximum of the all of the Stress Ratio's

0.6423

Governing Element: S-3

Governing Load Case 10 : EP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
K-10	...	-164.7	49.1	340.5	224.9	+ Tens
K-10	...	-164.7	-49.1	-340.5	-554.3	- Comp
K-9	...	-178.1	53.5	357.3	232.7	+ Tens
K-9	...	-178.1	-53.5	-357.3	-588.8	- Comp
K-8	...	-175.8	53.5	337.5	215.2	+ Tens
K-8	...	-175.8	-53.5	-337.5	-566.9	- Comp
K-7	...	-173.3	52.6	317.8	197.1	+ Tens



K-7	...	-173.3	-52.6	-317.8	-543.7	-	Comp
K-6	...	-171.2	52.6	298.1	179.4	+	Tens
K-6	...	-171.2	-52.6	-298.1	-521.9	-	Comp
K-5	...	-189.1	57.9	311.9	180.7	+	Tens
K-5	...	-189.1	-57.9	-311.9	-559.0	-	Comp
K-4	...	-186.5	56.9	289.9	160.3	+	Tens
K-4	...	-186.5	-56.9	-289.9	-533.3	-	Comp
K-3	...	-184.4	56.9	267.9	140.3	+	Tens
K-3	...	-184.4	-56.9	-267.9	-509.2	-	Comp
K-2	...	-181.4	55.9	241.6	116.1	+	Tens
K-2	...	-181.4	-55.9	-241.6	-478.8	-	Comp
K-1	...	-178.3	54.9	215.3	91.9	+	Tens
K-1	...	-178.3	-54.9	-215.3	-448.5	-	Comp
H-1	-46.6	-171.3	64.0	220.9	67.0	+	Tens
H-1	-46.6	-171.3	-64.0	-220.9	-502.8	-	Comp
S-1	-46.6	-161.5	52.6	220.4	64.8	+	Tens
S-1	-46.6	-161.5	-52.6	-220.4	-481.2	-	Comp
S-2	-46.6	-158.2	52.6	191.8	39.6	+	Tens
S-2	-46.6	-158.2	-52.6	-191.8	-449.3	-	Comp
S-3	-54.4	-166.9	60.8	192.3	31.9	+	Tens
S-3	-54.4	-166.9	-60.8	-192.3	-474.3	-	Comp
S-4	-54.4	-144.0	60.8	162.4	24.9	+	Tens
S-4	-54.4	-144.0	-60.8	-162.4	-421.5	-	Comp
S-5	-54.4	-121.7	39.7	134.5	...	+	Tens
S-5	-54.4	-121.7	-39.7	-134.5	-350.3	-	Comp
S-6	-54.4	-103.6	39.7	108.7	...	+	Tens
S-6	-54.4	-103.6	-39.7	-108.7	-306.3	-	Comp
S-7	-54.4	-86.7	39.7	84.8	...	+	Tens
S-7	-54.4	-86.7	-39.7	-84.8	-265.5	-	Comp
S-8	-54.4	-71.4	39.7	63.0	...	+	Tens
S-8	-54.4	-71.4	-39.7	-63.0	-228.3	-	Comp
S-9	-50.2	-55.5	36.7	45.6	...	+	Tens
S-9	-50.2	-55.5	-36.7	-45.6	-187.9	-	Comp
S-10	-50.2	-42.6	36.1	28.9	...	+	Tens
S-10	-50.2	-42.6	-36.1	-28.9	-157.8	-	Comp
S-11	-59.2	-26.3	42.6	17.8	...	+	Tens
S-11	-59.2	-26.3	-42.6	-17.8	-145.9	-	Comp
H-2	-59.2	-24.4	41.1	13.5	...	+	Tens
H-2	-59.2	-24.4	-41.1	-13.5	-138.2	-	Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: K-1

Total vessel (empty) weight	Fg	386761.312 kgf
Head Weight below section 2-2	DeltaFg	22070.721 kgf
Weight of vessel contents (liquid)	Ff	74491.055 kgf
Vessel Gross Weight	Fg + Ff	461252.375 kgf

Skirt Mean Diameter	Dz	4918.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm
Skirt Height	H	3000.000 mm

Additional Axial Force at section 1-1	F1	-7568.000 kgf
Moment at section 1-1	M1	587676.938 kgf-m.
Applied Moment - Bottom of Skirt	Mb	668664.500 kgf-m.
Weight at Bottom of Skirt	W	455896.906 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	4926.000 mm
Head Corroded Thickness	eb	14.000 mm

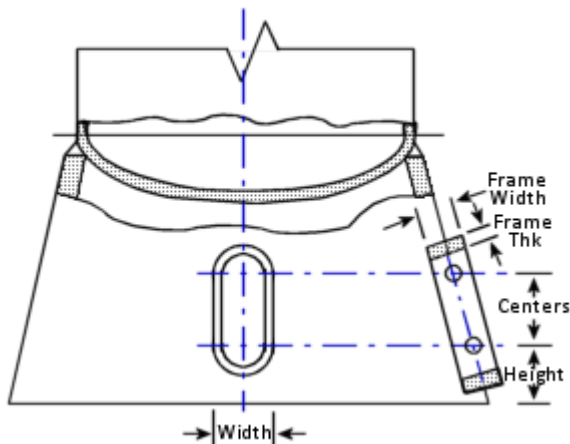
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 1231.0 \text{ mm}$$

$$K = D_i / (2 * h_i) = 4912.0 / (2 * 1231.0) = 1.9951$$

$$R = D_i (0.44 * K + 0.02) = 4912.0 * (0.44 * 1.995 + 0.02) = 4410.2656 \text{ mm}$$

$$r = D_i ((0.5 / K) - 0.08) = 4912.0 * ((0.5 / 1.995) - 0.08) = 838.04 \text{ mm}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	76607088.000 cm ⁴
Moment of Inertia	Iyy =	76607088.000 cm ⁴
Section Modulus	W =	311537824.000 mm ³
Section Metal Area	A =	2533.858 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(76607088, 76607088) / 2533.858)}$$

$$= 1738.776 \text{ mm}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1738.776/3000.0$$

$$= 0.580$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 587677 kgf-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -587677/.12462E+10 - 455896.906/2533.85815$$

$$= -227.082 \text{ kgf/cm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +587677/.12462E+10 - 455896.906/2533.85815$$

$$= -132.763 \text{ kgf/cm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -7568.0 - 386761.312 - 74491.055 + 4 \cdot 587677/4918.0$$

$$= 9159.5 \text{ kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -7568.0 - 386761.312 - 74491.055 - 4 \cdot 587677/4918.0$$

$$= -946800.2 \text{ kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (9159.497 + 22070.721 + 74491.055) / (\pi \cdot 4926.0 \cdot 14.0) + 410.0 \cdot 4926.0 / (4 \cdot 14.0)$$

$$= 416.57 \text{ kgf/cm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.9E+06 + 22070.721 + 74491.055) / (\pi \cdot 4926.0 \cdot 14.0) + 410.0 \cdot 4926.0 / (4 \cdot 14.0)$$

$$= -24.66 \text{ kgf/cm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (74491.055 + 22070.721) / (\pi \cdot 4926.0 \cdot 14.0) + 410.0 \cdot 4926.0 / (4 \cdot 14.0)$$

$$= 412.34 \text{ kgf/cm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 9159.497 / (\pi \cdot 4918.0 \cdot 16.4)$$

$$= 3.61 \text{ kgf/cm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -946800.250 / (\pi \cdot 4918.0 \cdot 16.4)$$

$$= -373.66 \text{ kgf/cm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(4926.0 + 14.0 - 4918.0 + 16.4) / (838.04 + 14.0)$$

$$= 0.9775$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.977) = 12.186 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(14.0^2 + 16.4^2 + 2 \cdot 14.0 \cdot 16.4 \cdot 0.977)}$$

$$= 15.115 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 15.11 \cdot 9159.5 = 138.44 \text{ kgf-m.}$$

$$M_q = a \cdot F_{zq} = 15.11 \cdot -946800.2 = -14310.59 \text{ kgf-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(14.0/16.4)^2$$

$$= 0.588$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.588 \cdot 6 \cdot 138.443 / (\pi \cdot 4926.0 \cdot 14.0^2)$$

$$= 16.12 \text{ kgf/cm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.588 \cdot 6 \cdot -14310.586 / (\pi \cdot 4926.0 \cdot 14.0^2)$$

$$= -1665.82 \text{ kgf/cm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.588 \cdot 6 \cdot 138.443 / (\pi \cdot 4918.0 \cdot 16.4^2)$$

$$= 11.76 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.588 \cdot 6 \cdot -14310.586 / (\pi \cdot 4918.0 \cdot 16.4^2)$$

$$= -1215.91 \text{ kgf/cm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.551/193.937 = 0.0028$$

$$y = 125 \cdot eb/D_b = 125 \cdot 14.0/4926.0 = 0.3553$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.355 + 3.994 = 4.3315$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (410.0 + 0.0) \cdot 4926.0 ((12.19/40.0) \cdot 4.332 - 1) / (4 \cdot 14.0)$$

$$= 117.55 \text{ kgf/cm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 416.57 - 16.115 + 117.554$$

$$= 518.01 \text{ kgf/cm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 416.57 + 16.115 - 117.554$$

$$= 315.13 \text{ kgf/cm}^2$$

Stress [σ_{1qitot}]:

$$\begin{aligned}
 &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\
 &= -24.663 - (-1665.816) + 117.554 \\
 &= 1758.71 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned}
 &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\
 &= -24.663 + (-1665.816) - 117.554 \\
 &= -1808.03 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned}
 &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\
 &= 412.342 + 16.115 + 117.554 \\
 &= 546.01 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 412.342 - 16.115 - 117.554 \\
 &= 278.67 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned}
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 412.342 + (-1665.816) + 117.554 \\
 &= -1135.92 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 412.342 - (-1665.816) - 117.554 \\
 &= 1960.60 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned}
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= 3.615 - 11.763 \\
 &= -8.15 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{3potot}]:

$$\begin{aligned}
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= 3.615 + 11.763 \\
 &= 15.38 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{3qitot}]:

$$\begin{aligned}
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -373.66 - (-1215.909) \\
 &= 842.25 \text{ kgf/cm}^2
 \end{aligned}$$

Stress [σ_{3qotot}]:

$$\begin{aligned}
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -373.66 + (-1215.909) \\
 &= -1589.57 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1/1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 1407.0(3 - (1/1.5)) \cdot (416.57/1407.0)^2 \\
 &= 4138.78 \text{ kgf/cm}^2
 \end{aligned}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1/1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 1407.0(3 - (1/1.5)) \cdot (-24.663/1407.0)^2 \\
 &= 4220.71 \text{ kgf/cm}^2
 \end{aligned}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1/1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 1407.0(3 - (1/1.5)) \cdot (412.342/1407.0)^2 \\
 &= 4140.44 \text{ kgf/cm}^2
 \end{aligned}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 1407.0(3 - (1/1.5) \cdot (412.342/1407.0)^2)$$

$$= 4140.44 \text{ kgf/cm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 1407.0(3 - (1/1.5) \cdot (3.615/1407.0)^2)$$

$$= 4220.99 \text{ kgf/cm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 1407.0(3 - (1/1.5) \cdot (-373.66/1407.0)^2)$$

$$= 4154.84 \text{ kgf/cm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	518.01	4138.78	Pass	46
Stot1po	315.13	4138.78	Pass	47
Stot1qi	1758.71	4220.71	Pass	48
Stot1qo	-1808.03	4220.71	Pass	49
Stot2pi	546.01	4140.44	Pass	50
Stot2po	278.67	4140.44	Pass	51
Stot2qi	-1135.92	4140.44	Pass	52
Stot2qo	1960.60	4140.44	Pass	53
Stot3pi	-8.15	4220.99	Pass	54
Stot3po	15.38	4220.99	Pass	55
Stot3qi	842.25	4154.84	Pass	56
Stot3qo	-1589.57	4154.84	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Packing is installed in the Field after being lifted.
Insulation is installed in the Shop.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	37121.043 mm
Center of Gravity of the Packing	44575.000 mm
Center of Gravity of the Liquid	27693.418 mm
Center of Gravity of the Insulation	41489.824 mm
Center of Gravity of the Lining	53939.160 mm
Center of Gravity of the Stiffening Rings	40990.348 mm
Center of Gravity of the Nozzles	38913.125 mm
Center of Gravity of the Added Weights (Operating)	41540.246 mm
Center of Gravity of the Added Weights (Empty)	37340.059 mm
Center of Gravity of Bare Shell New and Cold	27068.240 mm
Center of Gravity of Bare Shell Corroded	25816.584 mm
Vessel CG in the Operating Condition	35294.707 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	32398.143 mm
Vessel CG in the Test Condition	38812.855 mm

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**Skirt Data :**

Skirt Outside Diameter at Base	SOD, D6	4936.0000	mm
Skirt Thickness	STHK	22.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	44.0000	mm
Design Temperature of the Basing		82.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1407.13	kgf/cm ²
Basing Yield Stress		2473.72	kgf/cm ²
Inside Diameter of Basing	DI, D4	4752.0000	mm
Outside Diameter of Basing	DOU, D5	5272.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	1407.00	kgf/cm ²
Number of Bolts	RN, nb	32	
Diameter of Bolt Circle	DC, DBC	5104.0000	mm
Bolt Allowable Shear Stress		838.200	kgf/cm ²

Ultimate Comp. Strength of Concrete	FPC	421.0	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm ²
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	168.0000	mm
Width of Gussets at Base Plate	BWDT	168.0000	mm
Gusset Plate Elastic Modulus	E	2031470.0	kgf/cm ²
Gusset Plate Yield Stress	SY	2473.7	kgf/cm ²
Height of Gussets	HG, h1	256.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	0.0000	mm
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Dead Weight of Vessel	DW	284683.8	kgf
Operating Weight of Vessel	ROW	532172.4	kgf
Earthquake Moment on Basing	EQMOM	1048584.4	kgf-m.
Wind Moment on Basing	WIMOM	891706.4	kgf-m.

User Moment at the Base [Earthquake Case]		429834.6	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		429834.6	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 1478419/5104.0) - 518806) / 32$$

$$= 19994.6 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 1321541/5104.0) - 284684) / 32$$

$$= 23468.9 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5104.0) - 532172) / 32$$

$$= -16630.4 \text{ kgf}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

$$\text{Area Available in 1 Bolt (corroded)} \quad Abss : 11.5517 \text{ cm}^2$$

$$\text{Area Available in all the Bolts} \quad Abss \cdot RN : 369.6529 \text{ cm}^2$$

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.150	1.511	2.441	0.438	0.781	171992.3	456676.1	7726.4
4	0.225	0.263	1.297	2.605	0.454	0.777	166984.8	451668.5	7029.7
6	0.282	0.272	1.459	2.482	0.442	0.780	170701.5	455385.2	7542.2
8	0.277	0.275	1.446	2.492	0.443	0.780	170382.8	455066.6	7497.3
10	0.273	0.273	1.436	2.500	0.443	0.780	170144.9	454828.6	7463.9

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 170144.9 / (2.305 \cdot 5104.0 \cdot 2.5)$$

$$= 1156.905 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.0 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 32 \cdot 11.552 / (\pi \cdot 5104.0)$$

$$= 2.305 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 44.254[(2 \cdot 0.2735 \cdot 5104.0 + 260.0) / (2 \cdot 0.2735 \cdot 5104.0)]$$

$$= 48.375 \text{ kgf/cm}^2, \text{ Should be } \leq 229.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 168.0, b = 250.542, l/b = 0.670545

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(1709.068, 3082.142) = 3082.142 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 3082.142 / 1632.7)} + 0.0$$

$$= 33.655 \text{ mm}$$

Nomenclature:

a	= (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P	= Sa * Abss	Maximum Load on one Bolt
l		Radial Top Plate Width
lavg		Average Gusset Plate Width
g1	= Gamma 1	Constant Term f(b/l)
g2	= Gamma 2	Constant Term f(b/l)
g	= Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb		Allowable Bending Stress

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429

As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 16253.15 / (4 \cdot \pi) [1.3(\ln((2 \cdot 168.0 \cdot \sin(\pi \cdot 84.0/168.0)) / (\pi \cdot 40.0))) + 1] - [(0.7 - 0.044) \cdot 16253.15 / (4 \cdot \pi)]$$

$$= 2099.2126 \text{ kgf}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(2099.21)}{1649.15}} + 0.0$$

$$= 27.6360 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + C_a$$

$$= 16253.15 / (1265.526 \cdot 168.0) + 0.0$$

$$= 9.525 \text{ (not less than } 9.525 + 0.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 5104.0 / 32 = 501.084 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 \left(\frac{P \cdot a}{m \cdot (h + t_{ba})} \cdot 1.5 \cdot \text{Sktope} \right)^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76 (16253 \cdot 84.0 / (501.08 \cdot 300.0 \cdot 2111))^{2/3} \cdot 2468.01^{1/3} + C_a$$

$$= 13.556 + 1.6 = 15.156 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 31213.2 / (11.552 \cdot 32)$$

$$= 84.4 \text{ kgf/cm}^2. \text{ Must be less than } 838.2 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	33.6554	mm
Actual Basing Thickness as entered by user	44.0000	mm
Required Thickness of Chair Cap	27.6360	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	9.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	15.1561	mm
Given Thickness of Skirt	22.0000	mm
Required Gusset Height to meet local stress	101.1067	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 554.32 \cdot (22.0 - 1.6)$$

$$= 113.08 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 113 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707]$$

$$= 8.082 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 16253.2 / (170.0 + 2 \cdot (256.0 + 44.0))$$

$$= 21.108 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 16253.2 \cdot 84.0 / (170.0 \cdot (300.0) + 0.6667 \cdot (300.0)^2)$$

$$= 12.3 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{21.11^2 + 12.3^2}$$

$$= 24.43 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:



$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 24.43 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707] \\ &= 1.746 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 16253.2 / (2 \cdot 168.0) \\ &= 48.372 \text{ kgf/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 16253.2 \cdot 84.0 / (2 \cdot 300.0 \cdot 168.0) \\ &= 13.544 \text{ kgf/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{48.37^2 + 13.54^2} \\ &= 50.233 \text{ kgf/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 50.23 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707] \\ &= 3.59 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	8.0822	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.5903	mm

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: C-30"****From: 110**

Pressure for Reinforcement Calculations	P	566.874	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Elliptical Head	D	4900.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	20.0000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	18.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1362.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		300.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2950.0	mm
Centerline Distance to Tangent		2875.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: C-30"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 30.000 in.
 Actual Thickness Used in Calculation 0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 566.87 \cdot 4912.0 \cdot 0.8979 / (2 \cdot 1407.0 \cdot 1.0 - 0.2 \cdot 566.87)$
 $= 9.0640$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 566.87 \cdot 381.0 / (1407 \cdot 1.0 + 0.4 \cdot 566.87)$
 $= 1.5628$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.5655 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1468.0000 mm
 Parallel to Vessel Wall, opening length d 734.0000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 35.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.0/1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1407.0/1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1407.0/1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	66.530	33.308	NA
Area in Shell	A1	36.230	36.144	NA
Area in Nozzle Wall	A2	8.706	8.704	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		4.680	4.680	NA
Area in Element	A5	96.000	96.000	NA
TOTAL AREA AVAILABLE	Atot	145.616	145.528	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.



SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 867.7083 16.0000 mm
 Based on given Pad Diameter: 1362.0000 2.8189 mm
 Based on Shell or Nozzle Thickness: 846.5667 20.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (734.0 \cdot 9.064 \cdot 1.0 + 2 \cdot 14.0 \cdot 9.064 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 66.530 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 734.0 (1.0 \cdot 14.0 - 1.0 \cdot 9.064) - 2 \cdot 14.0$$

$$(1.0 \cdot 14.0 - 1.0 \cdot 9.064) \cdot (1 - 1.0)$$

$$= 36.230 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 35.0) \cdot (14.0 - 1.56) \cdot 1.0$$

$$= 8.706 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 18.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$$

$$= 4.680 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1362.0 - 762.0) \cdot 16.0 \cdot 1.0$$

$$= 96.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.5655 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 16.0618 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 16.0618 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9175 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[14.331, \max(16.0618, 7.5)]$$

$$= 14.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.5655, 14.3312)$$

$$= 14.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 279.7,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3237.8 kgf/cm ²	Passed
Occasional	: 74.3,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 130.3,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)



Govrn. thk, tg = 20.0, tr = 1.563, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.112$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 16.0, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.647$, Temp. Reduction = 20 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.647$, Temp. Reduction = 20 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 16.0, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.647$, Temp. Reduction = 20 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, tg = 20.0, tr = 9.064, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.647$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $566.87 / 1960.01 = 0.289$

Weld Size Calculations, Description: C-30"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$12.7260 = 0.7 * W_o \text{ mm}$
Pad Weld	$8.0000 = 0.5 * \text{TminPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr))S_v)$



$$= \max(0, (66.5297 - 36.2303 + 2 \cdot 14.0 \cdot 1.0 \cdot (1.0 \cdot 14.0 - 9.064)) / 1407)$$

$$= 44575.77 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / 707)^2 \cdot \text{fr2}) \cdot S_v$$

$$= (8.7061 + 96.0 + 4.68 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 153905.97 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v$$

$$= (8.7061 + 0.0 + 3.24 + (3.92)) \cdot 1407$$

$$= 22323.52 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S$$

$$= (8.7061 + 0.0 + 4.68 + 96.0 + (3.92)) \cdot 1407$$

$$= 159421.41 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot 18.0 \cdot 0.49 \cdot 1407$$

$$= 148538. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1362.0 \cdot 12.0 \cdot 0.49 \cdot 1407$$

$$= 176998. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 374.0) \cdot (20.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 162010. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 762.0 \cdot 16.0 \cdot 0.74 \cdot 1407$$

$$= 199398. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (20.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 174473. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (176998 + 162010) = 339008 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (148538 + 199398 + 174473 + 0) = 522408 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (176998 + 174473 + 0) = 351471 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 339007 kgf, must exceed W = 44575 kgf	or W1 = 153905 kgf
Path 2-2 = 522408 kgf, must exceed W = 44575 kgf	or W2 = 22323 kgf
Path 3-3 = 351470 kgf, must exceed W = 44575 kgf	or W3 = 159421 kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 745 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 16.4292 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 536.4291 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.695 %

Input Echo, WRC107/537 Item 1, Description: C-30" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	8819.999	mm
Vessel Thickness	Tv	20.000	mm
Design Temperature	T1	160.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.00	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.00	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	762.000	mm
Nozzle Thickness	Tn	20.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	16.000	mm
Diameter of Reinforcing Pad	Dpad	1362.000	mm
Design Internal Pressure	Dp	566.874	kPa
Include Pressure Thrust		No	

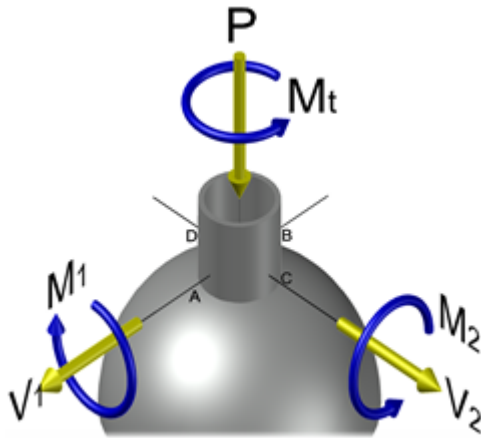
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	5734.0	kgf
Longitudinal Shear (SUS)	(V1) V1	3822.7	kgf
Circumferential Shear (SUS)	(Vc) V2	3822.7	kgf
Circumferential Moment (SUS)	(Mc) M1	8027.8	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	8027.8	kgf-m.
Torsional Moment (SUS)	Mt	11468.3	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(4423.0 (20.0 - 6.0))}$$

$$= 1583.176 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	5734.0	kgf
Circumferential Shear	(VC) V2	3822.7	kgf
Longitudinal Shear	(VL) V1	3822.7	kgf
Circumferential Moment	(MC) M1	8027.8	kgf-m.
Longitudinal Moment	(ML) M2	8027.8	kgf-m.
Torsional Moment	MT	11468.3	kgf-m.

Dimensionless Param: U = 1.04 TAU = 26.71 RHO = 4.00 (2.14)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 7	0.02190	(A,B,C,D)
$M(x) / P$	SP 7	0.00899	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.03556	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 7	0.01590	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.03556	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 7	0.01590	(A,B,C,D)
$N(y) * T / P$	SP 7	0.20021	(A,B,C,D)
$M(y) / P$	SP 7	0.02371	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.25159	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 7	0.05252	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.25159	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 7	0.05252	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-14.0	-14.0	-14.0	-14.0	-14.0	-14.0	-14.0	-14.0
Rad.	Bend. P	-34.3	34.3	-34.3	34.3	-34.3	34.3	-34.3	34.3
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-87.0	-87.0	87.0	87.0
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-233.4	233.4	233.4	-233.4
Rad.	Memb. ML	-87.0	-87.0	87.0	87.0	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-233.4	233.4	233.4	-233.4	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-368.7	166.8	272.1	-126.0	-368.7	166.8	272.1	-126.0
Tang. Memb. P	-127.6	-127.6	-127.6	-127.6	-127.6	-127.6	-127.6	-127.6
Tang. Bend. P	-90.6	90.6	-90.6	90.6	-90.6	90.6	-90.6	90.6
Tang. Memb. MC	0.0	0.0	0.0	0.0	-615.5	-615.5	615.5	615.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-770.9	770.9	770.9	-770.9
Tang. Memb. ML	-615.5	-615.5	615.5	615.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-770.9	770.9	770.9	-770.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-1604.6	118.4	1168.2	-192.3	-1604.6	118.4	1168.2	-192.3
Shear VC	10.6	10.6	-10.6	-10.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-10.6	-10.6	10.6	10.6
Shear MT	41.9	41.9	41.9	41.9	41.9	41.9	41.9	41.9
Tot. Shear	52.6	52.6	31.3	31.3	31.3	31.3	52.6	52.6
Str. Int.	1606.8	200.5	1169.3	204.7	1605.4	182.2	1171.3	221.3

Unitless Prm: U = 2.74 TAU = 0.00 (48.14) RHO = 0.00 (1.00)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.02175	(A,B,C,D)
$M(x) / P$	SR 2	0.00990	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SR 3	0.01691	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SR 3	0.01053	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SR 3	0.01691	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SR 3	0.01053	(A,B,C,D)
$N(y) * T / P$	SR 2	0.00650	(A,B,C,D)
$M(y) / P$	SR 2	0.00300	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SR 3	0.00507	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SR 3	0.00317	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SR 3	0.00507	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (kgf/cm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-63.6	-63.6	-63.6	-63.6	-63.6	-63.6	-63.6	-63.6
Rad.	Bend. P	-173.8	173.8	-173.8	173.8	-173.8	173.8	-173.8	173.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-278.4	-278.4	278.4	278.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1039.6	1039.6	1039.6	-1039.6
Rad.	Memb. ML	-278.4	-278.4	278.4	278.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1039.6	1039.6	1039.6	-1039.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1555.5	871.4	1080.6	-651.1	-1555.5	871.4	1080.6	-651.1
Tang. Memb. P		-19.0	-19.0	-19.0	-19.0	-19.0	-19.0	-19.0	-19.0
Tang.	Bend. P	-52.7	52.7	-52.7	52.7	-52.7	52.7	-52.7	52.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-83.4	-83.4	83.4	83.4
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-313.5	313.5	313.5	-313.5
Tang.	Memb. ML	-83.4	-83.4	83.4	83.4	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-313.5	313.5	313.5	-313.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-468.6	263.8	325.3	-196.5	-468.6	263.8	325.3	-196.5
Shear VC		12.8	12.8	-12.8	-12.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-12.8	-12.8	12.8	12.8
Shear MT		28.1	28.1	28.1	28.1	28.1	28.1	28.1	28.1



Tot. Shear	40.9	40.9	15.3	15.3	15.3	15.3	40.9	40.9
Str. Int.	1557.0	874.1	1080.9	651.6	1555.7	871.8	1082.8	654.7

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (kgf/cm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		425.5	425.5	425.5	425.5	425.5	425.5	425.5	425.5
Rad. Pl (SUS)		-100.9	-100.9	73.0	73.0	-100.9	-100.9	73.0	73.0
Rad. Q (SUS)		-267.8	267.8	199.1	-199.1	-267.8	267.8	199.1	-199.1
Long. Pm (SUS)		425.5	425.5	425.5	425.5	425.5	425.5	425.5	425.5
Long. Pl (SUS)		-743.1	-743.1	487.9	487.9	-743.1	-743.1	487.9	487.9
Long. Q (SUS)		-861.5	861.5	680.3	-680.3	-861.5	861.5	680.3	-680.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		10.6	10.6	-10.6	-10.6	-10.6	-10.6	10.6	10.6
Shear Q (SUS)		41.9	41.9	41.9	41.9	41.9	41.9	41.9	41.9
Pm (SUS)		425.5	425.5	425.5	425.5	425.5	425.5	425.5	425.5
Pm+Pl (SUS)		642.5	642.5	913.7	913.7	642.5	642.5	913.7	913.7
Pm+Pl+Q (Total)		1240.3	626.0	1594.8	311.9	1237.4	607.6	1596.8	328.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	425.47	1407.00	Passed
Pm+Pl (SUS)	913.69	2110.50	Passed
Pm+Pl+Q (TOTAL)	1596.75	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		911.7	911.7	911.7	911.7	911.7	911.7	911.7	911.7
Rad. Pl (SUS)		-342.0	-342.0	214.8	214.8	-342.0	-342.0	214.8	214.8
Rad. Q (SUS)		-1213.4	1213.4	865.9	-865.9	-1213.4	1213.4	865.9	-865.9
Long. Pm (SUS)		911.7	911.7	911.7	911.7	911.7	911.7	911.7	911.7
Long. Pl (SUS)		-102.4	-102.4	64.4	64.4	-102.4	-102.4	64.4	64.4
Long. Q (SUS)		-366.2	366.2	260.9	-260.9	-366.2	366.2	260.9	-260.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		12.8	12.8	-12.8	-12.8	-12.8	-12.8	12.8	12.8
Shear Q (SUS)		28.1	28.1	28.1	28.1	28.1	28.1	28.1	28.1
Pm (SUS)		911.7	911.7	911.7	911.7	911.7	911.7	911.7	911.7
Pm+Pl (SUS)		809.9	809.9	1127.6	1127.6	809.9	809.9	1127.6	1127.6
Pm+Pl+Q (Total)		1089.9	1785.8	1992.6	715.7	1087.2	1783.5	1994.5	718.9

**Vessel Stress Summation Comparison (kgf/cm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	911.70	1407.00	Passed
Pm+Pl (SUS)	1127.56	2110.50	Passed
Pm+Pl+Q (TOTAL)	1994.54	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: L1-2"****From: 120**

Pressure for Reinforcement Calculations	P	550.172	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	1958.3325	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.382	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 550.17 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 550.17)$
 $= 9.8166$ mm



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 550.17 \cdot 30.1625 / (1407 \cdot 1.0 + 0.4 \cdot 550.17)$
 $= 0.1201 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2657 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.8896	mm
Parallel to Vessel Wall, opening length	d	52.9448	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.2253	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: L1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 6.2657 mm
Wall Thickness per UG16(b),	tr16b	= 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.8166 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.8166 mm
Wall Thickness, shell/head, external pressure	trb2	= 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 7.5000 mm
Wall Thickness per table UG-45	tb3	= 9.4200 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.42, \max(15.8166, 7.5)]$
 $= 9.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(6.2657, 9.42)$
 $= 9.4200 \text{ mm}$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.69, tr = 0.12, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 9.69, tr = 0.12, c = 6.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = $550.17/1960.01 = 0.281$

Weld Size Calculations, Description: L1-2"

Intermediate Calc. for nozzle/shell Welds T_{min} 3.6901 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.5831 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1856 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 270.1856 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.94 mm

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**Input, Nozzle Desc: MH-4-24"****From: 130**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4200.0005	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-4-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3538 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1171.2000 mm
Parallel to Vessel Wall, opening length d 585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 35.0000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.776	32.981	NA
Area in Shell	A1	27.208	16.022	NA
Area in Nozzle Wall	A2	7.590	7.441	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	109.758	98.423	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 704.1432 18.0000 mm
Based on given Pad Diameter: 1009.6000 4.2544 mm
Based on Shell or Nozzle Thickness: 704.1432 18.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (585.6 \cdot 9.3538 \cdot 1.0 + 2 \cdot 12.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 54.776 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 585.6 (1.0 \cdot 14.0 - 1.0 \cdot 9.354) - 2 \cdot 12.0 \\
 &\quad (1.0 \cdot 14.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0) \\
 &= 27.208 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 35.0) \cdot (12.0 - 1.16) \cdot 1.0 \\
 &= 7.590 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3696 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[14.331, \max(15.3538, 7.5)] \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.3696, 14.3312) \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0



Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 18.0$, $c = 6.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.668$, Temp. Reduction = 18 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $tg = 18.0$, $tr = 1.156$, $c = 6.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $524.29 / 1960.01 = 0.267$

Weld Size Calculations, Description: MH-4-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 12.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 14.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$7.0000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (54.7761 - 27.2079 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 14.0 - 9.3538)) \cdot 1407)$
 $= 40357.41$ kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (7.5905 + 72.0 + 2.96 - 0.0 \cdot 1.0) \cdot 1407$
 $= 116148.38$ kgf

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.5905 + 0.0 + 1.0 + (3.36)) \cdot 1407$
 $= 16814.31$ kgf

Weld Load [W3]:

$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.5905 + 0.0 + 2.96 + 72.0 + (3.36)) \cdot 1407$



= 120875.89 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 298.8) \cdot (18.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 110944. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (20.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 139578. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (153069 + 110944) = 264013 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66017 + 179458 + 139578 + 0) = 385053 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (153069 + 139578 + 0) = 292647 \text{ kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 264012 \text{ kgf, must exceed } W = 40357 \text{ kgf} \quad \text{or } W1 = 116148 \text{ kgf}$$

$$\text{Path 2-2} = 385053 \text{ kgf, must exceed } W = 40357 \text{ kgf} \quad \text{or } W2 = 16814 \text{ kgf}$$

$$\text{Path 3-3} = 292647 \text{ kgf, must exceed } W = 40357 \text{ kgf} \quad \text{or } W3 = 120875 \text{ kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 780 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0337 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 539.0336 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 3.043 \%$$

This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 247.03 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 585.60 mm

**Input, Nozzle Desc: E1-36"****From: 140**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	36.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1400.0000	mm
Distance from Bottom/Left Tangent		7400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	18.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E1-36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	1.102	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$$

$$= 9.3538 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 524.29 \cdot 457.2 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$$

$$= 1.7347 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.7420 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1400.0000 mm

Parallel to Vessel Wall, opening length d 700.0000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 55.0000 mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	81.416	51.373	NA
Area in Shell	A1	109.342	96.363	NA
Area in Nozzle Wall	A2	22.292	22.284	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.240	3.240	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	134.874	121.887	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (870.4001 \cdot 9.3538 \cdot 1.0 + 2 \cdot 22.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 81.416 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 529.6(1.0 \cdot 30.0 - 1.0 \cdot 9.354) - 2 \cdot 22.0$$

$$(1.0 \cdot 30.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0)$$

$$= 109.342 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 55.0)(22.0 - 1.73)1.0$$



$$= 22.292 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 18.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 3.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 7.7420 \text{ mm}$$

$$\text{Wall Thickness per UG16(b)}, \quad tr16b = 7.5000 \text{ mm}$$

$$\text{Wall Thickness, shell/head, internal pressure} \quad trb1 = 15.3538 \text{ mm}$$

$$\text{Wall Thickness} \quad tb1 = \max(trb1, tr16b) = 15.3538 \text{ mm}$$

$$\text{Wall Thickness, shell/head, external pressure} \quad trb2 = 6.9206 \text{ mm}$$

$$\text{Wall Thickness} \quad tb2 = \max(trb2, tr16b) = 7.5000 \text{ mm}$$

$$\text{Wall Thickness per table UG-45} \quad tb3 = 14.3312 \text{ mm}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[14.331, \max(15.3538, 7.5)]$$

$$= 14.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.742, 14.3312)$$

$$= 14.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	184.4,	Allowable	:	1407.0 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	3333.1 kgf/cm ²	Passed
Occasional	:	51.6,	Allowable	:	1871.3 kgf/cm ²	Passed
Shear	:	84.6,	Allowable	:	984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

$$\text{Govrn. thk, } tg = 28.0, tr = 1.735, c = 6.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (tg - c) = 0.079, \text{Temp. Reduction} = 78 \text{ } ^\circ\text{C}$$

$$\text{Min Metal Temp. w/o impact per UCS-66, Curve A} \quad 22 \text{ } ^\circ\text{C}$$

$$\text{Min Metal Temp. at Required thickness (UCS 66.1)} \quad -104 \text{ } ^\circ\text{C}$$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

$$\text{Govrn. thk, } tg = 28.0, tr = 1.735, c = 6.0 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (tg - c) = 0.079, \text{Temp. Reduction} = 78 \text{ } ^\circ\text{C}$$

$$\text{Min Metal Temp. w/o impact per UCS-66, Curve D} \quad -32 \text{ } ^\circ\text{C}$$

$$\text{Min Metal Temp. at Required thickness (UCS 66.1)} \quad -104 \text{ } ^\circ\text{C}$$

$$\text{Gov. MDMT of the nozzle to shell joint welded assembly} : -104 \text{ } ^\circ\text{C}$$

Flange MDMT including Temperature reduction per UCS-66.1:

$$\text{Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)} \quad -18 \text{ } ^\circ\text{C}$$

$$\text{Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)} \quad -104 \text{ } ^\circ\text{C}$$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 524.29/1960.01 = 0.267$$

Weld Size Calculations, Description: E1-36"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	12.7260 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (81.4159 - 109.342 + 2 \cdot 22.0 \cdot 1.0 \cdot (1.0 \cdot 30.0 - 9.3538)) \cdot 1407)$$

$$= \max(0, -26510.40) \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (22.2919 + 0.0 + 3.24 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 35923.29 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (22.2919 + 0.0 + 3.24 + (13.2)) \cdot 1407$$

$$= 54495.66 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (22.2919 + 0.0 + 3.24 + 0.0 + (13.2)) \cdot 1407$$

$$= 54495.66 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 18.0 \cdot 0.49 \cdot 1407$$

$$= 178245. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 446.2) \cdot (28.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 303734. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 1407$$

$$= 239277. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (178245 + 303734) = 481980 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (178245 + 0 + 239277 + 0) = 417523 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (178245 + 239277 + 0) = 417523 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 481979 kgf, must exceed W = 0 kgf	or W1 = 35923 kgf
Path 2-2 = 417522 kgf, must exceed W = 0 kgf	or W2 = 54495 kgf
Path 3-3 = 417522 kgf, must exceed W = 0 kgf	or W3 = 54495 kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 43.0375 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 579.0374 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.159 %

This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 247.03 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 700.00 mm

Input Echo, WRC297 Item 1, Description: E1-36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4900.000	mm
Shell Thickness	36.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.00	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	28.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	524.29	kPa
Radial Load	P	6880.76	kgf
Circumferential Shear	Vc	4587.18	kgf
Longitudinal Shear	Vl	4587.18	kgf
Circumferential Moment	Mc	11560.09	kgf-m.
Longitudinal Moment	Ml	11560.09	kgf-m.
Torsional Moment	Mt	16514.42	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 914.4 + 2 \cdot 1.65 \cdot \sqrt{2471.0(36.0 - 6.0)} \\
 &= 1812.885 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 4942.000 mm
 Nozzle Outside Diameter (d) = 914.400 mm
 Vessel Thickness used (T) = 30.000 mm
 Nozzle Thickness used (t) = 22.000 mm
 T / t = 1.364
 d / t = 41.564

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.375$$

Nr/P = 0.048
 Mr/P = 0.103
 M0/P = 0.040
 N0/P = 0.132
 MrD/Mc = 0.221
 NrDL/Mc = 0.081
 M0d/Mc = 0.097
 N0DL/Mc = 0.158
 MrD/M1 = 0.102
 NrDL/M1 = 0.046
 M0D/M1 = 0.040
 N0DL/M1 = 0.216

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-101	-101	-101	-101
Outplane Bending (P)	-184	184	-184	184
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-302	-302	302	302
Outplane Bending (ML)	-337	337	337	-337
Normal Pressure Stress	434	440	434	440
Outplane Stress Summary	-491	558	789	489

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-36	-36	-36	-36
Inplane Bending (P)	-472	472	-472	472
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-64	-64	64	64
Inplane Bending (ML)	-856	856	856	-856
Inplane Pressure Stress	217	217	217	217
Inplane Stress Summary	-1213	1445	629	-139

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	41	41	41	41
Shear Stress Summary	52	52	31	31

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1217	1448	795	632

**Vessel Stresses (kgf/cm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-101	-101	-101	-101
Outplane Bending (P)	-184	184	-184	184
Outplane Membrane (Mc)	-222	-222	222	222
Outplane Bending (Mc)	-819	819	819	-819
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	217	217	217	217
Outplane Stress Summary	-1110	898	973	-296

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-36	-36	-36	-36
Inplane Bending (P)	-472	472	-472	472
Inplane Membrane (Mc)	-113	-113	113	113
Inplane Bending (Mc)	-1864	1864	1864	-1864
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	434	440	434	440
Inplane Stress Summary	-2052	2627	1903	-875

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	41	41	41	41
Shear Stress Summary	31	31	52	52

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2053	2627	1906	880

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-101	-101	-101	-101
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-302	-302	302	302
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	123	130	123	130
Hoop Stress Summary	-280	-273	325	331

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-672	672	-672	672
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-85	-85	85	85
Axial Bending (ML)	-1232	1232	1232	-1232
Axial Pressure Stress	61	61	61	61

Axial Stress Summary	-1940	1869	697	-424
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	14	14	-14	-14
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	57	57	57	57

Shear Stress Summary	71	71	42	42
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1943	2148	702	760

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-101	-101	-101	-101
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-222	-222	222	222
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	123	130	123	130

Hoop Stress Summary	-199	-193	244	251
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-672	672	-672	672
Axial Membrane (Mc)	-85	-85	85	85
Axial Bending (Mc)	-2835	2835	2835	-2835
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	61	61	61	61

Axial Stress Summary	-3542	3472	2300	-2026
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-14	-14	14	14
Shear due to Torsion	57	57	57	57

Shear Stress Summary	42	42	71	71
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



Two * Max Shear Stress 3543 3666 2302 2282

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	434	440	434	440	434	440	434	440
Circ. Pl (SUS)	-403	-403	201	201	-149	-149	77	77
Circ. Q (SUS)	-521	521	153	-153	-2336	2336	1392	-1392
Long. Pm (SUS)	217	217	217	217	217	217	217	217
Long. Pl (SUS)	-100	-100	28	28	-323	-323	121	121
Long. Q (SUS)	-1328	1328	384	-384	-1003	1003	635	-635
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	41	41	41	41	41	41	41	41
Pm (SUS)	434.0	440.0	434.0	440.0	434.0	440.0	434.0	440.0
Pm+Pl (SUS)	118.1	118.2	635.3	641.3	391.5	397.5	511.6	517.6
Pm+Pl+Q (Total)	1214	1447	793	630	2052	2627	1905	879

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	440.00	1407.00	Passed
Pm+Pl (SUS)	641.25	2110.50	Passed
Pm+Pl+Q (TOTAL)	2627.56	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	130	123	130	123	130	123	130
Circ. Pl (SUS)	-403	-403	201	201	-323	-323	121	121
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-96	-96	74	74	-96	-96	74	74
Long. Q (SUS)	-1904	1904	560	-560	-3507	3507	2163	-2163
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	14	14	-14	-14	-14	-14	14	14
Shear Q (SUS)	57	57	57	57	57	57	57	57
Pm (SUS)	123.0	130.0	123.0	130.0	123.0	130.0	123.0	130.0
Pm+Pl (SUS)	280.8	273.8	325.0	332.0	201.2	194.2	245.8	252.7
Pm+Pl+Q (Total)	1942	2146	699	760	3542	3666	2300	2283

Nozzle Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	1407.00	Passed
Pm+Pl (SUS)	331.99	2110.50	Passed
Pm+Pl+Q (TOTAL)	3666.01	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: E2-36"****From: 140**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	36.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1400.0000	mm
Distance from Bottom/Left Tangent		7400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		225.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	18.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E2-36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	1.102	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$$

$$= 9.3538 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 524.29 \cdot 457.2 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$$

$$= 1.7347 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.7420 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1400.0000 mm

Parallel to Vessel Wall, opening length d 700.0000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 55.0000 mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	81.416	51.373	NA
Area in Shell	A1	109.342	96.363	NA
Area in Nozzle Wall	A2	22.292	22.284	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.240	3.240	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	134.874	121.887	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (870.4001 \cdot 9.3538 \cdot 1.0 + 2 \cdot 22.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 81.416 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 529.6(1.0 \cdot 30.0 - 1.0 \cdot 9.354) - 2 \cdot 22.0$$

$$(1.0 \cdot 30.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0)$$

$$= 109.342 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 55.0)(22.0 - 1.73)1.0$$



$$= 22.292 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 18.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 3.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.7420 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[14.331, \max(15.3538, 7.5)]$$

$$= 14.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.742, 14.3312)$$

$$= 14.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 184.4,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3333.1 kgf/cm ²	Passed
Occasional	: 51.6,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 84.6,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 28.0, tr = 1.735, c = 6.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.079, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 28.0, tr = 1.735, c = 6.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.079, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 524.29/1960.01 = 0.267

Weld Size Calculations, Description: E2-36"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	12.7260 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (81.4159 - 109.342 + 2 \cdot 22.0 \cdot 1.0 \cdot (1.0 \cdot 30.0 - 9.3538)) \cdot 1407)$$

$$= \max(0, -26510.40) \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (22.2919 + 0.0 + 3.24 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 35923.29 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (22.2919 + 0.0 + 3.24 + (13.2)) \cdot 1407$$

$$= 54495.66 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (22.2919 + 0.0 + 3.24 + 0.0 + (13.2)) \cdot 1407$$

$$= 54495.66 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot 18.0 \cdot 0.49 \cdot 1407$$

$$= 178245. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 446.2) \cdot (28.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 303734. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 914.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 1407$$

$$= 239277. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (178245 + 303734) = 481980 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (178245 + 0 + 239277 + 0) = 417523 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (178245 + 239277 + 0) = 417523 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 481979 kgf, must exceed W = 0 kgf	or W1 = 35923 kgf
Path 2-2 = 417522 kgf, must exceed W = 0 kgf	or W2 = 54495 kgf
Path 3-3 = 417522 kgf, must exceed W = 0 kgf	or W3 = 54495 kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 43.0375 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 579.0374 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.159 %

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 324.11 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 700.00 mm

Input Echo, WRC297 Item 1, Description: E2-36" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	6.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	4900.000	mm	
Shell Thickness	36.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1407.00	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	6.0000	mm	
Nozzle Diameter	914.400	mm	
Nozzle Thickness	28.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	524.29	kPa
Radial Load	P	6880.76	kgf
Circumferential Shear	Vc	4587.18	kgf
Longitudinal Shear	Vl	4587.18	kgf
Circumferential Moment	Mc	11560.09	kgf-m.
Longitudinal Moment	Ml	11560.09	kgf-m.
Torsional Moment	Mt	16514.42	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 914.4 + 2 \cdot 1.65 \cdot \sqrt{2471.0(36.0 - 6.0)} \\
 &= 1812.885 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 4942.000 mm
 Nozzle Outside Diameter (d) = 914.400 mm
 Vessel Thickness used (T) = 30.000 mm
 Nozzle Thickness used (t) = 22.000 mm
 T / t = 1.364
 d / t = 41.564

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.375$$

Nr/P = 0.048
 Mr/P = 0.103
 M0/P = 0.040
 N0/P = 0.132
 MrD/Mc = 0.221
 NrDL/Mc = 0.081
 M0d/Mc = 0.097
 N0DL/Mc = 0.158
 MrD/M1 = 0.102
 NrDL/M1 = 0.046
 M0D/M1 = 0.040
 N0DL/M1 = 0.216

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-101	-101	-101	-101
Outplane Bending (P)	-184	184	-184	184
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-302	-302	302	302
Outplane Bending (ML)	-337	337	337	-337
Normal Pressure Stress	434	440	434	440
Outplane Stress Summary	-491	558	789	489

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-36	-36	-36	-36
Inplane Bending (P)	-472	472	-472	472
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-64	-64	64	64
Inplane Bending (ML)	-856	856	856	-856
Inplane Pressure Stress	217	217	217	217
Inplane Stress Summary	-1213	1445	629	-139

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	41	41	41	41
Shear Stress Summary	52	52	31	31

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1217	1448	795	632

**Vessel Stresses (kgf/cm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-101	-101	-101	-101
Outplane Bending (P)	-184	184	-184	184
Outplane Membrane (Mc)	-222	-222	222	222
Outplane Bending (Mc)	-819	819	819	-819
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	217	217	217	217
Outplane Stress Summary	-1110	898	973	-296

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-36	-36	-36	-36
Inplane Bending (P)	-472	472	-472	472
Inplane Membrane (Mc)	-113	-113	113	113
Inplane Bending (Mc)	-1864	1864	1864	-1864
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	434	440	434	440
Inplane Stress Summary	-2052	2627	1903	-875

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	41	41	41	41
Shear Stress Summary	31	31	52	52

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2053	2627	1906	880

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-101	-101	-101	-101
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-302	-302	302	302
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	123	130	123	130
Hoop Stress Summary	-280	-273	325	331

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-672	672	-672	672
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-85	-85	85	85
Axial Bending (ML)	-1232	1232	1232	-1232
Axial Pressure Stress	61	61	61	61

Axial Stress Summary	-1940	1869	697	-424
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	14	14	-14	-14
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	57	57	57	57

Shear Stress Summary	71	71	42	42
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1943	2148	702	760

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-101	-101	-101	-101
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-222	-222	222	222
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	123	130	123	130

Hoop Stress Summary	-199	-193	244	251
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-672	672	-672	672
Axial Membrane (Mc)	-85	-85	85	85
Axial Bending (Mc)	-2835	2835	2835	-2835
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	61	61	61	61

Axial Stress Summary	-3542	3472	2300	-2026
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-14	-14	14	14
Shear due to Torsion	57	57	57	57

Shear Stress Summary	42	42	71	71
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



Two * Max Shear Stress 3543 3666 2302 2282

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	434	440	434	440	434	440	434	440
Circ. Pl (SUS)	-403	-403	201	201	-149	-149	77	77
Circ. Q (SUS)	-521	521	153	-153	-2336	2336	1392	-1392
Long. Pm (SUS)	217	217	217	217	217	217	217	217
Long. Pl (SUS)	-100	-100	28	28	-323	-323	121	121
Long. Q (SUS)	-1328	1328	384	-384	-1003	1003	635	-635
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	41	41	41	41	41	41	41	41
Pm (SUS)	434.0	440.0	434.0	440.0	434.0	440.0	434.0	440.0
Pm+Pl (SUS)	118.1	118.2	635.3	641.3	391.5	397.5	511.6	517.6
Pm+Pl+Q (Total)	1214	1447	793	630	2052	2627	1905	879

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	440.00	1407.00	Passed
Pm+Pl (SUS)	641.25	2110.50	Passed
Pm+Pl+Q (TOTAL)	2627.56	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	130	123	130	123	130	123	130
Circ. Pl (SUS)	-403	-403	201	201	-323	-323	121	121
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-96	-96	74	74	-96	-96	74	74
Long. Q (SUS)	-1904	1904	560	-560	-3507	3507	2163	-2163
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	14	14	-14	-14	-14	-14	14	14
Shear Q (SUS)	57	57	57	57	57	57	57	57
Pm (SUS)	123.0	130.0	123.0	130.0	123.0	130.0	123.0	130.0
Pm+Pl (SUS)	280.8	273.8	325.0	332.0	201.2	194.2	245.8	252.7
Pm+Pl+Q (Total)	1942	2146	699	760	3542	3666	2300	2283

Nozzle Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	1407.00	Passed
Pm+Pl (SUS)	331.99	2110.50	Passed
Pm+Pl+Q (TOTAL)	3666.01	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: D-30"

From: 150

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1200.0000	mm
Distance from Bottom/Left Tangent		9600.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	18.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D-30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000	in.
Actual Thickness Used in Calculation	1.024	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$$

$$= 9.3538 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 524.29 \cdot 381.0 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$$

$$= 1.4456 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.5655 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1200.0000 mm

Parallel to Vessel Wall, opening length d 600.0000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 50.0000 mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 1407.0 / 1407.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	67.535	42.614	NA
Area in Shell	A1	79.569	67.855	NA
Area in Nozzle Wall	A2	18.554	18.435	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.240	3.240	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	101.363	89.529	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (722.0 \cdot 9.3538 \cdot 1.0 + 2 \cdot 20.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 67.535 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 478.0(1.0 \cdot 26.0 - 1.0 \cdot 9.354) - 2 \cdot 20.0$$

$$(1.0 \cdot 26.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0)$$

$$= 79.569 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 50.0)(20.0 - 1.45)1.0$$



$$= 18.554 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 18.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 3.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 7.5655 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 7.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 15.3538 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 15.3538 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 6.9206 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 7.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 14.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[14.331, \max(15.3538, 7.5)]$$

$$= 14.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.5655, 14.3312)$$

$$= 14.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 26.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	194.0,	Allowable	:	1407.0 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	3323.5 kgf/cm ²	Passed
Occasional	:	47.0,	Allowable	:	1871.3 kgf/cm ²	Passed
Shear	:	93.9,	Allowable	:	984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $tg = 26.0$, $tr = 1.446$, $c = 6.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr * E*/(tg - c) = 0.072$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 20 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $tg = 26.0$, $tr = 1.446$, $c = 6.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr * E*/(tg - c) = 0.072$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 524.29/1960.01 = 0.267$$

Weld Size Calculations, Description: D-30"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	12.7260 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$$

$$= \max(0, (67.5348 - 79.5686 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 26.0 - 9.3538)) \cdot 1407)$$

$$= \max(0, -7563.12) \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (18.5544 + 0.0 + 3.24 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 30664.73 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (18.5544 + 0.0 + 3.24 + (10.4)) \cdot 1407$$

$$= 45297.51 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (18.5544 + 0.0 + 3.24 + 0.0 + (10.4)) \cdot 1407$$

$$= 45297.51 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot 18.0 \cdot 0.49 \cdot 1407$$

$$= 148538. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 371.0) \cdot (26.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 229586. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 1407$$

$$= 199398. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (148538 + 229586) = 378124 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (148538 + 0 + 199398 + 0) = 347935 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (148538 + 199398 + 0) = 347935 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 378123 kgf, must exceed W = 0 kgf	or W1 = 30664 kgf
Path 2-2 = 347935 kgf, must exceed W = 0 kgf	or W2 = 45297 kgf
Path 3-3 = 347935 kgf, must exceed W = 0 kgf	or W3 = 45297 kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 29.8059 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 561.8059 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.533 %

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 478.28 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 600.00 mm

Input Echo, WRC297 Item 1, Description: D-30" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4900.000	mm
Shell Thickness	32.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.00	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	6.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	26.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	524.29	kPa
Radial Load	P	5733.99	kgf
Circumferential Shear	Vc	3822.66	kgf
Longitudinal Shear	Vl	3822.66	kgf
Circumferential Moment	Mc	8027.83	kgf-m.
Longitudinal Moment	Ml	8027.83	kgf-m.
Torsional Moment	Mt	11468.29	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 762.0 + 2 \cdot 1.65 \cdot \sqrt{2469.0(32.0 - 6.0)} \\
 &= 1598.106 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 4938.000 mm
 Nozzle Outside Diameter (d) = 762.000 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 20.000 mm
 T / t = 1.300
 d / t = 38.100

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.127$$

Nr/P = 0.053
 Mr/P = 0.113
 M0/P = 0.043
 N0/P = 0.131
 MrD/Mc = 0.231
 NrDL/Mc = 0.083
 M0d/Mc = 0.099
 N0DL/Mc = 0.149
 MrD/Ml = 0.112
 NrDL/Ml = 0.051
 M0D/Ml = 0.043
 N0DL/Ml = 0.214

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-110	-110	-110	-110
Outplane Bending (P)	-219	219	-219	219
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-334	-334	334	334
Outplane Bending (ML)	-403	403	403	-403
Normal Pressure Stress	502	507	502	507
Outplane Stress Summary	-565	686	910	546

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-44	-44	-44	-44
Inplane Bending (P)	-576	576	-576	576
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-79	-79	79	79
Inplane Bending (ML)	-1049	1049	1049	-1049
Inplane Pressure Stress	251	251	251	251
Inplane Stress Summary	-1498	1753	759	-188

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	48	48	48	48
Shear Stress Summary	60	60	36	36

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1502	1756	918	738

**Vessel Stresses (kgf/cm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-110	-110	-110	-110
Outplane Bending (P)	-219	219	-219	219
Outplane Membrane (Mc)	-232	-232	232	232
Outplane Bending (Mc)	-923	923	923	-923
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	251	251	251	251
Outplane Stress Summary	-1235	1050	1076	-331

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-44	-44	-44	-44
Inplane Bending (P)	-576	576	-576	576
Inplane Membrane (Mc)	-130	-130	130	130
Inplane Bending (Mc)	-2158	2158	2158	-2158
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	502	507	502	507
Inplane Stress Summary	-2406	3067	2170	-989

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	48	48	48	48
Shear Stress Summary	36	36	60	60

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2407	3068	2173	994

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-110	-110	-110	-110
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-334	-334	334	334
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	112	119	112	119
Hoop Stress Summary	-332	-325	335	342

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	-12	-12	-12	-12
Axial Bending (P)	-747	747	-747	747
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-95	-95	95	95
Axial Bending (ML)	-1373	1373	1373	-1373
Axial Pressure Stress	56	56	56	56

Axial Stress Summary	-2171	2069	765	-486
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	15	15	-15	-15
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	62	62	62	62

Shear Stress Summary	78	78	46	46
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	2175	2400	770	834

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-110	-110	-110	-110
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-232	-232	232	232
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	112	119	112	119

Hoop Stress Summary	-230	-224	234	240
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-12	-12	-12	-12
Axial Bending (P)	-747	747	-747	747
Axial Membrane (Mc)	-95	-95	95	95
Axial Bending (Mc)	-2987	2987	2987	-2987
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	56	56	56	56

Axial Stress Summary	-3786	3684	2379	-2101
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-15	-15	15	15
Shear due to Torsion	62	62	62	62

Shear Stress Summary	46	46	78	78
----------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



Two * Max Shear Stress 3787 3909 2382 2347

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	502	507	502	507	502	507	502	507
Circ. Pl (SUS)	-444	-444	224	224	-174	-174	86	86
Circ. Q (SUS)	-622	622	184	-184	-2734	2734	1582	-1582
Long. Pm (SUS)	251	251	251	251	251	251	251	251
Long. Pl (SUS)	-123	-123	35	35	-342	-342	122	122
Long. Q (SUS)	-1625	1625	473	-473	-1142	1142	704	-704
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	48	48	48	48	48	48	48	48
Pm (SUS)	502.0	507.0	502.0	507.0	502.0	507.0	502.0	507.0
Pm+Pl (SUS)	130.0	130.1	726.3	731.3	419.7	424.7	588.7	593.7
Pm+Pl+Q (Total)	1500	1756	918	737	2407	3067	2173	994

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	507.00	1407.00	Passed
Pm+Pl (SUS)	731.32	2110.50	Passed
Pm+Pl+Q (TOTAL)	3067.64	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	119	112	119	112	119	112	119
Circ. Pl (SUS)	-444	-444	224	224	-342	-342	122	122
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	-107	-107	83	83	-107	-107	83	83
Long. Q (SUS)	-2120	2120	626	-626	-3734	3734	2240	-2240
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	62	62	62	62	62	62	62	62
Pm (SUS)	112.0	119.0	112.0	119.0	112.0	119.0	112.0	119.0
Pm+Pl (SUS)	332.8	325.8	337.1	344.1	231.2	224.3	236.3	243.2
Pm+Pl+Q (Total)	2174	2398	770	835	3785	3907	2381	2347

Nozzle Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	119.00	1407.00	Passed
Pm+Pl (SUS)	344.10	2110.50	Passed
Pm+Pl+Q (TOTAL)	3907.13	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH-3-24"

From: 160

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	3699.9985	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-3-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3538 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1171.2000 mm
Parallel to Vessel Wall, opening length d 585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 30.0000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.776	31.293	NA
Area in Shell	A1	15.496	7.686	NA
Area in Nozzle Wall	A2	6.506	6.378	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	96.962	89.025	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 775.2341 18.0000 mm
Based on given Pad Diameter: 1009.6000 7.4535 mm
Based on Shell or Nozzle Thickness: 775.2341 18.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (585.6 \cdot 9.3538 \cdot 1.0 + 2 \cdot 12.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 54.776 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 585.6(1.0 \cdot 12.0 - 1.0 \cdot 9.354) - 2 \cdot 12.0 \\
 &\quad (1.0 \cdot 12.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0) \\
 &= 15.496 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 30.0) \cdot (12.0 - 1.16) \cdot 1.0 \\
 &= 6.506 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3696 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[14.331, \max(15.3538, 7.5)] \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.3696, 14.3312) \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0



Thickness Ratio = $tr * E * / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E * = 1.0$
 Thickness Ratio = $tr * E * / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E * = 1.0$
 Thickness Ratio = $tr * E * / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29/1960.01 = 0.267$

Weld Size Calculations, Description: MH-3-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 12.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (54.7761 - 15.4959 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 12.0 - 9.3538)) \cdot 1407)$
 $= 56160.80 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (6.5061 + 72.0 + 2.96 - 0.0 \cdot 1.0) \cdot 1407$
 $= 114622.70 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (6.5061 + 0.0 + 1. + (2.88)) \cdot 1407$
 $= 14613.27 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (6.5061 + 0.0 + 2.96 + 72.0 + (2.88)) \cdot 1407$
 $= 118674.85 \text{ kgf}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 298.8) \cdot (18.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 110944. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 119639. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (153069 + 110944) = 264013 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66017 + 179458 + 119639 + 0) = 365113 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (153069 + 119639 + 0) = 272707 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 264012 kgf, must exceed W = 56160 kgf	or W1 = 114622 kgf
Path 2-2 = 365113 kgf, must exceed W = 56160 kgf	or W2 = 14613 kgf
Path 3-3 = 272707 kgf, must exceed W = 56160 kgf	or W3 = 118674 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0337 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.0338 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 3.043 \%$$

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**Input, Nozzle Desc: UC-2-24"****From: 160**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	3699.9985	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12800.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC-2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3538 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1171.2000 mm
Parallel to Vessel Wall, opening length d 585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 30.0000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.776	31.293	NA
Area in Shell	A1	15.496	7.686	NA
Area in Nozzle Wall	A2	6.506	6.378	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	96.962	89.025	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 775.2341 18.0000 mm
Based on given Pad Diameter: 1009.6000 7.4535 mm
Based on Shell or Nozzle Thickness: 775.2341 18.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (585.6 \cdot 9.3538 \cdot 1.0 + 2 \cdot 12.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 54.776 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 585.6(1.0 \cdot 12.0 - 1.0 \cdot 9.354) - 2 \cdot 12.0 \\
 &\quad (1.0 \cdot 12.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0) \\
 &= 15.496 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 30.0) \cdot (12.0 - 1.16) \cdot 1.0 \\
 &= 6.506 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3696 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[14.331, \max(15.3538, 7.5)] \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.3696, 14.3312) \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

$$\begin{aligned}
 &\text{Govrn. thk, } tg = 18.0, tr = 1.156, c = 6.0 \text{ mm, } E* = 1.0 \\
 &\text{Thickness Ratio} = tr \cdot E* / (tg - c) = 0.096, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}
 \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

$$\begin{aligned}
 &\text{Govrn. thk, } tg = 18.0, tr = 1.156, c = 6.0 \text{ mm, } E* = 1.0 \\
 &\text{Thickness Ratio} = tr \cdot E* / (tg - c) = 0.096, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}
 \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } tg = 18.0, tr = 1.156, c = 6.0 \text{ mm, } E* = 1.0$$



Thickness Ratio = $tr * E * / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E * = 1.0$
 Thickness Ratio = $tr * E * / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E * = 1.0$
 Thickness Ratio = $tr * E * / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29/1960.01 = 0.267$

Weld Size Calculations, Description: UC-2-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 12.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * t_n * f_{r1} * (E1 * t - tr)) * S_v)$
 $= \max(0, (54.7761 - 15.4959 + 2 * 12.0 * 1.0 * (1.0 * 12.0 - 9.3538)) * 1407)$
 $= 56160.80 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 * f_{r2}) * S_v$
 $= (6.5061 + 72.0 + 2.96 - 0.0 * 1.0) * 1407$
 $= 114622.70 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 * t_n * t * f_{r1})) * S_v$
 $= (6.5061 + 0.0 + 1. + (2.88)) * 1407$
 $= 14613.27 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 * t_n * t * f_{r1})) * S$
 $= (6.5061 + 0.0 + 2.96 + 72.0 + (2.88)) * 1407$
 $= 118674.85 \text{ kgf}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 298.8) \cdot (18.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 110944. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 119639. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (153069 + 110944) = 264013 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66017 + 179458 + 119639 + 0) = 365113 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (153069 + 119639 + 0) = 272707 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 264012 kgf, must exceed W = 56160 kgf	or W1 = 114622 kgf
Path 2-2 = 365113 kgf, must exceed W = 56160 kgf	or W2 = 14613 kgf
Path 3-3 = 272707 kgf, must exceed W = 56160 kgf	or W3 = 118674 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0337 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.0338 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 3.043 \%$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 585.60 mm

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Input, Nozzle Desc: MH-2-24"

From: 180

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4199.9985	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		18550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3538 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1171.2000 mm
Parallel to Vessel Wall, opening length d 585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 30.0000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.776	32.965	NA
Area in Shell	A1	15.496	4.342	NA
Area in Nozzle Wall	A2	6.506	6.378	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	96.962	85.680	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 775.2341 18.0000 mm
Based on given Pad Diameter: 1009.6000 7.4535 mm
Based on Shell or Nozzle Thickness: 775.2341 18.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (585.6 \cdot 9.3538 \cdot 1.0 + 2 \cdot 12.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 54.776 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 585.6(1.0 \cdot 12.0 - 1.0 \cdot 9.354) - 2 \cdot 12.0 \\
 &\quad (1.0 \cdot 12.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0) \\
 &= 15.496 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 30.0) \cdot (12.0 - 1.16) \cdot 1.0 \\
 &= 6.506 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3696 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[14.331, \max(15.3538, 7.5)] \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.3696, 14.3312) \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29 / 1960.01 = 0.267$

Weld Size Calculations, Description: MH-2-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 12.0000$ mm
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 12.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (54.7761 - 15.4959 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 12.0 - 9.3538)) \cdot 1407)$
 $= 56160.80 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (6.5061 + 72.0 + 2.96 - 0.0 \cdot 1.0) \cdot 1407$
 $= 114622.70 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (6.5061 + 0.0 + 1. + (2.88)) \cdot 1407$
 $= 14613.27 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (6.5061 + 0.0 + 2.96 + 72.0 + (2.88)) \cdot 1407$
 $= 118674.85 \text{ kgf}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 298.8) \cdot (18.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 110944. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 119639. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (153069 + 110944) = 264013 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66017 + 179458 + 119639 + 0) = 365113 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (153069 + 119639 + 0) = 272707 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 264012 kgf, must exceed W = 56160 kgf	or W1 = 114622 kgf
Path 2-2 = 365113 kgf, must exceed W = 56160 kgf	or W2 = 14613 kgf
Path 3-3 = 272707 kgf, must exceed W = 56160 kgf	or W3 = 118674 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0337 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.0338 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 3.043 \%$$

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**Input, Nozzle Desc: UC-1-24"****From: 180**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4900.00	mm
Design Length of Section	L	4199.9985	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		19600.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	6.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: UC-1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2456.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3538 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1171.2000 mm
Parallel to Vessel Wall, opening length d 585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 30.0000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.776	32.965	NA
Area in Shell	A1	15.496	4.342	NA
Area in Nozzle Wall	A2	6.506	6.378	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	96.962	85.680	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 775.2341 18.0000 mm
Based on given Pad Diameter: 1009.6000 7.4535 mm
Based on Shell or Nozzle Thickness: 775.2341 18.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (585.6 \cdot 9.3538 \cdot 1.0 + 2 \cdot 12.0 \cdot 9.3538 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 54.776 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 585.6 (1.0 \cdot 12.0 - 1.0 \cdot 9.354) - 2 \cdot 12.0 \\
 &\quad (1.0 \cdot 12.0 - 1.0 \cdot 9.3538) \cdot (1 - 1.0) \\
 &= 15.496 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 30.0) \cdot (12.0 - 1.16) \cdot 1.0 \\
 &= 6.506 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.3696 mm
Wall Thickness per UG16(b),	tr16b = 7.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.3538 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.3538 mm
Wall Thickness, shell/head, external pressure	trb2 = 6.9206 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 7.5000 mm
Wall Thickness per table UG-45	tb3 = 14.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[14.331, \max(15.3538, 7.5)] \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(7.3696, 14.3312) \\
 &= 14.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.156, c = 6.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.096$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $tg = 18.0$, $tr = 9.354$, $c = 6.0$ mm, $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.779$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -43 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29/1960.01 = 0.267$

Weld Size Calculations, Description: UC-1-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 12.0000$ mm
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 12.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (54.7761 - 15.4959 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 12.0 - 9.3538)) \cdot 1407)$
 $= 56160.80 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (6.5061 + 72.0 + 2.96 - 0.0 \cdot 1.0) \cdot 1407$
 $= 114622.70 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (6.5061 + 0.0 + 1. + (2.88)) \cdot 1407$
 $= 14613.27 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (6.5061 + 0.0 + 2.96 + 72.0 + (2.88)) \cdot 1407$
 $= 118674.85 \text{ kgf}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 298.8) \cdot (18.0 - 6.0) \cdot 0.7 \cdot 1407$$

$$= 110944. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 6.0) \cdot 0.74 \cdot 1407$$

$$= 119639. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (153069 + 110944) = 264013 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66017 + 179458 + 119639 + 0) = 365113 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (153069 + 119639 + 0) = 272707 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 264012 kgf, must exceed W = 56160 kgf	or W1 = 114622 kgf
Path 2-2 = 365113 kgf, must exceed W = 56160 kgf	or W2 = 14613 kgf
Path 3-3 = 272707 kgf, must exceed W = 56160 kgf	or W3 = 118674 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 671 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0337 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.0338 mm

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 3.043 \%$$

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**Input, Nozzle Desc: MH-1-24"****From: 200**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4906.00	mm
Design Length of Section	L	4200.0049	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		25300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgmv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $524.29 \cdot 2454.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
= 9.3462 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 304.8 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.1564 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3696 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1151.2001 mm
Parallel to Vessel Wall, opening length d 575.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.5000 mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.0 / 1407.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	53.797	32.395	NA
Area in Shell	A1	21.031	10.038	NA
Area in Nozzle Wall	A2	10.298	10.160	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	72.000	72.000	NA
TOTAL AREA AVAILABLE	Atot	106.289	95.158	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 717.9749 18.0000 mm
Based on given Pad Diameter: 1009.6000 4.8769 mm
Based on Shell or Nozzle Thickness: 748.9391 14.0000 mm

Area Required [A]:



$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (575.6 \cdot 9.3462 \cdot 1.0 + 2 \cdot 17.0 \cdot 9.3462 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 53.797 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 575.6(1.0 \cdot 13.0 - 1.0 \cdot 9.346) - 2 \cdot 17.0 \\
 &\quad (1.0 \cdot 13.0 - 1.0 \cdot 9.3462) \cdot (1 - 1.0) \\
 &= 21.031 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 32.5) \cdot (17.0 - 1.16) \cdot 1.0 \\
 &= 10.298 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0 \\
 &= 2.960 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (1009.6 - 609.6) \cdot 18.0 \cdot 1.0 \\
 &= 72.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.3696 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.3462 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.3462 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9199 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[9.331, \max(10.3462, 2.5)] \\
 &= 9.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(2.3696, 9.3312) \\
 &= 9.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 18.0, tr = 1.156, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.068, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 18.0, tr = 1.156, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.068, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.156, c = 1.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.068$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $tg = 14.0$, $tr = 9.346$, $c = 1.0$ mm, $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $tg = 14.0$, $tr = 9.346$, $c = 1.0$ mm, $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29 / 1960.01 = 0.267$

Weld Size Calculations, Description: MH-1-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 17.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.5000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (53.7969 - 21.0311 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.3462)) \cdot 1407)$
 $= 47849.30 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (10.2983 + 72.0 + 2.96 - 0.0 \cdot 1.0) \cdot 1407$
 $= 119958.27 \text{ kgf}$

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (10.2983 + 0.0 + 1.0 + (4.42)) \cdot 1407$
 $= 22115.63 \text{ kgf}$

Weld Load [W3]:

$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (10.2983 + 0.0 + 2.96 + 72.0 + (4.42)) \cdot 1407$
 $= 126177.22 \text{ kgf}$



Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1009.6 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 153069. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 296.3) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407$$

$$= 155855. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 18.0 \cdot 0.74 \cdot 1407$$

$$= 179458. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 1407$$

$$= 129608. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (153069 + 155855) = 308924 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66017 + 179458 + 129608 + 0) = 375083 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (153069 + 129608 + 0) = 282677 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 308924 kgf, must exceed W = 47849 kgf or W1 = 119958 kgf

Path 2-2 = 375083 kgf, must exceed W = 47849 kgf or W2 = 22115 kgf

Path 3-3 = 282677 kgf, must exceed W = 47849 kgf or W3 = 126177 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 618 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 19.0103 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 533.0102 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.043 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 575.60 mm

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**Input, Nozzle Desc: A1-20"****From: 210**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4906.00	mm
Design Length of Section	L	3708.8286	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	2100.0000	mm
Distance from Bottom/Left Tangent		28599.99	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		148.20	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1208.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		350.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

**Reinforcement CALCULATION, Description: A1-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 20.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 524.29 \cdot 2454.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
 $= 9.3462 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 524.29 \cdot 254.0 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
 $= 0.9637 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6402 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1940.6304 mm
 Parallel to Vessel Wall, opening length d 970.3152 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	90.688	51.889	NA
Area in Shell	A1	35.453	22.363	NA
Area in Nozzle Wall	A2	15.901	15.230	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		3.400	3.400	NA
Area in Element	A5	112.000	112.000	NA
TOTAL AREA AVAILABLE	Atot	166.754	152.993	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 29.24 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 732.5863 16.0000 mm



Based on given Pad Diameter: 1207.9999 5.1334 mm
Based on Shell or Nozzle Thickness: 1296.5858 14.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (970.3152 \cdot 9.3462 \cdot 1.0 + 2 \cdot 17.0 \cdot 9.3462 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 90.688 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 970.315 (1.0 \cdot 13.0 - 1.0 \cdot 9.346) - 2 \cdot 17.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.3462) \cdot (1 - 1.0)$$

$$= 35.453 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$

$$= (2 \cdot 32.5) \cdot (17.0 - 0.96) \cdot 1.0 / \sin(41.0)$$

$$= 15.901 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 12.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$$

$$= 3.400 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1739.9156 - 1039.9158) \cdot 16.0 \cdot 1.0$$

$$= 112.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 18.0, tr = 0.964, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.057, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -30 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 14.0, tr = 9.346, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -34 °C



Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 14.0, tr = 9.346, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D

-48 °C

Gov. MDMT of the Nozzle

: -48 °C

Gov. MDMT of the Reinforcement Pad

: -30 °C

Gov. MDMT of the nozzle to shell joint welded assembly :

-30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)

-18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)

-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $524.29/1960.01 = 0.267$ Weld Size Calculations, Description: A1-20"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = $0.7 \cdot W_o$ mm
Pad Weld	6.5000 = $0.5 \cdot T_{minPad}$	9.8980 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (90.6879 - 35.4531 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.3462)) \cdot 1407)$$

$$= 79463.15 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (15.901 + 112.0 + 3.4 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 184740.22 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (15.901 + 0.0 + 1.44 + (4.42)) \cdot 1407$$

$$= 30617.67 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (15.901 + 0.0 + 3.4 + 112.0 + (4.42)) \cdot 1407$$

$$= 190959.14 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1039.9158 \cdot 12.0 \cdot 0.49 \cdot 1407$$

$$= 135142. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1208.0 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 183149. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:



$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 502.5577) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407 \\
 &= 264348. \text{ kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 1039.9158 \cdot 16.0 \cdot 0.74 \cdot 1407 \\
 &= 272122. \text{ kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 1039.9158 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 1407 \\
 &= 221099. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (183149 + 264348) = 447497 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (135142 + 272122 + 221099 + 0) = 628362 \text{ kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (183149 + 221099 + 0) = 404248 \text{ kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 447497 kgf, must exceed W = 79463 kgf	or W1 = 184740 kgf
Path 2-2 = 628362 kgf, must exceed W = 79463 kgf	or W2 = 30617 kgf
Path 3-3 = 404247 kgf, must exceed W = 79463 kgf	or W3 = 190959 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 618 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

Note : Checking Nozzle in plane parallel to the vessel axis.**Reinforcement CALCULATION, Description: A1-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 524.29 \cdot 2454.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29) \\
 &= 9.3462 \text{ mm}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\
 &= 524.29 \cdot 254.0 / (1407 \cdot 1.0 + 0.4 \cdot 524.29) \\
 &= 0.9637 \text{ mm}
 \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.6402 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	948.0001 mm
Parallel to Vessel Wall, opening length	d	474.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm



The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	44.301	25.348	NA
Area in Shell	A1	17.319	10.924	NA
Area in Nozzle Wall	A2	10.424	9.984	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.440	1.440	NA
Area in Element	A5	70.400	70.400	NA
TOTAL AREA AVAILABLE	Atot	99.582	92.748	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 602.4917 16.0000 mm

Based on given Pad Diameter: 1208.0000 3.4361 mm

Based on Shell or Nozzle Thickness: 615.9905 14.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (474.0 \cdot 9.3462 \cdot 1.0 + 2 \cdot 17.0 \cdot 9.3462 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 44.301 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 474.0(1.0 \cdot 13.0 - 1.0 \cdot 9.346) - 2 \cdot 17.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.346) \cdot (1 - 1.0)$$

$$= 17.319 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 32.5) \cdot (17.0 - 0.96) \cdot 1.0$$

$$= 10.424 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 12.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$

$$= 1.440 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (948.0 - 508.0) \cdot 16.0 \cdot 1.0$$

$$= 70.400 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.6402 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 10.3462 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 10.3462 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 1.9199 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 9.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.331, \max(10.3462, 2.5)]$$

$$= 9.3312 \text{ mm}$$



Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.6402, 9.3312)$$

$$= 9.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 212.5,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3305.0 kgf/cm ²	Passed
Occasional	: 36.0,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 113.4,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: A1-20"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm
Pad Weld	6.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (44.3011 - 17.3189 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.3462)) \cdot 1407)$$

$$= 39711.88 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (10.4236 + 70.4 + 1.44 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 115744.73 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (10.4236 + 0.0 + 1.44 + (4.42)) \cdot 1407$$

$$= 22910.98 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (10.4236 + 0.0 + 1.44 + 70.4 + (4.42)) \cdot 1407$$

$$= 121963.66 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 12.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1208.0 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 183149. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 245.5) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407$$



= 129134. kgf

Tension, Pad Groove Weld [Tpgw]:

= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$

= $(3.1416/2) \cdot 508.0 \cdot 16.0 \cdot 0.74 \cdot 1407$

= 132932. kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$

= $(3.1416/2.0) \cdot 508.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 1407$

= 108007. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (183149 + 129134) = 312283 kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (66017 + 132932 + 108007 + 0) = 306956 kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (183149 + 108007 + 0) = 291156 kgf

Summary of Failure Path Calculations:

Path 1-1 = 312283 kgf, must exceed W = 39711 kgf or W1 = 115744 kgf

Path 2-2 = 306955 kgf, must exceed W = 39711 kgf or W2 = 22910 kgf

Path 3-3 = 291156 kgf, must exceed W = 39711 kgf or W3 = 121963 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 618 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 577.9073 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1604.7882 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 3.673 %

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: A1-20" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4906.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.00 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.00 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	18.0000	mm



Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1208.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	524.29	kPa
Radial Load	P	3822.66	kgf
Circumferential Shear	Vc	2548.44	kgf
Longitudinal Shear	Vl	2548.44	kgf
Circumferential Moment	Mc	3567.93	kgf-m.
Longitudinal Moment	Ml	3567.93	kgf-m.
Torsional Moment	Mt	5097.03	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(2468.5(14.0 - 1.0))}$$

$$= 1099.156 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4937.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	17.000	mm
T / t	=	1.706	
d / t	=	29.882	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.343$$

Nr/P	=	0.083
Mr/P	=	0.128
M0/P	=	0.064
N0/P	=	0.160
MrD/Mc	=	0.225
NrDL/Mc	=	0.112
M0d/Mc	=	0.128
N0DL/Mc	=	0.128
MrD/Ml	=	0.149
NrDL/Ml	=	0.085
M0D/Ml	=	0.077
N0DL/Ml	=	0.250

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-72	-72	-72	-72
Outplane Bending (P)	-175	175	-175	175
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-208	-208	208	208



Outplane Bending (ML)	-383	383	383	-383
Normal Pressure Stress	449	455	449	455

Outplane Stress Summary	-390	732	793	382
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	-37	-37	-37	-37
Inplane Bending (P)	-348	348	-348	348
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-70	-70	70	70
Inplane Bending (ML)	-744	744	744	-744
Inplane Pressure Stress	224	224	224	224

Inplane Stress Summary	-976	1209	654	-138
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	43	43	43	43

Shear Stress Summary	54	54	32	32
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	981	1215	800	524
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-72	-72	-72	-72
Outplane Bending (P)	-175	175	-175	175
Outplane Membrane (Mc)	-107	-107	107	107
Outplane Bending (Mc)	-641	641	641	-641
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	224	224	224	224

Outplane Stress Summary	-772	862	725	-207
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-37	-37	-37	-37
Inplane Bending (P)	-348	348	-348	348
Inplane Membrane (Mc)	-93	-93	93	93
Inplane Bending (Mc)	-1124	1124	1124	-1124
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	449	455	449	455

Inplane Stress Summary	-1154	1797	1282	-265
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Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	43	43	43	43

Shear Stress Summary	32	32	54	54
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1157	1798	1287	298
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-72	-72	-72	-72
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-208	-208	208	208
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	86	92	86	92

Hoop Stress Summary	-195	-188	222	228
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-685	685	-685	685
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-114	-114	114	114
Axial Bending (ML)	-1547	1547	1547	-1547
Axial Pressure Stress	43	43	43	43

Axial Stress Summary	-2319	2147	1005	-719
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	73	73	73	73

Shear Stress Summary	92	92	55	55
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	2323	2343	1009	953
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-72	-72	-72	-72
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Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-107	-107	107	107
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	86	92	86	92

Hoop Stress Summary	-93	-87	120	127
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-685	685	-685	685
Axial Membrane (Mc)	-114	-114	114	114
Axial Bending (Mc)	-2458	2458	2458	-2458
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	43	43	43	43

Axial Stress Summary	-3229	3058	1915	-1629
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	73	73	73	73

Shear Stress Summary	55	55	92	92
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	3230	3147	1920	1766

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{2460.5(14.0 - 1.0)}$$

$$= 1098.197 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4921.000 mm
Nozzle Outside Diameter	(d) =	1208.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	367.000 mm
T / t	=	0.035
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.776$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.048
Mr/P	=	0.072
M0/P	=	0.019
N0/P	=	0.020
MrD/Mc	=	0.241
NrDL/Mc	=	0.106
M0d/Mc	=	0.072
N0DL/Mc	=	0.043



MrD/M1 = 0.049
 NrDL/M1 = 0.055
 M0D/M1 = 0.015
 N0DL/M1 = 0.022

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-46	-46	-46	-46
Outplane Bending (P)	-261	261	-261	261
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-37	-37	37	37
Outplane Bending (ML)	-155	155	155	-155
Normal Pressure Stress	1006	1011	1006	1011

Outplane Stress Summary	506	1344	892	1109
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-108	-108	-108	-108
Inplane Bending (P)	-972	972	-972	972
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-95	-95	95	95
Inplane Bending (ML)	-509	509	509	-509
Inplane Pressure Stress	503	503	503	503

Inplane Stress Summary	-1183	1781	26	954
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17

Shear Stress Summary	27	27	6	6
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1690	1783	892	1110
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-46	-46	-46	-46
Outplane Bending (P)	-261	261	-261	261
Outplane Membrane (Mc)	-74	-74	74	74
Outplane Bending (Mc)	-753	753	753	-753
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	503	503	503	503

Outplane Stress Summary	-631	1397	1023	39
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Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-108	-108	-108	-108
Inplane Bending (P)	-972	972	-972	972
Inplane Membrane (Mc)	-185	-185	185	185
Inplane Bending (Mc)	-2524	2524	2524	-2524
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1006	1011	1006	1011
Inplane Stress Summary	-2784	4214	2634	-462

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	17	17	17	17
Shear Stress Summary	6	6	27	27

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	2784	4214	2635	504

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	449	455	449	455	449	455	449	455
Circ. Pl (SUS)	-280	-280	136	136	-130	-130	56	56
Circ. Q (SUS)	-558	558	208	-208	-1472	1472	776	-776
Long. Pm (SUS)	224	224	224	224	224	224	224	224
Long. Pl (SUS)	-107	-107	33	33	-179	-179	35	35
Long. Q (SUS)	-1092	1092	396	-396	-816	816	466	-466
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	43	43	43	43	43	43	43	43
Pm (SUS)	449.0	455.0	449.0	455.0	449.0	455.0	449.0	455.0
Pm+Pl (SUS)	170.9	176.7	585.3	591.3	319.4	325.4	505.4	511.4
Pm+Pl+Q (Total)	979	1214	800	526	1155	1798	1286	296

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	455.00	1407.00	Passed
Pm+Pl (SUS)	591.30	2110.50	Passed
Pm+Pl+Q (TOTAL)	1798.16	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1006	1011	1006	1011	1006	1011	1006	1011
Circ. Pl (SUS)	-83	-83	-9	-9	-293	-293	77	77
Circ. Q (SUS)	-416	416	-106	106	-3496	3496	1552	-1552
Long. Pm (SUS)	503	503	503	503	503	503	503	503
Long. Pl (SUS)	-203	-203	-13	-13	-120	-120	28	28
Long. Q (SUS)	-1481	1481	-463	463	-1014	1014	492	-492
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1006	1011	1006	1011	1006	1011	1006	1011
Pm+Pl (SUS)	923	928	997	1002	713	718	1083	1088
Pm+Pl+Q (Total)	1688	1782	891	1108	2783	4214	2635	505

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1011.00	1407.00	Passed
Pm+Pl (SUS)	1088.18	2110.50	Passed
Pm+Pl+Q (TOTAL)	4214.02	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	92	86	92	86	92	86	92
Circ. Pl (SUS)	-280	-280	136	136	-179	-179	35	35
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-128	-128	100	100	-128	-128	100	100
Long. Q (SUS)	-2232	2232	862	-862	-3143	3143	1773	-1773
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	73	73	73	73	73	73	73	73
Pm (SUS)	86.0	92.0	86.0	92.0	86.0	92.0	86.0	92.0
Pm+Pl (SUS)	196.9	191.1	225.9	231.7	107.4	104.0	153.1	154.7
Pm+Pl+Q (Total)	2320	2342	1008	953	3228	3146	1920	1766

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Co2 Stripper

PV Elite 28

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FileName : Co2 Stripper_Op_R4

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Nozzle Calcs.: A1-20"

Nozl: 40 12:22pm Mar 30,2026



HEXAGON

Pm (SUS)	92.00	1407.00	Passed
Pm+Pl (SUS)	231.65	2110.50	Passed
Pm+Pl+Q (TOTAL)	3228.96	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: A2-20"****From: 210**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.00	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.00	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	4906.00	mm
Design Length of Section	L	3708.8286	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	2100.0000	mm
Distance from Bottom/Left Tangent		28599.99	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		328.20	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1000.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1208.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		350.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

**Reinforcement CALCULATION, Description: A2-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 20.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 524.29 \cdot 2454.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29)$
 $= 9.3462 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 524.29 \cdot 254.0 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
 $= 0.9637 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6402 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1940.6304 mm
 Parallel to Vessel Wall, opening length d 970.3152 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.0 / 1407.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	90.688	51.889	NA
Area in Shell	A1	35.453	22.363	NA
Area in Nozzle Wall	A2	15.901	15.230	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		3.400	3.400	NA
Area in Element	A5	112.000	112.000	NA
TOTAL AREA AVAILABLE	Atot	166.754	152.993	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 29.24 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 732.5863 16.0000 mm



Based on given Pad Diameter: 1207.9999 5.1334 mm
Based on Shell or Nozzle Thickness: 1296.5858 14.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (970.3152 \cdot 9.3462 \cdot 1.0 + 2 \cdot 17.0 \cdot 9.3462 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 90.688 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 970.315 (1.0 \cdot 13.0 - 1.0 \cdot 9.346) - 2 \cdot 17.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.3462) \cdot (1 - 1.0)$$

$$= 35.453 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$

$$= (2 \cdot 32.5) \cdot (17.0 - 0.96) \cdot 1.0 / \sin(41.0)$$

$$= 15.901 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 12.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$$

$$= 3.400 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1739.9156 - 1039.9158) \cdot 16.0 \cdot 1.0$$

$$= 112.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 18.0, tr = 0.964, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.057, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -30 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 14.0, tr = 9.346, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.719, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -34 °C



Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 14.0, tr = 9.346, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.719$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve D

-48 °C

Gov. MDMT of the Nozzle

: -48 °C

Gov. MDMT of the Reinforcement Pad

: -30 °C

Gov. MDMT of the nozzle to shell joint welded assembly :

-30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)

-18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)

-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $524.29/1960.01 = 0.267$ Weld Size Calculations, Description: A2-20"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = $0.7 \cdot W_o$ mm
Pad Weld	6.5000 = $0.5 \cdot T_{minPad}$	9.8980 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (90.6879 - 35.4531 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.3462)) \cdot 1407)$$

$$= 79463.15 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (15.901 + 112.0 + 3.4 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 184740.22 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (15.901 + 0.0 + 1.44 + (4.42)) \cdot 1407$$

$$= 30617.67 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (15.901 + 0.0 + 3.4 + 112.0 + (4.42)) \cdot 1407$$

$$= 190959.14 \text{ kgf}$$
Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1039.9158 \cdot 12.0 \cdot 0.49 \cdot 1407$$

$$= 135142. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1208.0 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 183149. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:



$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 502.5577) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407 \\
 &= 264348. \text{ kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 1039.9158 \cdot 16.0 \cdot 0.74 \cdot 1407 \\
 &= 272122. \text{ kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 1039.9158 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 1407 \\
 &= 221099. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (183149 + 264348) = 447497 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (135142 + 272122 + 221099 + 0) = 628362 \text{ kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (183149 + 221099 + 0) = 404248 \text{ kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 447497 kgf, must exceed W = 79463 kgf	or W1 = 184740 kgf
Path 2-2 = 628362 kgf, must exceed W = 79463 kgf	or W2 = 30617 kgf
Path 3-3 = 404247 kgf, must exceed W = 79463 kgf	or W3 = 190959 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 618 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

Note : Checking Nozzle in plane parallel to the vessel axis.**Reinforcement CALCULATION, Description: A2-20"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$\begin{aligned}
 &= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 524.29 \cdot 2454.0 / (1407 \cdot 1.0 - 0.6 \cdot 524.29) \\
 &= 9.3462 \text{ mm}
 \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned}
 &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\
 &= 524.29 \cdot 254.0 / (1407 \cdot 1.0 + 0.4 \cdot 524.29) \\
 &= 0.9637 \text{ mm}
 \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.6402 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	948.0001 mm
Parallel to Vessel Wall, opening length	d	474.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm



The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	44.301	25.348	NA
Area in Shell	A1	17.319	10.924	NA
Area in Nozzle Wall	A2	10.424	9.984	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.440	1.440	NA
Area in Element	A5	70.400	70.400	NA
TOTAL AREA AVAILABLE	Atot	99.582	92.748	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 602.4917 16.0000 mm
 Based on given Pad Diameter: 1208.0000 3.4361 mm
 Based on Shell or Nozzle Thickness: 615.9905 14.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (474.0 \cdot 9.3462 \cdot 1.0 + 2 \cdot 17.0 \cdot 9.3462 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 44.301 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 474.0(1.0 \cdot 13.0 - 1.0 \cdot 9.346) - 2 \cdot 17.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.346) \cdot (1 - 1.0)$$

$$= 17.319 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 32.5) \cdot (17.0 - 0.96) \cdot 1.0$$

$$= 10.424 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 12.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$

$$= 1.440 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (948.0 - 508.0) \cdot 16.0 \cdot 1.0$$

$$= 70.400 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.6402 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 10.3462 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 10.3462 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.9199 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 9.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.331, \max(10.3462, 2.5)]$$

$$= 9.3312 \text{ mm}$$



Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.6402, 9.3312)$$

$$= 9.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 212.5,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3305.0 kgf/cm ²	Passed
Occasional	: 36.0,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 113.4,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: A2-20"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = 0.7 * Wo mm
Pad Weld	6.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (44.3011 - 17.3189 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.3462)) \cdot 1407)$$

$$= 39711.88 \text{ kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (10.4236 + 70.4 + 1.44 - 0.0 \cdot 1.0) \cdot 1407$$

$$= 115744.73 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (10.4236 + 0.0 + 1.44 + (4.42)) \cdot 1407$$

$$= 22910.98 \text{ kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (10.4236 + 0.0 + 1.44 + 70.4 + (4.42)) \cdot 1407$$

$$= 121963.66 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 12.0 \cdot 0.49 \cdot 1407$$

$$= 66017. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1208.0 \cdot 14.0 \cdot 0.49 \cdot 1407$$

$$= 183149. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 245.5) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407$$



= 129134. kgf

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 508.0 \cdot 16.0 \cdot 0.74 \cdot 1407$$

$$= 132932. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (14.0 - 1.0) \cdot 0.74 \cdot 1407$$

$$= 108007. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (183149 + 129134) = 312283 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (66017 + 132932 + 108007 + 0) = 306956 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (183149 + 108007 + 0) = 291156 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 312283 kgf, must exceed W = 39711 kgf or W1 = 115744 kgf
 Path 2-2 = 306955 kgf, must exceed W = 39711 kgf or W2 = 22910 kgf
 Path 3-3 = 291156 kgf, must exceed W = 39711 kgf or W3 = 121963 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 618 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 577.9073 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1604.7882 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.673 %

This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 60.13 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 474.00 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: A2-20" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4906.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.00 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.00 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	18.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	1407.00 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1208.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	524.29	kPa
Radial Load	P	3822.66	kgf
Circumferential Shear	Vc	2548.44	kgf
Longitudinal Shear	Vl	2548.44	kgf
Circumferential Moment	Mc	3567.93	kgf-m.
Longitudinal Moment	Ml	3567.93	kgf-m.
Torsional Moment	Mt	5097.03	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(2468.5(14.0 - 1.0))}$$

$$= 1099.156 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4937.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	17.000	mm
T / t	=	1.706	
d / t	=	29.882	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.343$$

Nr/P	=	0.083
Mr/P	=	0.128
M0/P	=	0.064
N0/P	=	0.160
MrD/Mc	=	0.225
NrDL/Mc	=	0.112
M0d/Mc	=	0.128
N0DL/Mc	=	0.128
MrD/Ml	=	0.149
NrDL/Ml	=	0.085
M0D/Ml	=	0.077
N0DL/Ml	=	0.250

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-72	-72	-72	-72



Outplane Bending (P)	-175	175	-175	175
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-208	-208	208	208
Outplane Bending (ML)	-383	383	383	-383
Normal Pressure Stress	449	455	449	455

Outplane Stress Summary	-390	732	793	382
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-37	-37	-37	-37
Inplane Bending (P)	-348	348	-348	348
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-70	-70	70	70
Inplane Bending (ML)	-744	744	744	-744
Inplane Pressure Stress	224	224	224	224

Inplane Stress Summary	-976	1209	654	-138
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	43	43	43	43

Shear Stress Summary	54	54	32	32
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	981	1215	800	524

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-72	-72	-72	-72
Outplane Bending (P)	-175	175	-175	175
Outplane Membrane (Mc)	-107	-107	107	107
Outplane Bending (Mc)	-641	641	641	-641
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	224	224	224	224

Outplane Stress Summary	-772	862	725	-207
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-37	-37	-37	-37
Inplane Bending (P)	-348	348	-348	348
Inplane Membrane (Mc)	-93	-93	93	93
Inplane Bending (Mc)	-1124	1124	1124	-1124
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	449	455	449	455



Inplane Stress Summary	-1154	1797	1282	-265
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	43	43	43	43

Shear Stress Summary	32	32	54	54
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1157	1798	1287	298
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-72	-72	-72	-72
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-208	-208	208	208
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	86	92	86	92

Hoop Stress Summary	-195	-188	222	228
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-685	685	-685	685
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-114	-114	114	114
Axial Bending (ML)	-1547	1547	1547	-1547
Axial Pressure Stress	43	43	43	43

Axial Stress Summary	-2319	2147	1005	-719
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	18	18	-18	-18
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	73	73	73	73

Shear Stress Summary	92	92	55	55
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	2323	2343	1009	953
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	-72	-72	-72	-72
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-107	-107	107	107
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	86	92	86	92
Hoop Stress Summary	-93	-87	120	127

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-685	685	-685	685
Axial Membrane (Mc)	-114	-114	114	114
Axial Bending (Mc)	-2458	2458	2458	-2458
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	43	43	43	43
Axial Stress Summary	-3229	3058	1915	-1629

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-18	-18	18	18
Shear due to Torsion	73	73	73	73
Shear Stress Summary	55	55	92	92

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	3230	3147	1920	1766

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(2460.5(14.0 - 1.0))}$$

$$= 1098.197 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4921.000 mm
Nozzle Outside Diameter	(d) =	1208.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	367.000 mm
T / t	=	0.035
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.776$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.048
Mr/P	=	0.072
M0/P	=	0.019
N0/P	=	0.020



MrD/Mc = 0.241
 NrDL/Mc = 0.106
 MØd/Mc = 0.072
 NØDL/Mc = 0.043
 MrD/Ml = 0.049
 NrDL/Ml = 0.055
 MØD/Ml = 0.015
 NØDL/Ml = 0.022

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-46	-46	-46	-46
Outplane Bending (P)	-261	261	-261	261
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-37	-37	37	37
Outplane Bending (ML)	-155	155	155	-155
Normal Pressure Stress	1006	1011	1006	1011
Outplane Stress Summary	506	1344	892	1109

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-108	-108	-108	-108
Inplane Bending (P)	-972	972	-972	972
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-95	-95	95	95
Inplane Bending (ML)	-509	509	509	-509
Inplane Pressure Stress	503	503	503	503
Inplane Stress Summary	-1183	1781	26	954

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	27	27	6	6

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1690	1783	892	1110

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-46	-46	-46	-46
Outplane Bending (P)	-261	261	-261	261
Outplane Membrane (Mc)	-74	-74	74	74
Outplane Bending (Mc)	-753	753	753	-753
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	503	503	503	503



Outplane Stress Summary -631 1397 1023 39

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-108	-108	-108	-108
Inplane Bending (P)	-972	972	-972	972
Inplane Membrane (Mc)	-185	-185	185	185
Inplane Bending (Mc)	-2524	2524	2524	-2524
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1006	1011	1006	1011

Inplane Stress Summary -2784 4214 2634 -462

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	17	17	17	17

Shear Stress Summary 6 6 27 27

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2784	4214	2635	504

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	449	455	449	455	449	455	449	455
Circ. Pl (SUS)	-280	-280	136	136	-130	-130	56	56
Circ. Q (SUS)	-558	558	208	-208	-1472	1472	776	-776
Long. Pm (SUS)	224	224	224	224	224	224	224	224
Long. Pl (SUS)	-107	-107	33	33	-179	-179	35	35
Long. Q (SUS)	-1092	1092	396	-396	-816	816	466	-466
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	43	43	43	43	43	43	43	43
Pm (SUS)	449.0	455.0	449.0	455.0	449.0	455.0	449.0	455.0
Pm+Pl (SUS)	170.9	176.7	585.3	591.3	319.4	325.4	505.4	511.4
Pm+Pl+Q (Total)	979	1214	800	526	1155	1798	1286	296

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	455.00	1407.00	Passed



Pm+Pl (SUS)	591.30	2110.50	Passed
Pm+Pl+Q (TOTAL)	1798.16	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1006	1011	1006	1011	1006	1011	1006	1011
Circ. Pl (SUS)	-83	-83	-9	-9	-293	-293	77	77
Circ. Q (SUS)	-416	416	-106	106	-3496	3496	1552	-1552
Long. Pm (SUS)	503	503	503	503	503	503	503	503
Long. Pl (SUS)	-203	-203	-13	-13	-120	-120	28	28
Long. Q (SUS)	-1481	1481	-463	463	-1014	1014	492	-492
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1006	1011	1006	1011	1006	1011	1006	1011
Pm+Pl (SUS)	923	928	997	1002	713	718	1083	1088
Pm+Pl+Q (Total)	1688	1782	891	1108	2783	4214	2635	505

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1011.00	1407.00	Passed
Pm+Pl (SUS)	1088.18	2110.50	Passed
Pm+Pl+Q (TOTAL)	4214.02	4221.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	86	92	86	92	86	92	86	92
Circ. Pl (SUS)	-280	-280	136	136	-179	-179	35	35
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	43	43	43	43	43	43	43	43
Long. Pl (SUS)	-128	-128	100	100	-128	-128	100	100
Long. Q (SUS)	-2232	2232	862	-862	-3143	3143	1773	-1773
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	18	18	-18	-18	-18	-18	18	18
Shear Q (SUS)	73	73	73	73	73	73	73	73
Pm (SUS)	86.0	92.0	86.0	92.0	86.0	92.0	86.0	92.0
Pm+Pl (SUS)	196.9	191.1	225.9	231.7	107.4	104.0	153.1	154.7
Pm+Pl+Q (Total)	2320	2342	1008	953	3228	3146	1920	1766

**Nozzle Stress Summation Comparison (kgf/cm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.00	1407.00	Passed
Pm+Pl (SUS)	231.65	2110.50	Passed
Pm+Pl+Q (TOTAL)	3228.96	4221.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: B-36"****From: 230**

Pressure for Reinforcement Calculations	P	524.294	kPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.71	kPa
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Elliptical Head	D	4906.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1407.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	18.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.00	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.00	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1314.4000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B-36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.709 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $524.29 \cdot 4908.0 \cdot 0.8996 / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 524.29)$
= 8.3916 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $524.29 \cdot 457.2 / (1407 \cdot 1.0 + 0.4 \cdot 524.29)$
= 1.7347 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.8786 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1760.7998 mm
Parallel to Vessel Wall, opening length d 880.3999 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1407.0/1407.1)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1407.0/1407.1)
= 1.000

Weld Strength Reduction Factor [fr4]:
= min(1, Sp/Sv)
= min(1, 1407.0/1407.1)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1., 1.)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	73.879	39.871	NA
Area in Shell	A1	22.965	17.103	NA
Area in Nozzle Wall	A2	8.395	8.316	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		4.477	4.477	NA
Area in Element	A5	71.993	71.993	NA
TOTAL AREA AVAILABLE	Atot	107.830	101.889	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 1125.7677 18.0000 mm
Based on given Pad Diameter: 1314.3999 9.5116 mm
Based on Shell or Nozzle Thickness: 1231.4515 12.0000 mm



Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (880.3999 \cdot 8.3916 \cdot 1.0 + 2 \cdot 17.0 \cdot 8.3916 \cdot 1.0 \cdot (1 - 1.))$$

$$= 73.879 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 880.4(1.0 \cdot 11.0 - 1.0 \cdot 8.392) - 2 \cdot 17.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 8.3916) \cdot (1 - 1.)$$

$$= 22.965 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 27.5) \cdot (17.0 - 1.73) \cdot 0.9999$$

$$= 8.395 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (2.5175) \cdot 1. + (0.0) \cdot 1. + 14.0^2 \cdot 1.$$

$$= 4.477 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1314.3999 - 914.4) \cdot 18.0 \cdot 0.9999$$

$$= 71.993 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.8786 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.3225 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.3225 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9191 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.331, \max(10.3225, 2.5)]$$

$$= 9.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.8786, 9.3312)$$

$$= 9.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 237.1,	Allowable	: 1407.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3280.4 kgf/cm ²	Passed
Occasional	: 67.9,	Allowable	: 1871.3 kgf/cm ²	Passed
Shear	: 107.4,	Allowable	: 984.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**

Govrn. thk, tg = 18.0, tr = 1.735, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.102, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

**Nozzle Neck to Pad Weld for the Nozzle, Curve: D**

Govrn. thk, tg = 18.0, tr = 1.735, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.0, tr = 1.735, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 8.392, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.763$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:D, Curve:B)

Govrn. thk, tg = 12.0, tr = 8.392, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.763$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-36 °C
Gov. MDMT of the Reinforcement Pad	:	-36 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $524.29 / 1960.01 = 0.267$

Weld Size Calculations, Description: B-36"

Intermediate Calc. for nozzle/shell Welds	Tmin	17.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$12.7260 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.5000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (73.8795 - 22.9647 + 2 \cdot 17.0 \cdot 0.9999 \cdot (1.0 \cdot 11.0 - 8.3916)) 1407)$
 $= 72891.64 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



$$\begin{aligned}
 &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\
 &= (8.3951 + 71.9931 + 4.4771 - 0.0 \cdot 1.) \cdot 1407 \\
 &= 119416.73 \text{ kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\
 &= (8.3951 + 0.0 + 3.2397 + (3.7396)) \cdot 1407 \\
 &= 21633.90 \text{ kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\
 &= (8.3951 + 0.0 + 4.4771 + 71.9931 + (3.7396)) \cdot 1407 \\
 &= 124678.91 \text{ kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.1416/2.0) \cdot 914.4 \cdot 18.0 \cdot 0.49 \cdot 1407 \\
 &= 178245. \text{ kgf}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.1416/2.0) \cdot 1314.4 \cdot 14.0 \cdot 0.49 \cdot 1407 \\
 &= 199281. \text{ kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.1416 \cdot 448.7) \cdot (18.0 - 1.0) \cdot 0.7 \cdot 1407 \\
 &= 236019. \text{ kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.1416/2) \cdot 914.4 \cdot 18.0 \cdot 0.74 \cdot 1407 \\
 &= 269187. \text{ kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 914.4 \cdot (12.0 - 1.0) \cdot 0.74 \cdot 1407 \\
 &= 164519. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (199281 + 236019) = 435299 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (178245 + 269187 + 164519 + 0) = 611951 \text{ kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (199281 + 164519 + 0) = 363799 \text{ kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 435299 kgf, must exceed W = 72891 kgf	or W1 = 119416 kgf
Path 2-2 = 611950 kgf, must exceed W = 72891 kgf	or W2 = 21633 kgf
Path 3-3 = 363799 kgf, must exceed W = 72891 kgf	or W3 = 124678 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 617 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 kPa

The Drop for this Nozzle is : 23.6831 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 635.6831 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

Input Echo, WRC107/537 Item 1, Description: B-36" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	8830.800	mm
Vessel Thickness	Tv	12.000	mm
Design Temperature	T1	160.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	18.000	mm
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.00	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	18.000	mm
Diameter of Reinforcing Pad	Dpad	1314.400	mm
Design Internal Pressure	Dp	524.294	kPa
Include Pressure Thrust		No	

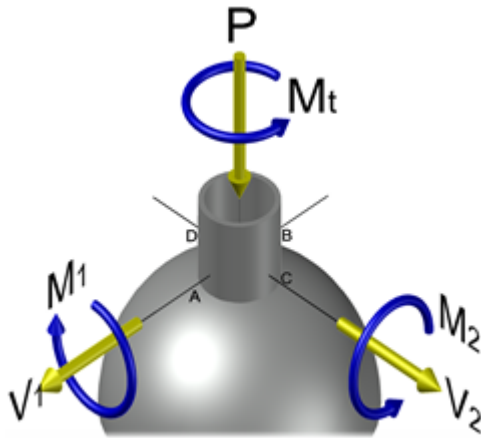
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	6880.8	kgf
Longitudinal Shear (SUS)	(V1) V1	4587.2	kgf
Circumferential Shear (SUS)	(Vc) V2	4587.2	kgf
Circumferential Moment (SUS)	(Mc) M1	11560.1	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	11560.1	kgf-m.
Torsional Moment (SUS)	Mt	16514.4	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 914.4 + 2 \cdot 1.65 \cdot \sqrt{(4421.9 (12.0 - 1.0))}$$

$$= 1642.205 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	6880.8	kgf
Circumferential Shear	(VC)	V2	4587.2	kgf
Longitudinal Shear	(VL)	V1	4587.2	kgf
Circumferential Moment	(MC)	M1	11560.1	kgf-m.
Longitudinal Moment	(ML)	M2	11560.1	kgf-m.
Torsional Moment		MT	16514.4	kgf-m.

Dimensionless Param: U = 1.28 TAU = 26.39 RHO = 4.00 (1.71)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 7	0.02142	(A,B,C,D)
$M(x) / P$	SP 7	0.00661	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.02856	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 7	0.01022	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.02856	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 7	0.01022	(A,B,C,D)
$N(y) * T / P$	SP 7	0.15784	(A,B,C,D)
$M(y) / P$	SP 7	0.01607	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.18197	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 7	0.03065	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.18197	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 7	0.03065	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-17.5	-17.5	-17.5	-17.5	-17.5	-17.5	-17.5	-17.5
Rad.	Bend. P	-32.4	32.4	-32.4	32.4	-32.4	32.4	-32.4	32.4
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-109.5	-109.5	109.5	109.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-235.1	235.1	235.1	-235.1
Rad.	Memb. ML	-109.5	-109.5	109.5	109.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-235.1	235.1	235.1	-235.1	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-394.6	140.5	294.7	-110.6	-394.6	140.5	294.7	-110.6
Tang. Memb. P	-129.1	-129.1	-129.1	-129.1	-129.1	-129.1	-129.1	-129.1
Tang. Bend. P	-78.9	78.9	-78.9	78.9	-78.9	78.9	-78.9	78.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-697.8	-697.8	697.8	697.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-705.1	705.1	705.1	-705.1
Tang. Memb. ML	-697.8	-697.8	697.8	697.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-705.1	705.1	705.1	-705.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-1610.9	-42.9	1194.9	-57.6	-1610.9	-42.9	1194.9	-57.6
Shear VC	11.0	11.0	-11.0	-11.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-11.0	-11.0	11.0	11.0
Shear MT	43.4	43.4	43.4	43.4	43.4	43.4	43.4	43.4
Tot. Shear	54.4	54.4	32.3	32.3	32.3	32.3	54.4	54.4
Str. Int.	1613.3	213.2	1196.0	126.0	1611.8	194.5	1198.1	144.6

Unitless Prm: U = 2.98 TAU = 0.00 (38.16) RHO = 0.00 (0.65)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01053	(A,B,C,D)
N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00317	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (kgf/cm²)

Type of Stress		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-123.7	-123.7	-123.7	-123.7	-123.7	-123.7	-123.7	-123.7
Rad.	Bend. P	-337.8	337.8	-337.8	337.8	-337.8	337.8	-337.8	337.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-732.7	-732.7	732.7	732.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-2736.1	2736.1	2736.1	-2736.1
Rad.	Memb. ML	-732.7	-732.7	732.7	732.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-2736.1	2736.1	2736.1	-2736.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-3930.3	2217.5	3007.4	-1789.3	-3930.3	2217.5	3007.4	-1789.3
Tang. Memb. P		-37.0	-37.0	-37.0	-37.0	-37.0	-37.0	-37.0	-37.0
Tang. Bend. P		-102.4	102.4	-102.4	102.4	-102.4	102.4	-102.4	102.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-219.5	-219.5	219.5	219.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-825.2	825.2	825.2	-825.2
Tang. Memb. ML		-219.5	-219.5	219.5	219.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-825.2	825.2	825.2	-825.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-1184.1	671.0	905.3	-540.3	-1184.1	671.0	905.3	-540.3
Shear VC		20.2	20.2	-20.2	-20.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-20.2	-20.2	20.2	20.2
Shear MT		55.3	55.3	55.3	55.3	55.3	55.3	55.3	55.3



Tot. Shear	75.5	75.5	35.1	35.1	35.1	35.1	75.5	75.5
Str. Int.	3932.4	2221.2	3008.0	1790.3	3930.8	2218.3	3010.1	1793.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (kgf/cm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		407.1	407.1	407.1	407.1	407.1	407.1	407.1	407.1
Rad. Pl (SUS)		-127.0	-127.0	92.0	92.0	-127.0	-127.0	92.0	92.0
Rad. Q (SUS)		-267.5	267.5	202.7	-202.7	-267.5	267.5	202.7	-202.7
Long. Pm (SUS)		407.1	407.1	407.1	407.1	407.1	407.1	407.1	407.1
Long. Pl (SUS)		-826.9	-826.9	568.6	568.6	-826.9	-826.9	568.6	568.6
Long. Q (SUS)		-784.0	784.0	626.2	-626.2	-784.0	784.0	626.2	-626.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		11.0	11.0	-11.0	-11.0	-11.0	-11.0	11.0	11.0
Shear Q (SUS)		43.4	43.4	43.4	43.4	43.4	43.4	43.4	43.4
Pm (SUS)		407.1	407.1	407.1	407.1	407.1	407.1	407.1	407.1
Pm+Pl (SUS)		700.2	700.2	976.0	976.0	700.2	700.2	976.0	976.0
Pm+Pl+Q (Total)		1221.2	562.5	1603.1	364.8	1218.0	553.1	1605.2	383.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	407.11	1407.13	Passed
Pm+Pl (SUS)	976.00	2110.70	Passed
Pm+Pl+Q (TOTAL)	1605.24	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3
Rad. Pl (SUS)		-856.4	-856.4	609.0	609.0	-856.4	-856.4	609.0	609.0
Rad. Q (SUS)		-3073.9	3073.9	2398.3	-2398.3	-3073.9	3073.9	2398.3	-2398.3
Long. Pm (SUS)		1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3
Long. Pl (SUS)		-256.5	-256.5	182.5	182.5	-256.5	-256.5	182.5	182.5
Long. Q (SUS)		-927.5	927.5	722.8	-722.8	-927.5	927.5	722.8	-722.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		20.2	20.2	-20.2	-20.2	-20.2	-20.2	20.2	20.2
Shear Q (SUS)		55.3	55.3	55.3	55.3	55.3	55.3	55.3	55.3
Pm (SUS)		1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3	1073.3
Pm+Pl (SUS)		817.4	817.4	1683.3	1683.3	817.4	817.4	1683.3	1683.3
Pm+Pl+Q (Total)		2859.1	3294.5	4081.2	1251.0	2857.5	3291.6	4083.3	1258.1

**Vessel Stress Summation Comparison (kgf/cm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1073.26	1407.13	Passed
Pm+Pl (SUS)	1683.26	2110.70	Passed
Pm+Pl+Q (TOTAL)	4083.35	4221.40	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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